



**Politecnico
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Project 1

Quarter Car Modeling and Control

Driver Assistance Systems Design

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Contents

1	Introduction - 1st Project	1
2	Quarter-Car Model – Theoretical Background	1
3	Exercise 1 – Frequency-Response Analysis	2
3.I	Theory	2
3.II	Implementation	3
3.III	Results	4
4	Exercise 2.1 – Road-Profile Excitation	5
4.I	Theory	5
4.II	Implementation – Simulink model	6
4.III	Results	7
5	Exercise 2.2 – “Real-World” Skyhook on a Semi-Active Suspension	7
5.I	Theory	7
5.II	Implementation	8
5.III	Results	9
6	Exercise 3 (Part 1) – Nonlinear Semi-Active Suspension	10
6.I	Theory	10
6.II	Implementation	11
6.III	Results – K_p tuning	12
6.IV	Results – I_c tuning at fixed $K_p = 0.3$	13
7	Exercise 3 (Part 2) – Bode of the Linearised System	15
7.I	Theory	15
7.II	Implementation	15
7.III	Results	16
8	Conclusions	16
9	Introduction - 2nd Project	18
10	Lateral Dynamics – Theoretical Background	18
11	Part 1 – Lateral Dynamics	19
11.I	Cornering stiffness extraction	19
11.II	Stability derivatives and state-space at 80 km/h	20
11.III	Root loci versus speed	20
11.IV	Step steer at 80 km/h	21
11.V	Steady-state gains	22
12	Part 2 – Lane-Keeping Control	23

12.I	Exercise 1 – DST and DSTP open-loop models	24
12.II	Exercise 2 – PDF and PIDF lane-keeping controllers	25
12.III	Exercise 3 – Stanley kinematic controller	31
13	Controllers Benchmark	33
14	Conclusions	34

1 Introduction - 1st Project

The vertical dynamics of a road vehicle dominates ride comfort, road holding and tyre wear. The classical *quarter-car* model – two masses linked by a spring–damper element and connected to the ground through the tyre stiffness and damping – captures the essential phenomena with only two degrees of freedom, while remaining simple enough to support analytical insight and controller design.

This project investigates the quarter-car model in three progressive steps:

- **Exercise 1** – frequency-domain analysis of the linear model under passive, skyhook and groundhook configurations;
- **Exercise 2** – time-domain simulation under random road excitation, and implementation of a *real-world* skyhook law on a semi-active suspension;
- **Exercise 3** – nonlinear semi-active suspension with a current-controlled magnetorheological damper, parameter tuning under bump and random profiles, and Bode analysis of the linearized system around two extreme operating points.

The same nominal vehicle parameters are used throughout, summarised in Table 2.

Symbol	Value	Description
m_s	400 kg	Sprung mass (chassis)
m_u	50 kg	Unsprung mass (wheel + suspension)
K	25 000 N/m	Suspension stiffness
P	150 000 N/m	Tyre vertical stiffness
c	1500 Ns/m	Passive damping coefficient
c_t	0 Ns/m	Tyre damping coefficient

Table 1: Quarter-car nominal parameters.

2 Quarter-Car Model – Theoretical Background

The 2-DOF passive quarter-car equations of motion are

$$m_s \ddot{z}_s + c(\dot{z}_s - \dot{z}_u) + K(z_s - z_u) = 0, \quad (1)$$

$$m_u \ddot{z}_u + c(\dot{z}_u - \dot{z}_s) + c_t(\dot{z}_u - \dot{r}) + K(z_u - z_s) + P(z_u - r) = 0, \quad (2)$$

where z_s and z_u are the vertical displacements of the sprung and unsprung masses and r is the road profile.

Skyhook and groundhook. An *ideal* skyhook damper c_s would connect the sprung mass to an inertial reference above the vehicle, removing low-frequency oscillations of

the chassis. An *ideal* groundhook damper c_g would connect the unsprung mass to the ground, reducing tyre-force fluctuations and improving handling. Both are physically unfeasible, but the same dynamic effect can be reproduced by an actuator placed between the two masses producing the force

$$F = -(c_s + c)\dot{z}_s + (c + c_g)\dot{z}_u. \quad (3)$$

Damping ratios. Defining the critical damping $c_{cr} = 2\sqrt{Km_s}$, all damping coefficients are normalised:

$$\zeta = \frac{c}{c_{cr}}, \quad \zeta_t = \frac{c_t}{c_{cr}}, \quad \zeta_s = \frac{c_s}{c_{cr}}, \quad \zeta_g = \frac{c_g}{c_{cr}}. \quad (4)$$

The control logic is conveniently expressed through two parameters,

$$\zeta_m = \frac{\zeta_s + \zeta_g}{2}, \quad \zeta_d = \frac{\zeta_s - \zeta_g}{2}, \quad \zeta_0 = \zeta + \zeta_m, \quad (5)$$

where $\zeta_d > 0$ corresponds to skyhook behaviour and $\zeta_d < 0$ to groundhook behaviour.

State-space representation. Selecting $\mathbf{x} = [z_s, \dot{z}_s, z_u, \dot{z}_u, r]^T$ and input $u = \dot{r}$, the system becomes

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu, \quad \mathbf{y} = C\mathbf{x} + Du, \quad (6)$$

with $\mathbf{y} = [\ddot{z}_s, F_t, F_{act}]^T$ (comfort, handling, actuator force). The road state r is appended in order to use $\dot{r}$ as a true input; this introduces a nil eigenvalue in A . The matrices used in the MATLAB scripts are generated symbolically in `symbolic_state_space_gen.m`.

3 Exercise 1 – Frequency-Response Analysis

3.1 Theory

Three transfer functions are required by the assignment:

$$\frac{\ddot{Z}_s}{R}, \quad \frac{F_t}{PR}, \quad \frac{F_{act}}{PR}. \quad (7)$$

Since the input of the state-space model is $\dot{r}$, MATLAB's `bode` computes $X/\dot{R}$. Because $\dot{R}(j\omega) = j\omega R(j\omega)$, the conversion

$$\left| \frac{X}{R} \right| = \omega \cdot \left| \frac{X}{\dot{R}} \right| \quad (8)$$

is required, and the second and third transfer functions are further normalised by P . Each working condition is also analysed via the eigenvalues of A , which describe damping ratios and natural frequencies of the closed-loop system.

3.II Implementation

Three damping configurations are looped over:

```
1 switch damp
2     case 1    % passive damper
3         zeta0 = zeta;    zeta_d = 0;
4     case 2    % skyhook
5         zeta0 = 0.43;    zeta_d = 0.25;
6     case 3    % groundhook
7         zeta0 = 0.43;    zeta_d = -0.25;
8 end
```

For each case the A, B, C, D matrices are built according to Section 10 and a frequency vector $f = \text{linspace}(10^{-3}, 100, 1000)$ rad/s is fed to `bode`. The magnitudes returned are multiplied by ω (and divided by P for the second and third outputs):

```
1 sys = ss(A,B,C,D);
2 [mag, phase, wout] = bode(sys, f);
3 if n == 1
4     plot(wout, squeeze(mag(n,:,:)) .* wout)    % zddot_s / r
5 else
6     plot(wout, squeeze(mag(n,:,:)) .* wout / P)    % F / (P*r)
7 end
```

The eigenvalues of A are then computed by `eig(A)` and plotted on the complex plane.

3.III Results

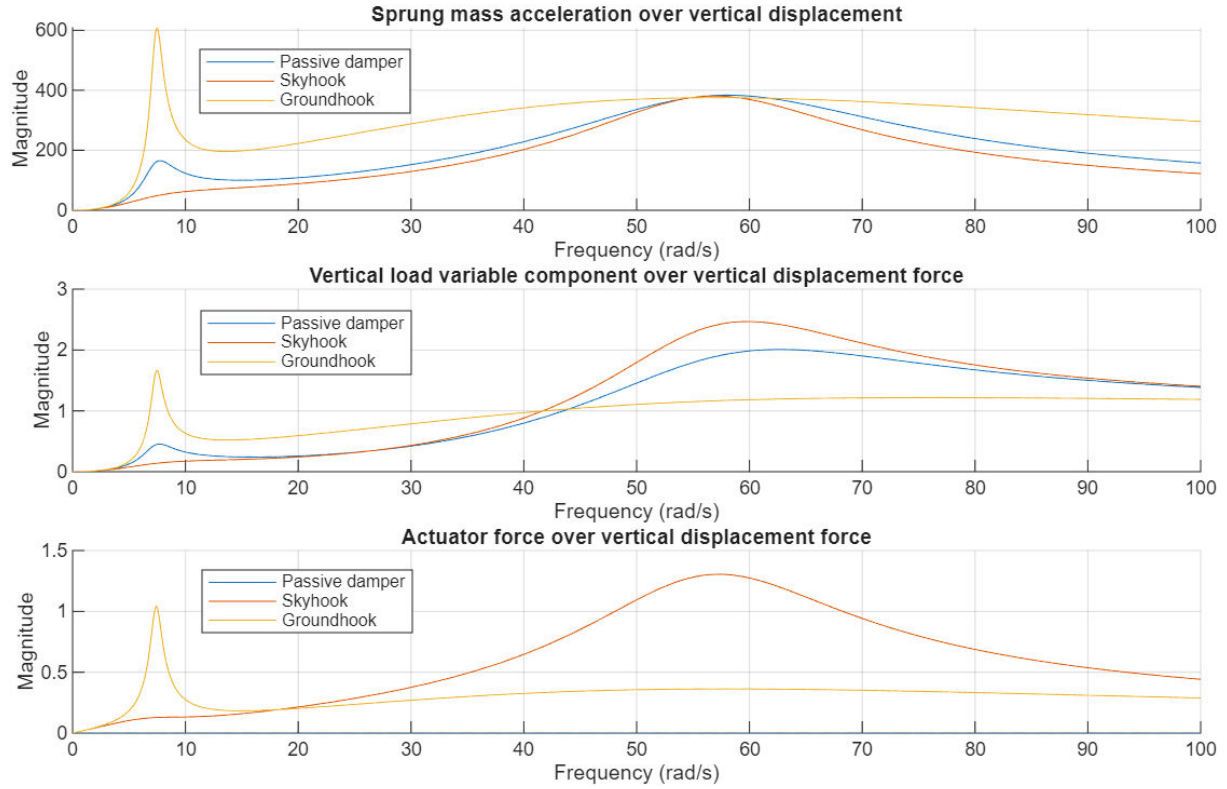


Figure 1: Frequency response of the quarter-car model under passive damper, skyhook and groundhook configurations: sprung mass acceleration $\ddot{z}_s/r$ (top), normalised tyre-force variation $F_t/(Pr)$ (middle) and normalised actuator force $F_{act}/(Pr)$ (bottom).

Figure 1 highlights the three classical trade-offs:

- Around the *sprung mass* natural frequency (≈ 7 rad/s) the skyhook strongly attenuates the peak of $\ddot{z}_s/r$, while the groundhook amplifies it. This confirms the comfort-oriented nature of skyhook control.
- Around the *unsprung mass* resonance (≈ 55 rad/s) the groundhook flattens the tyre-force peak, while the skyhook keeps it almost as large as in the passive case – a clear handling improvement of groundhook.
- The actuator force is null in the passive case and exhibits a peak at one of the two resonances depending on which working condition is chosen (sprung resonance for skyhook, unsprung resonance for groundhook), giving a quick estimate of the actuator authority required.



Figure 2: Eigenvalues (pole map) of the linearised quarter-car under the three working conditions.

The pole map in Figure 2 shows two pairs of complex poles for the passive damper (one for the chassis mode at ~ 7 rad/s, one for the wheel mode at ~ 56 rad/s). The **skyhook** configuration shifts the chassis poles further into the left half plane (more damping on $\ddot{z}_s$) while leaving the wheel poles almost unchanged. The **groundhook**, conversely, pushes the wheel poles deep to the left ($\sigma \approx -44$) and relaxes the chassis poles, which become almost critically damped on the real axis. The nil pole at the origin in each case comes from the augmented road state r .

4 Exercise 2.1 – Road-Profile Excitation

4.1 Theory

The road profile is now a stochastic disturbance representative of a real surface. The model is fed with band-limited white noise filtered to obtain a frequency content consistent with vehicle speed: the spatial PSD of the road becomes a temporal PSD that scales with V (the longitudinal speed). The actuator force-velocity diagram

$$F_{act} \quad \text{vs} \quad v_{rel} = \dot{z}_s - \dot{z}_u \quad (9)$$

characterises the type of damper: a linear viscous damper would lie on a straight line, while skyhook/groundhook produce dispersed clouds because the force depends on absolute velocities rather than on the relative one.

4.II Implementation – Simulink model

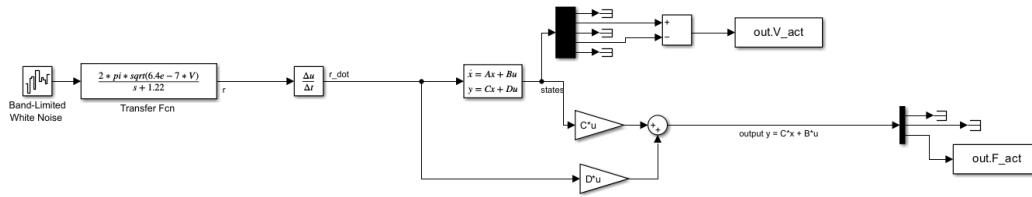


Figure 3: Simulink model Exercise2SIM.slx used in Exercise 2.1.

The Simulink model in Figure 3 feeds a *Band-Limited White Noise* block through the road shaping filter

$$G_r(s) = \frac{2\pi\sqrt{6.4 \cdot 10^{-7} V}}{s + 1.22}, \quad (10)$$

whose gain scales with the vehicle speed V . The derivative block produces $\dot{r}$, which is the input of the state-space block implementing the (A, B, C, D) matrices written by the MATLAB script. The outputs are demultiplexed: the relative velocity is computed as $\dot{z}_s - \dot{z}_u$ and logged together with the actuator force through V_{act} and F_{act} To-Workspace blocks. The MATLAB driver sweeps the three vehicle speeds and the three damping configurations:

```

1 speeds = [10 30 50];
2 for n = 1:length(speeds)
3     V = speeds(n);
4     for damp = 1:3
5         % set zeta0, zeta_d, build A,B,C,D
6         out = sim("Exercise2SIM.slx");
7         plot(out.V_act, out.F_act, '.');
8     end
9 end

```

4.III Results

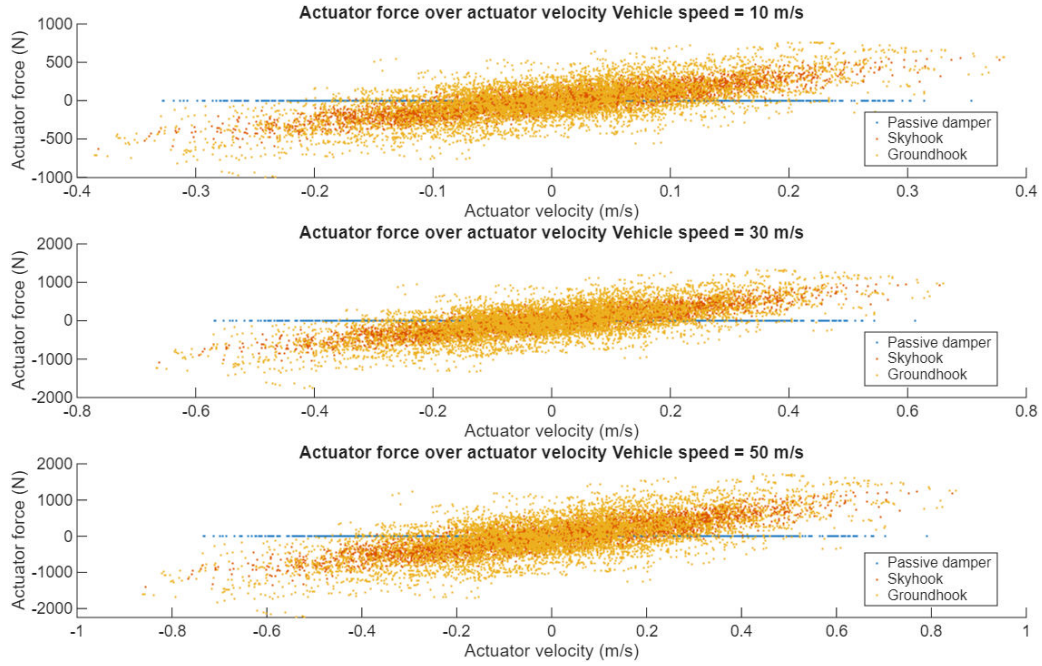


Figure 4: Actuator force vs. relative velocity, for the three control strategies at $V = 10, 30, 50$ m/s.

In Figure 4 the passive damper produces a flat line at 0 N, since the actuator is not implemented in this configuration. The skyhook and groundhook clouds are noticeably broader: the same relative velocity can correspond to widely different forces depending on the absolute sprung and unsprung velocities. As the speed increases the road excitation becomes more energetic, and both the relative velocity range and the actuator force span grow accordingly. This confirms that the actuator must deliver larger forces at higher speeds, information that is critical for component sizing.

5 Exercise 2.2 – “Real-World” Skyhook on a Semi-Active Suspension

5.1 Theory

A semi-active damper can only dissipate energy: the force must be opposite to the relative velocity. The *real-world skyhook* control law selects between an active-like force and a purely passive one according to a sign condition:

$$F = \begin{cases} -(c_s + c)\dot{z}_s + c\dot{z}_u, & \text{if } \dot{z}_s(\dot{z}_s - \dot{z}_u) > 0, \\ -c(\dot{z}_s - \dot{z}_u), & \text{otherwise.} \end{cases} \quad (11)$$

The first branch is active when the chassis is moving in the same direction as the damper would naturally push it. For example, when the chassis is moving upwards and simultaneously the damper elongates, or when the chassis is moving downwards and the damper compresses; the second branch keeps the damper passive otherwise. The constraint $c v_{rel} \leq F \leq c_{max} v_{rel}$ is then imposed by a saturation, with $c_{max} = 4000$ Ns/m.

5.II Implementation

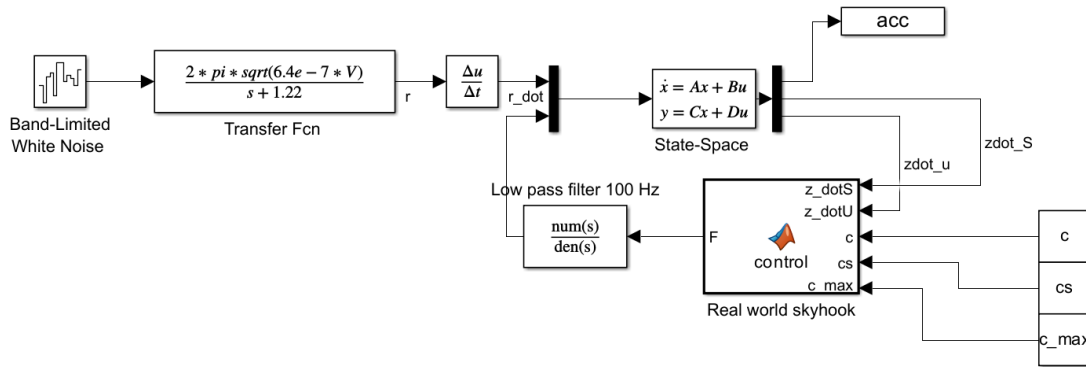


Figure 5: Simulink model `exercise2_2SIM.slx`: the real-world skyhook law is implemented inside the *MATLAB Function* block and applied as a second input to the state-space, after a 100 Hz low-pass filter that prevents chattering.

In the model of Figure 5 the state-space block uses the passive matrices (no damping inside A), while the actuator force F enters B as a separate input. A MATLAB Function block implements (11) using the sprung and unsprung velocities fed back from the state-space. A first-order low-pass filter with cut frequency 100 Hz smooths the discontinuity introduced by the switching law. The MATLAB driver sweeps the tuning parameter c_s :

```

1 cs_tune = [0, 500, 1000, 2000, 3000, 4000];
2 for i = 1:length(cs_tune)
3     cs = cs_tune(i);
4     out = sim('exercise2_2SIM.slx');
5     rms_acc(i) = sqrt(mean(out.acc.^2));
6 end

```

5.III Results

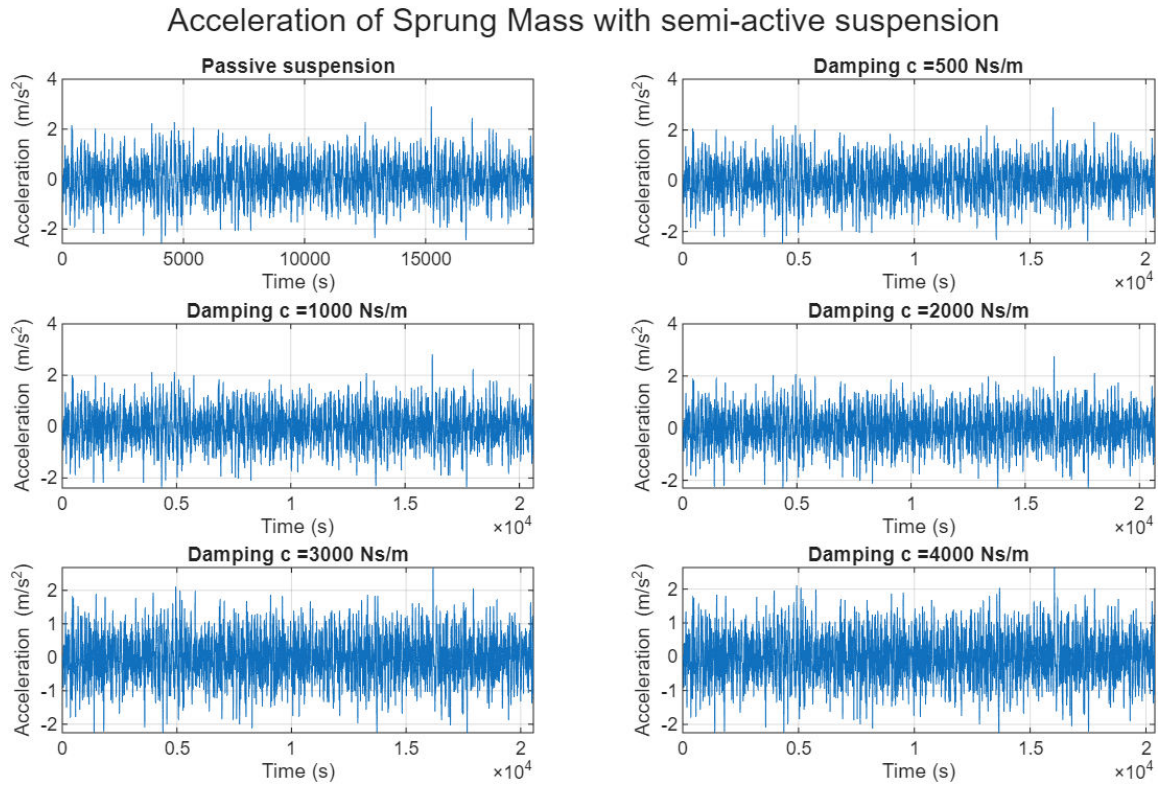


Figure 6: Sprung-mass acceleration time history for $c_s = 0$ (passive) and five semi-active configurations.

```

-----REAL WORLD SKYHOOK IMPLEMENTATION - RESULTS-----
RMS Acceleration of Sprung Mass with damping c: 0 Ns/m is: 0.72 m/s^2

RMS Acceleration of Sprung Mass with damping c: 500 Ns/m is: 0.69 m/s^2
RMS Acceleration of Sprung Mass is reduced by: 4.04 %

RMS Acceleration of Sprung Mass with damping c: 1000 Ns/m is: 0.68 m/s^2
RMS Acceleration of Sprung Mass is reduced by: 6.43 %

RMS Acceleration of Sprung Mass with damping c: 2000 Ns/m is: 0.66 m/s^2
RMS Acceleration of Sprung Mass is reduced by: 8.20 %

RMS Acceleration of Sprung Mass with damping c: 3000 Ns/m is: 0.67 m/s^2
RMS Acceleration of Sprung Mass is reduced by: 7.88 %

RMS Acceleration of Sprung Mass with damping c: 4000 Ns/m is: 0.67 m/s^2
RMS Acceleration of Sprung Mass is reduced by: 7.15 %

-----MAXIMUM AND MINIMUM ACCELERATIONS-----
The worst RMS acceleration is: 0.72 m/s^2
obtained with a damping cs= 0 Ns/m
The best RMS acceleration is: 0.66 m/s^2
obtained with a damping cs= 2000 Ns/m

```

Figure 7: RMS sprung-mass acceleration and percentage reduction versus passive case.

Figures 6 and 7 show that even a modest semi-active correction reduces the RMS acceleration: from 0.72 m/s^2 (passive, $c_s = 0$) to a minimum of 0.66 m/s^2 at $c_s = 2000 \text{ Ns/m}$, that is a **8.2%** comfort improvement. Increasing c_s further is counter-productive: the curves at $c_s = 3000$ and 4000 Ns/m show slightly higher RMS values, because the saturation activates more often and the switching law spends more time in the passive branch. The optimum is therefore $c_s \approx 2000 \text{ Ns/m}$, which sits roughly half-way between the passive coefficient and the prescribed c_{max} .

6 Exercise 3 (Part 1) – Nonlinear Semi-Active Suspension

6.1 Theory

A magneto-rheological (MR) damper provides a nonlinear force $u(I, v_{wc})$ as a function of the control current I and the piston velocity $v_{wc} = \dot{z}_u - \dot{z}_s$. The nonlinear quarter-car equations are

$$m_u \ddot{z}_u = -u(I, v_{wc}) - K(z_u - z_s) - P(z_u - r) - c_t(\dot{z}_u - \dot{r}), \quad (12)$$

$$m_s \ddot{z}_s = u(I, v_{wc}) + K(z_u - z_s). \quad (13)$$

The outputs are the chassis vertical acceleration $y_1 = \frac{c}{m_s}(\dot{z}_u - \dot{z}_s) + \frac{K}{m_s}(z_u - z_s)$ and the dynamic tyre-road force $y_2 = P(z_u - r)$. A proportional feedback law manipulates the current,

$$I = \text{sat}(I_c - K_p y_1^d), \quad I \in [0.29, 1.6] \text{ A}, \quad (14)$$

where y_1^d is y_1 delayed by a Unit Delay block with $\tau = 0.005 \text{ s}$, I_c is the upper-bound current and K_p is the tuning gain. The closed-loop adapts the damping in real time: when the chassis accelerates upward, the current is reduced, the damper softens and the disturbance is filtered; when the chassis accelerates downward, the current is increased and the damper hardens.

6.11 Implementation

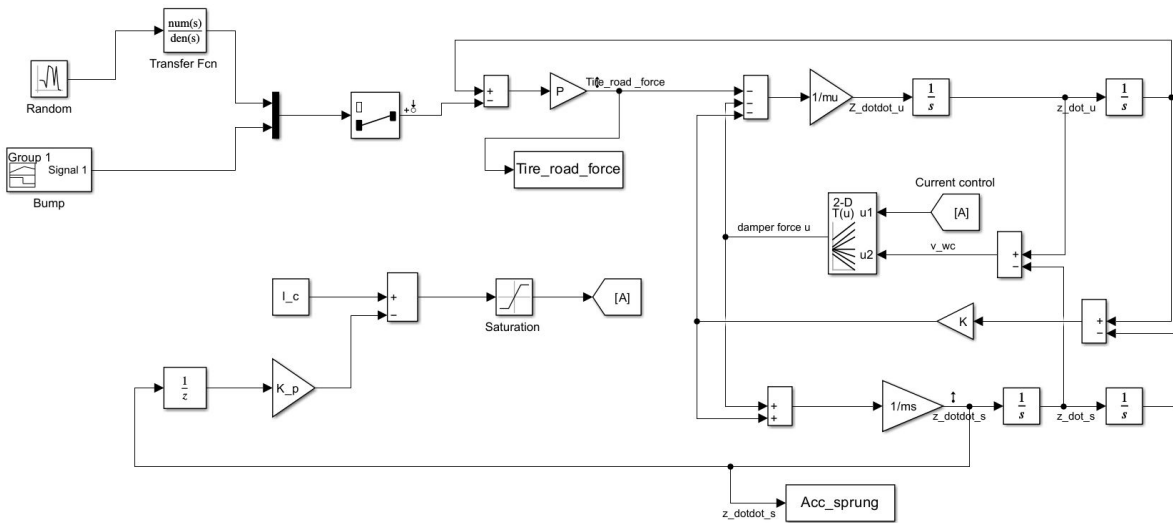


Figure 8: Simulink model `exercise_3sim.slx`. From left to right: random/bump selector, 2-DOF nonlinear quarter-car (four integrators), 2-D lookup table for the MR damper, current control loop with Unit Delay, gain K_p and Saturation.

The model in Figure 8 contains four integrators for the states $\dot{z}_s, z_s, \dot{z}_u, z_u$, the algebraic relations for y_1 and y_2 , the 2-D lookup table that returns the actuator force, and the discrete control branch on the right. The selector at the input switches between a random profile (band-limited noise shaped as in Exercise 2.1) and a bump shape, controlled by the workspace variable `inp`.

K_p tuning at $I_c = 1.6$ A. The proportional gain is swept in $K_p \in [0.2, 0.6]$ and four performance indicators are recorded for each road profile: RMS and maximum chassis acceleration, RMS and maximum tyre-ground force.

```

1 K_p_tuning = linspace(0.2,0.6,20);
2 for k = 1:2           % 1 = random, 2 = bump
3     inp = k;
4     for i = 1:length(K_p_tuning)
5         K_p = K_p_tuning(i);
6         sim("exercise_3sim.slx");
7         acc_RMS(i) = sqrt(mean(Acc_sprung.^2));
8         force_RMS(i) = sqrt(mean(Tire_road_force.^2));
9         acc_Max(i) = max(Acc_sprung);
10        force_Max(i) = max(Tire_road_force);
11    end
12 end

```

6.III Results – K_p tuning

K_p optimization for random road profile

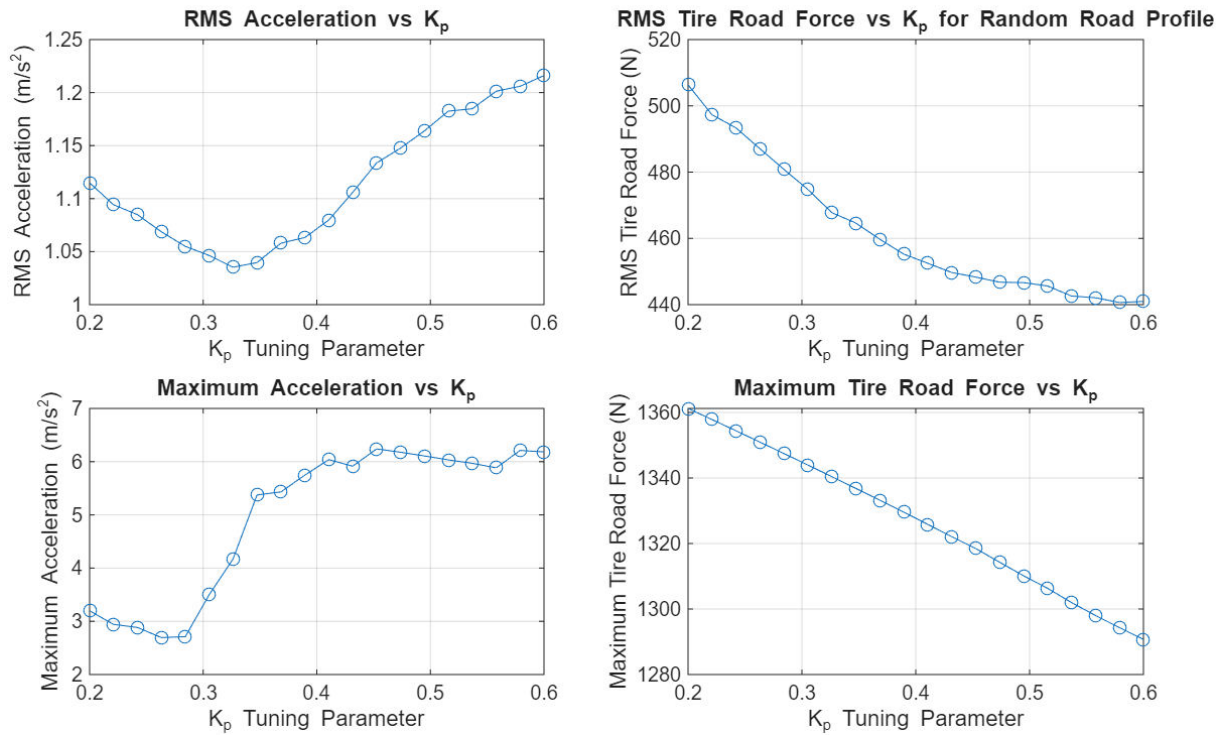


Figure 9: K_p optimisation for a random road profile, $I_c = 1.6$ A.

K_p optimization for bump road profile

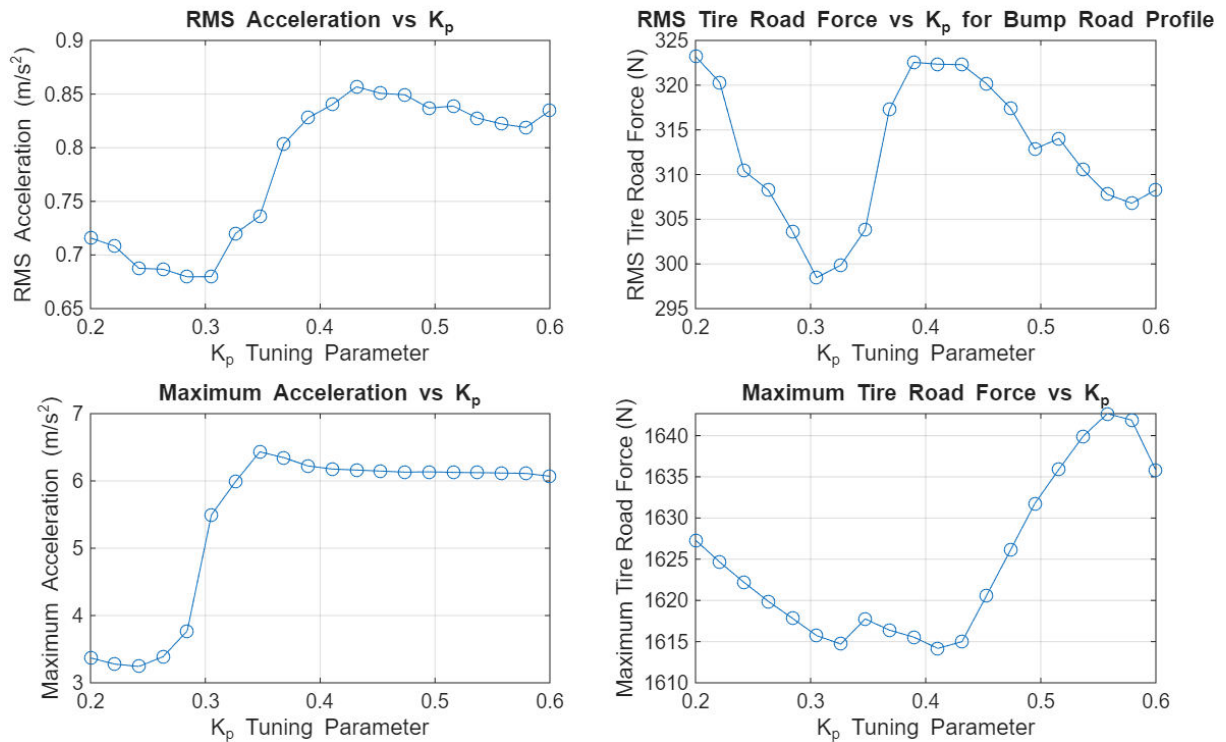


Figure 10: K_p optimisation for a bump road profile, $I_c = 1.6$ A.

For the **random profile** (Figure 9) the four metrics point to different optima:

- RMS acceleration minimised at $K_p = 0.33$ (1.04 m/s^2);
- RMS tyre force monotonically decreasing, lowest at $K_p = 0.58$ ($\approx 440 \text{ N}$);
- maximum acceleration minimum at $K_p = 0.26$ (2.69 m/s^2), then jumps up sharply for $K_p \geq 0.3$;
- maximum tyre force monotonically decreasing.

The classic *comfort-vs-handling* trade-off emerges: comfort metrics favour low K_p (soft damper, large body excursion absorbed by the suspension), while handling metrics prefer high K_p (firm damper, smaller wheel hop).

For the **bump profile** (Figure 10) the four optima are much closer: $K_p \approx 0.24$ – 0.31 minimises both RMS metrics; the maximum tyre force has its lowest value at $K_p \approx 0.41$. The transient nature of a bump excitation aligns comfort and handling indicators on a common interval. A reasonable single value covering both scenarios is $K_p \approx 0.30$: near-optimal on the bump and close to the RMS-acceleration optimum on the random profile, accepting a sub-optimal but still acceptable RMS force.

6.IV Results – I_c tuning at fixed $K_p = 0.3$

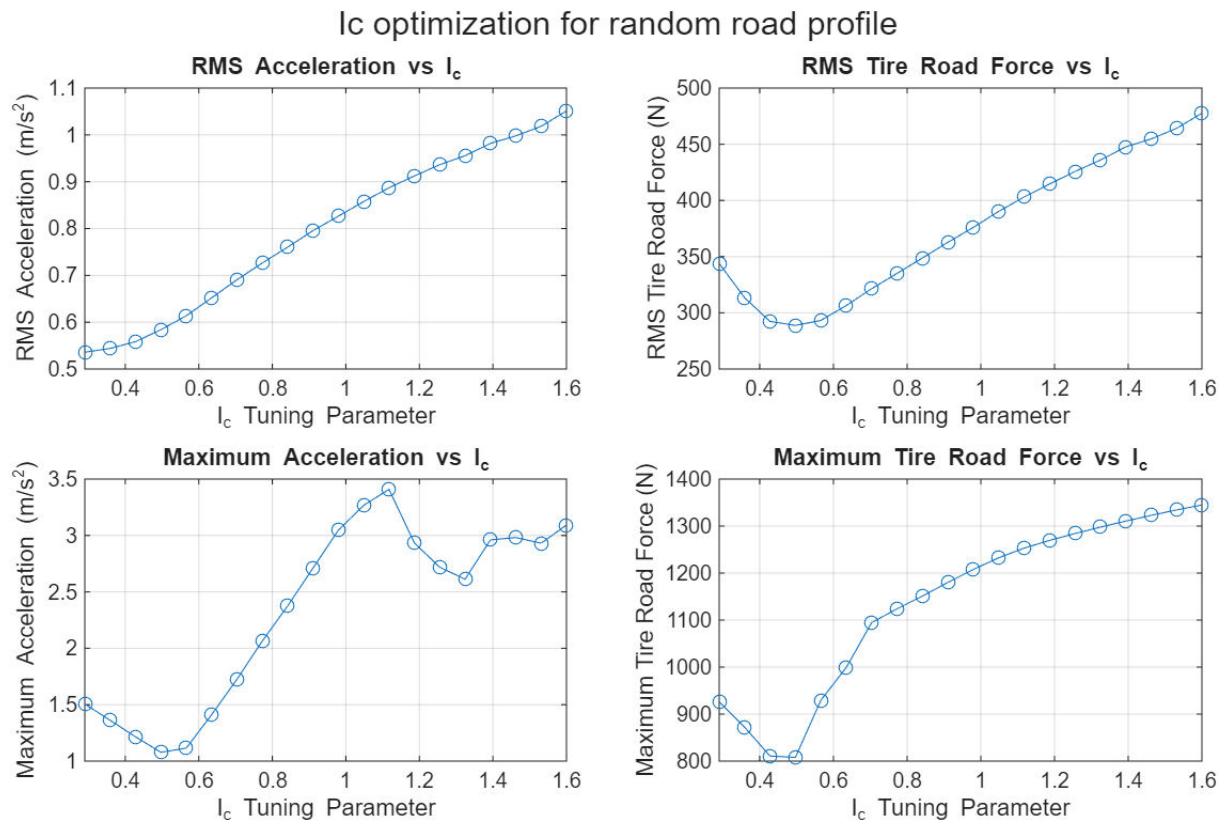


Figure 11: I_c optimisation for a random road profile, $K_p = 0.3$.

I_c optimization for bump road profile

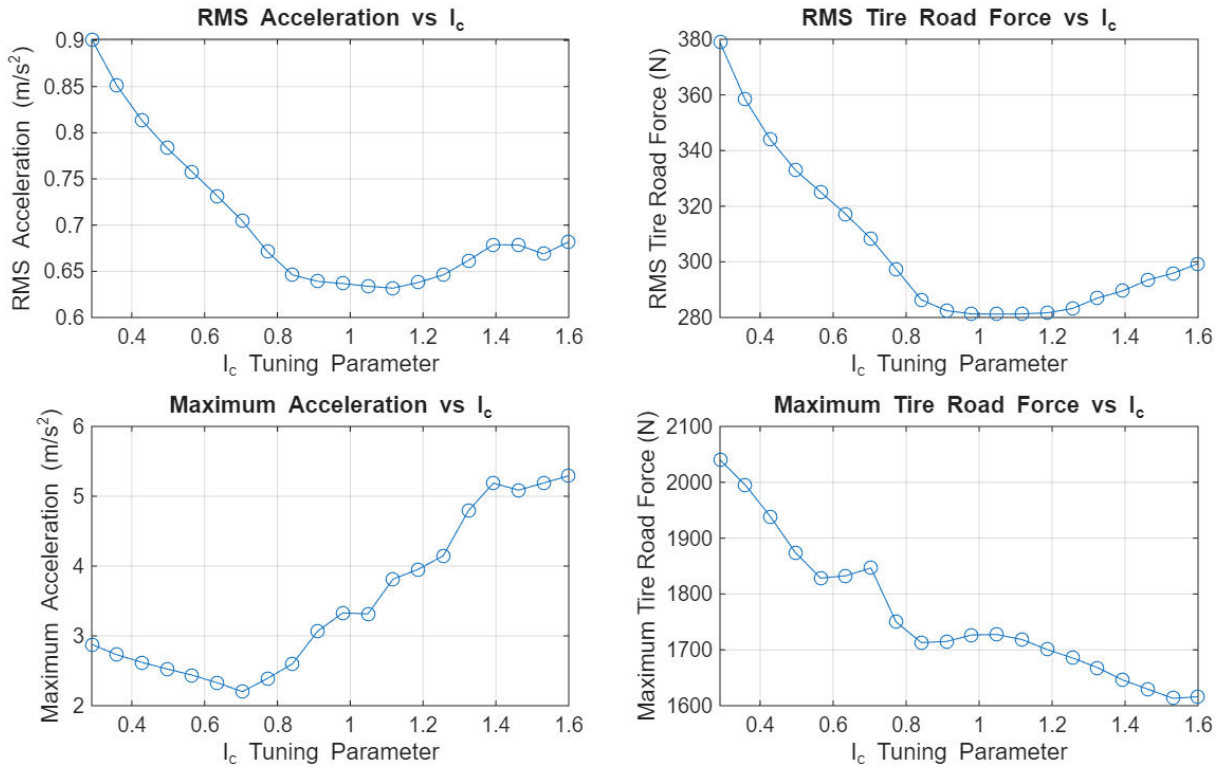


Figure 12: I_c optimisation for a bump road profile, $K_p = 0.3$.

For the **random profile** (Figure 11) the RMS acceleration grows almost linearly with I_c , from 0.53 to 1.05 m/s²: a softer baseline damper is better for comfort. The RMS tyre force has a clear minimum at $I_c \approx 0.5$ A; below this point the damper is too compliant and the wheel oscillates excessively, above it the sprung-mass motion injects more energy into the tyre. The maximum indicators show a similar non-monotonic behaviour with a minimum around 0.5 A.

For the **bump profile** (Figure 12) the picture changes: the RMS acceleration is minimised around $I_c \approx 1.0$ A, and so is the RMS tyre force. A bump is a deterministic, low-energy impulse, and extra damping helps to absorb it quickly without amplifying tyre hopping.

Take-away. The optimal current depends strongly on the type of excitation. A practical implementation therefore requires either a road-profile classifier (e.g., a low-pass-filtered estimator of road harshness) or, alternatively, a robust compromise such as $I_c \approx 0.5$ A on random pavement and $I_c \approx 1.0$ A in transient manoeuvres.

7 Exercise 3 (Part 2) – Bode of the Linearised System

7.I Theory

The nonlinear system can be linearised in Simulink through the *Linear Analysis Tool*: input/output perturbation points are added on the road derivative and on the two outputs y_1, y_2 ; Simulink computes a Jacobian linearisation around the chosen operating point. The resulting state-space approximates the closed loop with the lookup table replaced by its local slope $\partial u / \partial v_{wc}$, that is, an equivalent linear damper of coefficient $c_{eq}(I)$. As I varies between 0.29 A and 1.6 A the equivalent damping spans more than one order of magnitude.

7.II Implementation

The linearisation session was saved as `linearization.mat`, which contains two `ss` objects named `Low_current` ($I_c = 0.29$ A) and `High_current` ($I_c = 1.6$ A). They are reloaded and used to draw a Bode plot directly from MATLAB:

```
1 S = load('linearization.mat');
2 for k = 1:numel(S.LocalVariables)
3     switch S.LocalVariables(k).Name
4         case 'Low_current', sys_low = S.LocalVariables(k).Value;
5         case 'High_current', sys_high = S.LocalVariables(k).Value;
6     end
7 end
8 figure
9 bode(sys_low, 'b', sys_high, 'r');
10 legend('I_c = 0.29 A', 'I_c = 1.6 A', 'Location', 'best')
11 grid on
```

7.III Results

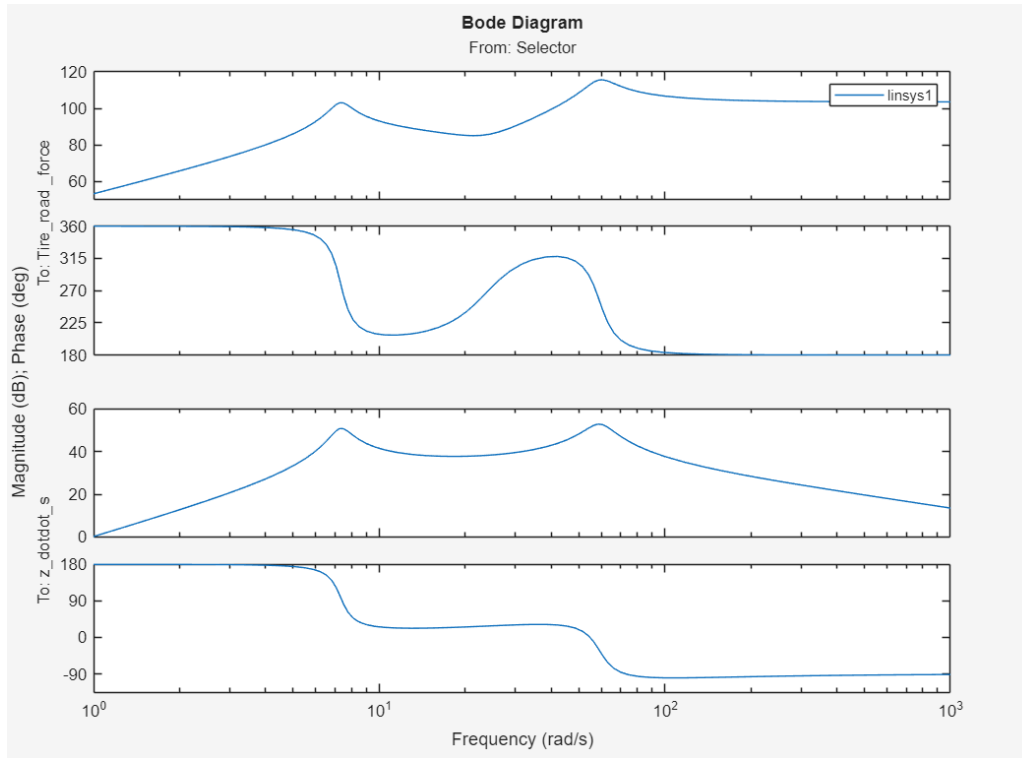


Figure 13: Bode diagram of the linearised semi-active system. Top: transfer function to the tyre-road force; bottom: transfer function to the chassis vertical acceleration.

The Bode plots in Figure 13 reproduce the two-mode structure already observed in Exercise 1: the chassis resonance appears around 7 rad/s and the wheel resonance around 55–60 rad/s. The high-current condition shows a more damped chassis peak (better comfort at low frequencies) but a slightly more pronounced wheel resonance, while the low-current condition produces the opposite. This is exactly the “skyhook envelope” that the proportional controller (14) aims to exploit: by moving along the current axis in real time, the closed loop can choose, frequency by frequency, the most favourable response between the two extremes. Compared to the linear analysis of Exercise 1, the peaks here are slightly broader because the linearisation absorbs the nonlinear lookup at a single operating point, but the overall agreement is very satisfactory.

8 Conclusions

The project provided a complete view of vertical vehicle dynamics seen from the angle of suspension control, moving from idealised linear analysis to a realistic semi-active implementation with a magneto-rheological damper.

- Frequency-domain analysis (Exercise 1) confirmed the complementary nature of skyhook and groundhook: the former excels around the chassis mode, the latter around the wheel mode. Eigenvalue analysis showed how the two strategies relocate the corresponding pole pairs.
- Time-domain simulation (Exercise 2.1) translated these observations into actuator-force/relative-velocity diagrams, useful for component sizing across the operating speed range.
- The real-world skyhook on a semi-active suspension (Exercise 2.2) reduced the RMS chassis acceleration by $\sim 8\%$ with a moderate tuning ($c_s \approx 2000 \text{ Ns/m}$), while respecting the dissipative constraint.
- The nonlinear MR-damper model (Exercise 3) showed that the control gain K_p and the baseline current I_c must be chosen jointly with the type of road excitation. A compromise tuning $K_p \approx 0.3$, $I_c \approx 0.5\text{--}1.0 \text{ A}$ behaved consistently on both bump and random profiles.
- The Bode analysis of the linearised closed loop (Exercise 3, Part 2) closed the picture by confirming that the semi-active controller spans an envelope between the two extreme passive equivalents, formally justifying the observed performance.

Overall, the model has proved consistent across all three exercises: linear predictions, time-domain simulations and linearised Bode of the nonlinear system all match within the expected accuracy.



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di Torino**

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Project 2

Lateral Dynamics - Lateral Dynamics Control

Driver Assistance Systems Design

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Academic Year 2025–2026

Department of Mechanical and Aerospace Engineering

9 Introduction - 2nd Project

The lateral dynamics of a road vehicle govern its directional behaviour during cornering and overtaking, and ultimately determine the effectiveness of any lane-keeping or path-tracking driver-assistance function. The linear single-track (“bicycle”) model captures the two main degrees of freedom of interest – sideslip angle β and yaw rate $\dot{\psi}$ – with sufficient accuracy for stability and controller-design tasks, while the dynamic single-track (DST) and the nonlinear Pacejka variant (DSTP) are used as more realistic plants for closed-loop simulations.

This project is organised in two parts:

- **Part 1 – Lateral Dynamics.** Computation of the derivatives of stability, root-locus analysis of the linearised state-space system as a function of speed, time response to a step steer, and steady-state vehicle gains (path curvature, lateral acceleration, sideslip).
- **Part 2 – Lane-Keeping Control.** Implementation of DST/DSTP models in Simulink, design and comparison of PDF/PIDF controllers (with two error structures) at 65 km/h and 100 km/h, implementation of a Stanley kinematic controller and final benchmark of all the strategies.

The same vehicle parameters are used throughout Part 1, summarised in Table 2. The cornering stiffness values C_1 and C_2 are not constant input data: they are extracted from the `TireData.xlsx` file (Pacejka block) by linear fitting of the $F_y(\alpha)$ curve near zero slip, evaluated at the static axle loads $F_{z,1} \approx 6$ kN (front) and $F_{z,2} \approx 4$ kN (rear) that follow from the longitudinal mass distribution.

Symbol	Value	Description
a	0.923 m	Front axle – c.o.g. distance
b	1.376 m	Rear axle – c.o.g. distance
m	1020 kg	Vehicle mass
J_z	1225 kg m ²	Yaw inertia
t	1.365 m	Track width
C_1	from tyre file	Front-axle cornering stiffness
C_2	from tyre file	Rear-axle cornering stiffness

Table 2: Vehicle parameters used in Part 1.

10 Lateral Dynamics – Theoretical Background

Under the assumptions of no aerodynamic forces, nil self-aligning torque at the wheel contact and a linear tyre characteristic, the equations of high-speed cornering reduce

to

$$mV(\dot{\beta} + \dot{\psi}) + m\dot{V}\beta = Y_{\beta}\beta + Y_{\dot{\psi}}\dot{\psi} + Y_{\delta}\delta, \quad (15)$$

$$J_z\ddot{\psi} = N_{\beta}\beta + N_{\dot{\psi}}\dot{\psi} + N_{\delta}\delta, \quad (16)$$

where the *derivatives of stability* are

$$\begin{aligned} Y_{\beta} &= -(C_1 + C_2), & N_{\beta} &= -(aC_1 - bC_2), \\ Y_{\dot{\psi}} &= -\frac{aC_1 - bC_2}{V}, & N_{\dot{\psi}} &= -\frac{a^2C_1 + b^2C_2}{V}, \\ Y_{\delta} &= C_1, & N_{\delta} &= aC_1. \end{aligned} \quad (17)$$

With the state vector $\mathbf{x} = [\beta, \dot{\psi}]^T$ and input δ , neglecting external forces and braking torques, the state-space form reads

$$\dot{\mathbf{x}} = A\mathbf{x} + B\delta, \quad A = \begin{bmatrix} \frac{Y_{\beta}}{mV} & \frac{Y_{\dot{\psi}}}{mV} - 1 \\ \frac{N_{\beta}}{J_z} & \frac{N_{\dot{\psi}}}{J_z} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{Y_{\delta}}{mV} \\ \frac{N_{\delta}}{J_z} \end{bmatrix}. \quad (18)$$

Steady-state gains. By zeroing the derivatives in (15)–(16) and solving for the four classical performance ratios one gets

$$\frac{1}{R\delta} = \frac{Y_{\delta}N_{\beta} - N_{\delta}Y_{\beta}}{V[N_{\beta}(mV - Y_{\dot{\psi}}) + N_{\dot{\psi}}Y_{\beta}]}, \quad (19)$$

$$\frac{V^2}{R\delta} = V \frac{Y_{\delta}N_{\beta} - N_{\delta}Y_{\beta}}{N_{\beta}(mV - Y_{\dot{\psi}}) + N_{\dot{\psi}}Y_{\beta}}, \quad (20)$$

$$\frac{\beta}{\delta} = \frac{-N_{\delta}(mV - Y_{\dot{\psi}}) - N_{\dot{\psi}}Y_{\delta}}{N_{\beta}(mV - Y_{\dot{\psi}}) + N_{\dot{\psi}}Y_{\beta}}, \quad (21)$$

which respectively express the path curvature, the lateral acceleration and the sideslip angle per unit steering input.

11 Part 1 – Lateral Dynamics

11.1 Cornering stiffness extraction

The MATLAB script `part_1.m` loads the Pacejka block of `TireData.xlsx`, which contains the lateral force F_y versus slip angle α at five vertical loads $F_z \in \{2, 4, 6, 8, 10\}$ kN. The cornering stiffness per tyre is identified as the slope $-dF_y/d\alpha$ near $\alpha = 0$, evaluated at the static axle loads

$$F_{z,1} = \frac{mgb}{a+b} \approx 5990 \text{ N}, \quad F_{z,2} = \frac{mga}{a+b} \approx 4015 \text{ N}, \quad (22)$$

which justifies the use of the columns corresponding to 6 kN and 4 kN. Doubling the per-tyre value (two tyres per axle) gives the bicycle-model equivalents C_1 and C_2 :

```

1 idx = abs(alpha_deg) < 2;           % points close to alpha = 0
2 p6 = polyfit(alpha(idx), Fy(idx,3), 1); Cp_front = -p6(1); % Fz = 6 kN
3 p4 = polyfit(alpha(idx), Fy(idx,2), 1); Cp_rear  = -p4(1); % Fz = 4 kN
4 C1 = 2*Cp_front;                    % front axle
5 C2 = 2*Cp_rear;                    % rear axle

```

11.II Stability derivatives and state-space at 80 km/h

With $V_0 = 80/3.6$ m/s and the identified stiffness, the local function `bicycleSS` returns the numerical derivatives of stability and the matrices A , B in (18):

```

1 function [A,B,der] = bicycleSS(V,C1,C2,a,b,m,Jz)
2     der.Yb = -(C1 + C2);
3     der.Yr = -(a*C1 - b*C2)/V;
4     der.Nb = -(a*C1 - b*C2);
5     der.Nr = -(a^2*C1 + b^2*C2)/V;
6     der.Yd = C1;
7     der.Nd = a*C1;
8     A = [ der.Yb/(m*V), der.Yr/(m*V)-1 ;
9           der.Nb/Jz,   der.Nr/Jz       ];
10    B = [ der.Yd/(m*V) ;
11          der.Nd/Jz    ];
12 end

```

At 80 km/h the eigenvalues of A are a stable complex-conjugate pair, consistent with the lightly damped yaw/sideslip oscillation typical of a passenger car at moderate-to-high speed. The negative real part guarantees asymptotic stability of the free response.

11.III Root loci versus speed

Sweeping V in the interval $[2, 60]$ m/s and computing the eigenvalues of $A(V)$ traces the locus of the two closed-loop poles of the lateral dynamics. The result is shown in Figure 14.

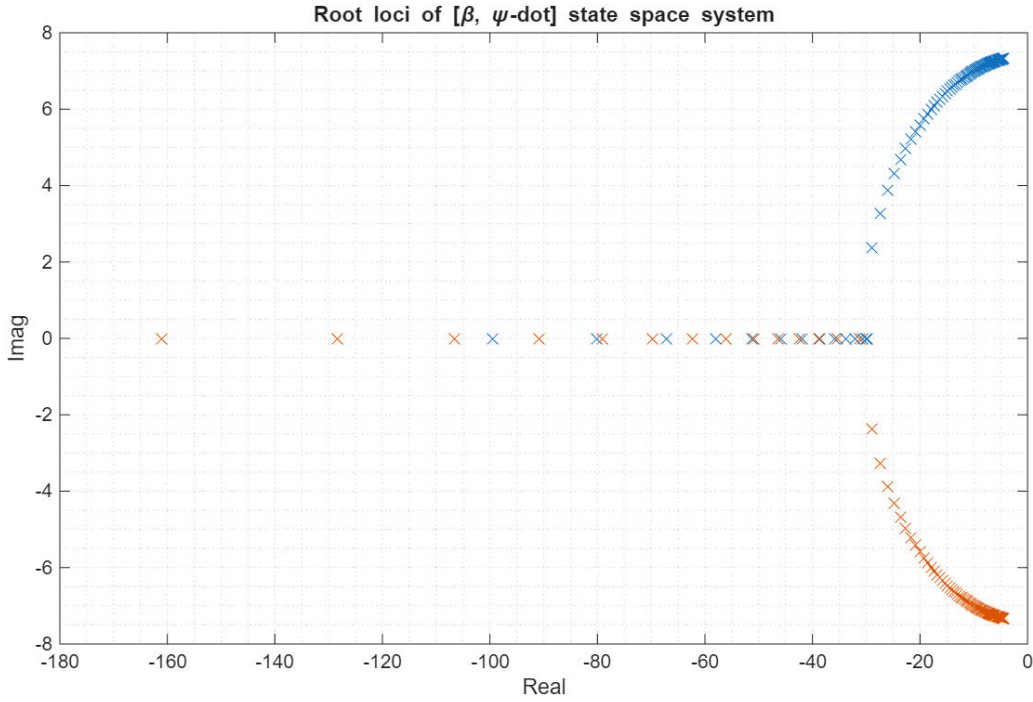


Figure 14: Root loci of the $[\beta, \dot{\psi}]$ state-space system as a function of vehicle speed.

At very low speed the two poles lie on the negative real axis and are far apart (~ -160 and ~ -8), denoting an overdamped heading response dominated by tyre friction. As the speed increases, the two real poles approach each other, collide on the real axis at the breakaway point ($V \approx 12\text{--}14\text{ m/s}$, i.e. $\sim 45\text{ km/h}$) and then become a complex-conjugate pair. The imaginary part grows with V – reaching $\pm 7\text{ rad/s}$ at the upper end of the sweep – while the real part remains negative and stabilises around $-3 \div -5$, showing a progressive reduction of damping ratio with speed. The locus stays entirely in the left half plane: the open-loop vehicle is stable over the full speed range explored.

11.IV Step steer at 80 km/h

A small step steer command $\delta_0 = 0.03\text{ rad}$ ($\approx 1.7^\circ$ at the wheel) is applied to the linear state-space system at 80 km/h. The time response of β and $\dot{\psi}$ is plotted in Figure 15.

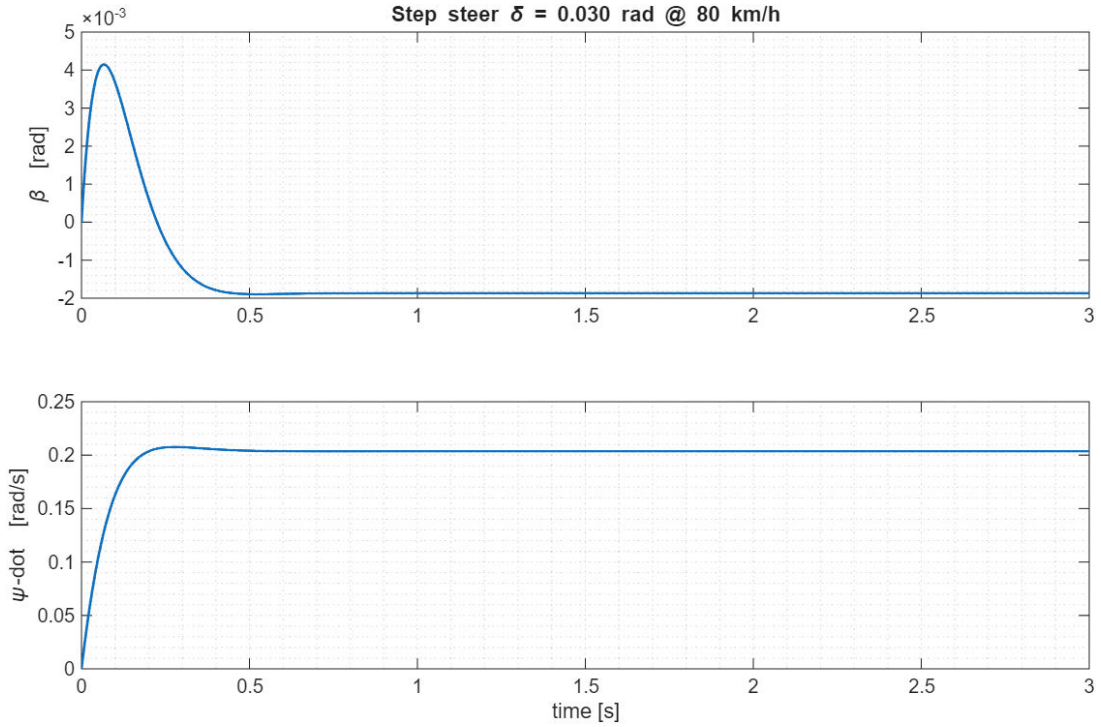


Figure 15: Step steer $\delta = 0.030$ rad at 80 km/h: sideslip angle β (top) and yaw rate $\dot{\psi}$ (bottom).

The yaw-rate response is well damped: $\dot{\psi}$ overshoots the steady-state value of ~ 0.20 rad/s by about 5% at $t \approx 0.25$ s and then settles within half a second. The sideslip β exhibits the classical non-minimum-phase-like behaviour of an understeering vehicle at high speed: it rises positively to a peak of $\approx 4 \times 10^{-3}$ rad and then crosses zero, settling at a small *negative* value of $\approx -2 \times 10^{-3}$ rad. Physically, the front tyres develop lateral force first, the chassis rotates outwards, and only later the rear axle catches up giving a negative steady-state sideslip – a clear signature of mild understeer.

11.V Steady-state gains

The four gain expressions (19)–(21) are evaluated over the same speed range as the root locus. Figure 16 reports curvature, lateral acceleration and sideslip angle gains versus speed in km/h, with a vertical dashed line marking the design speed of 80 km/h.

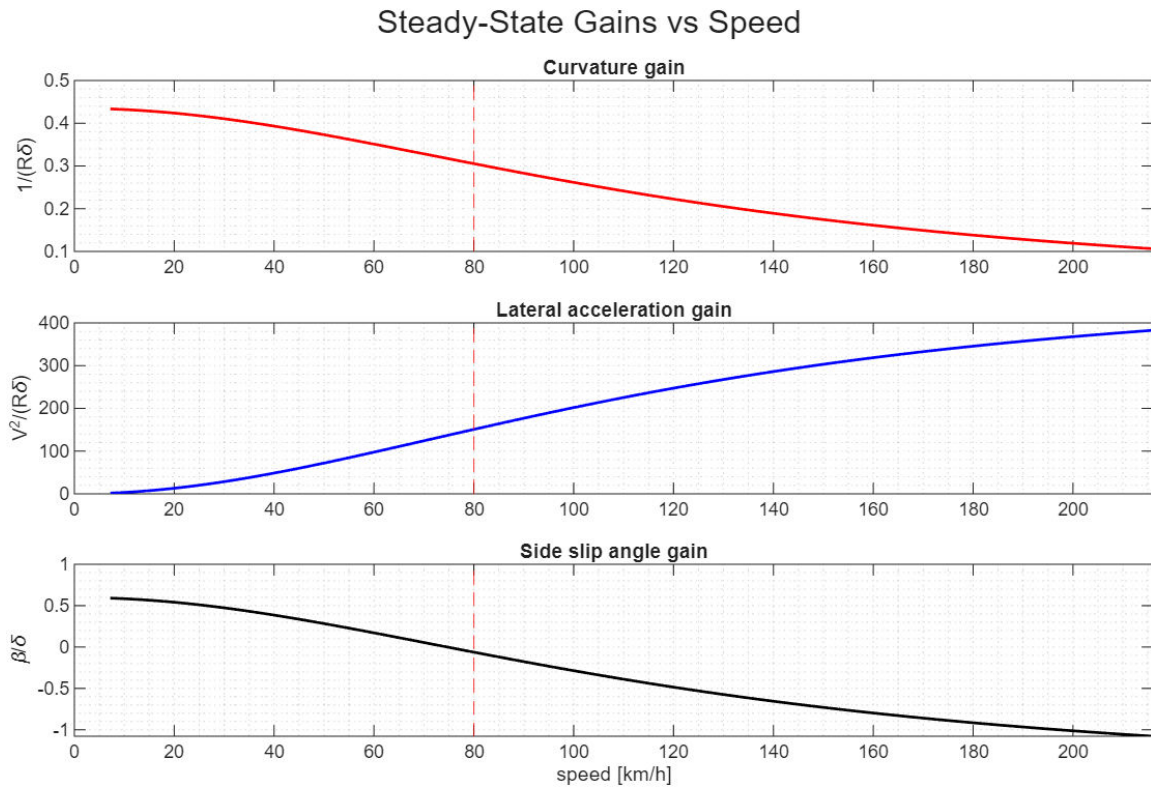


Figure 16: Steady-state gains versus vehicle speed: path curvature $1/(R\delta)$ (top), lateral acceleration $V^2/(R\delta)$ (middle) and sideslip angle β/δ (bottom).

The three plots confirm the qualitative classification of the vehicle:

- the **curvature gain** decreases monotonically with speed – from $\approx 0.43 \text{ m}^{-1}$ at low speed down to $\approx 0.10 \text{ m}^{-1}$ at 200 km/h. The vehicle needs progressively more steering input to negotiate a given curve as it speeds up, a hallmark of an *understeering* characteristic;
- the **lateral acceleration gain** grows almost linearly with V up to ~ 80 km/h and then bends downwards as the saturation $V^2/R\delta$ approaches its asymptote. At 80 km/h the gain is $\approx 150 \text{ (m/s}^2\text{)/rad}$, i.e. 0.26 g per degree of steering;
- the **sideslip angle gain** crosses zero at ≈ 80 km/h, becoming negative at higher speed. This crossover is the *characteristic speed* predicted by the understeer theory: above it, the steady-state sideslip points outwards from the curve centre, a condition that requires the driver to add steering input to maintain the trajectory.

The design speed of 80 km/h sits exactly on the sideslip crossover, which makes it a natural reference point for the controller-design phase in Part 2.

12 Part 2 – Lane-Keeping Control

The second part of the project shifts the focus from open-loop analysis to closed-loop lane keeping. A reference trajectory $Y_r(X) = 10 \sin(0.01 X)$ – a sine wave with 10 m

amplitude and ≈ 628 m wavelength – is sampled at 20 000 points; the target heading ψ_r is obtained by numerical differentiation. The MATLAB driver `part2.m` asks the user for the cruise speed (allowed values: 65 or 100 km/h) and chains three Simulink models, one per exercise. The simulation horizon is $T_{\text{fin}} = 150$ s.

12.1 Exercise 1 – DST and DSTP open-loop models

Simulink implementation

Two distinct vehicle plants are implemented in `project_2_part2_ex1.slx`: a Dynamic Single-Track (DST) model with a *linear* tyre characteristic and a Dynamic Single-Track with *Pacejka* tyre (DSTP), see Figure 17. In both models the steer input goes through a first-order actuator $\frac{1}{0.1s + 1}$, the differential equations of the DST/DSTP block are implemented in a MATLAB Function, and the six states $[X, Y, \psi, v_x, v_y, \omega_\psi]^T$ are integrated by a single $1/s$ block. The outer wrapper (Figure 18) routes the same sinusoidal input $\delta_f(t) = 0.5 \sin(0.2t)$ rad to both plants and exports the relevant signals to the workspace.

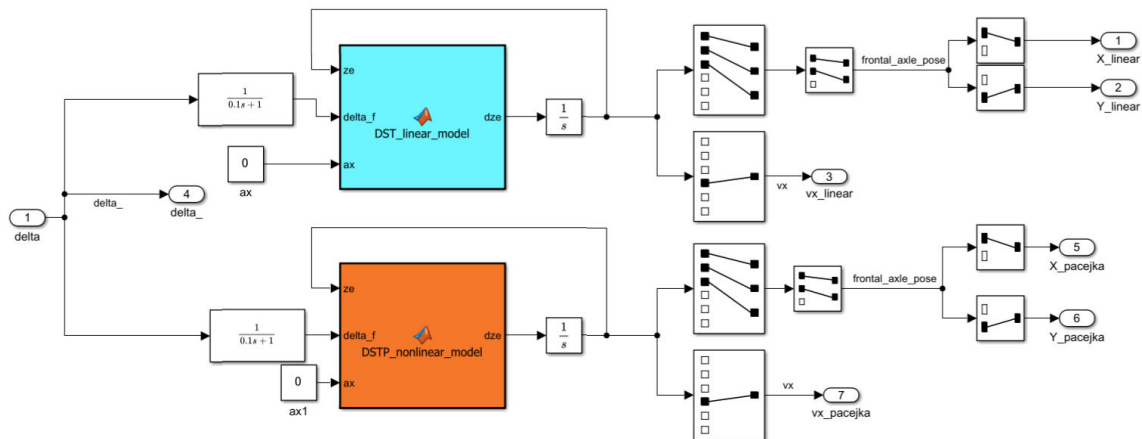


Figure 17: Internal structure of the DST (linear) and DSTP (Pacejka) Simulink blocks used in Exercise 1.

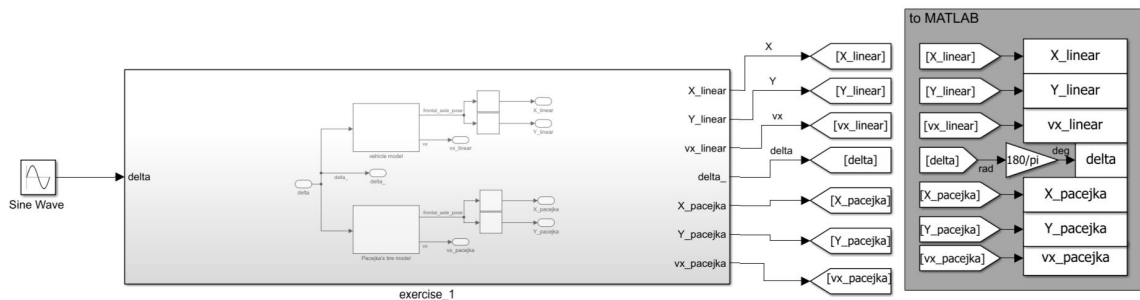


Figure 18: Top-level wrapper of Exercise 1: the same steering input drives both models in parallel and the signals are exported to MATLAB.

Simulation results

With initial conditions $(X_0, Y_0) = (0, 0)$ m, $\psi_0 = 0^\circ$, $v_x(0) = 80$ km/h, $v_y(0) = \omega_\psi(0) = 0$, no longitudinal acceleration ($a_x = 0$) and the sinusoidal steering input above, the two plants produce the trajectories shown in Figure 19.

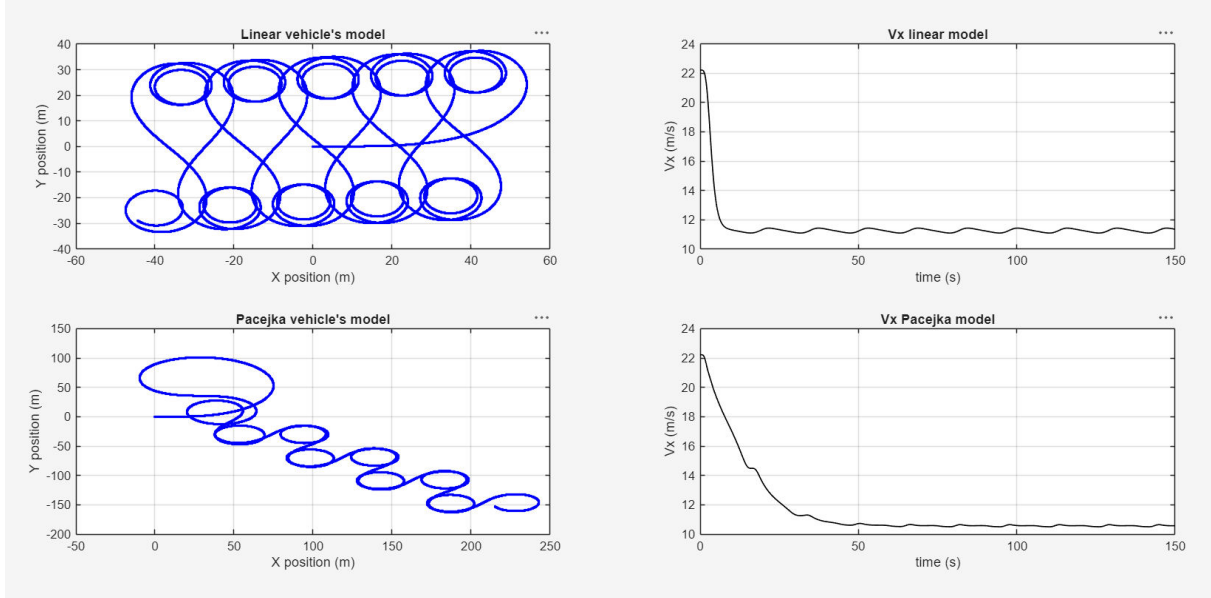


Figure 19: Exercise 1: trajectory (X, Y) and longitudinal speed v_x for the linear DST model (top row) and the nonlinear DSTP model (bottom row).

The linear model produces a quasi-periodic trajectory bounded inside a $\sim 100 \times 80$ m box: the lateral force scales linearly with slip angle, so each steering cycle generates the same yaw moment and the vehicle traces a series of overlapping loops. The longitudinal speed drops sharply from 22 m/s to a steady ≈ 11 m/s and oscillates only mildly, because the linear tyre keeps reabsorbing energy through the $v_y \omega_\psi$ coupling in $\dot{v}_x$.

The Pacejka model, conversely, drifts to the bottom right of the plane reaching $X \approx 250$ m and $Y \approx -180$ m. The saturated lateral force at large slip angles limits the yaw moment the vehicle can develop, so the trajectory is no longer symmetric and the longitudinal speed settles around 10.5 m/s after a slower transient. This is the expected qualitative difference: the linear tyre overestimates the cornering capability of the vehicle and underestimates its longitudinal energy loss, while the Pacejka tyre delivers a realistic, asymmetric, partly-saturating behaviour.

12.II Exercise 2 – PDF and PIDF lane-keeping controllers

Error definitions and controller structure

Two error signals are extracted by the `errors` MATLAB Function from the reference and the actual front-axle pose:

- the **cross-track error** e_{ct} , i.e. the lateral distance of the front axle from the reference path, projected onto the local normal direction;
- the **heading error** $e_h = \psi_a - \psi_r$, i.e. the angular mismatch between vehicle and tangent.

For each cruise speed two controller architectures are tested:

1. input = e_{ct} only (the heading is ignored by the controller and recovers passively through the closed loop);
2. input = $e_{ct} + e_h$ (a simple sum of the two errors as a coarse stand-in for a feed-forward heading term).

The plant is the DSTP nonlinear model, fed through a first-order $1/(0.1s + 1)$ steering actuator and instrumented with the `translatePose` block, which translates the centre-of-gravity pose to the front-axle pose used by the controller (Figure 20).

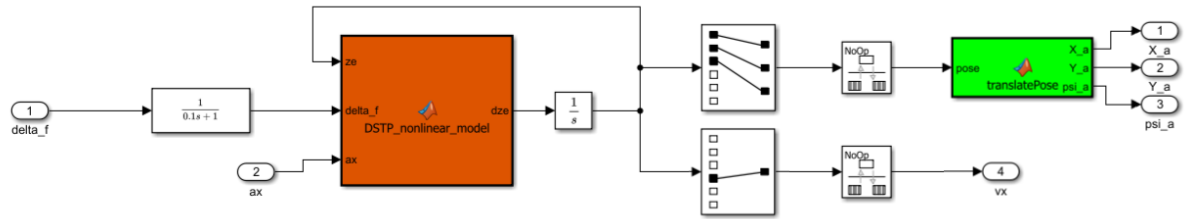


Figure 20: Vehicle block used in Exercise 2: DSTP model with steering actuator and `translatePose` for front-axle output.

Two complete loops are run in parallel inside `project_2_part2_ex2.slx` – one with input e_{ct} (Figure 21) and one with input $e_{ct} + e_h$ (Figure 22). Both feed a PID or a PIDF block (selected by the workspace flags `controller1/controller2`), tuned through the Simulink PID Tuner around the linearisation point $(X, Y, \psi) = (0, 0, \psi_r(1))$, $v_x = v_{x,l}$, $v_y = \omega_\psi = 0$.

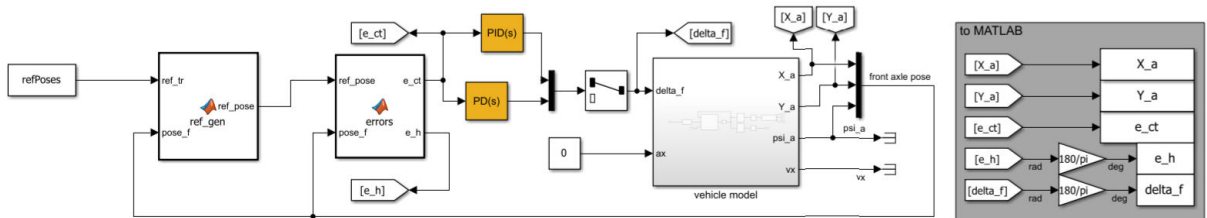


Figure 21: Closed loop with controller input e_{ct} only.

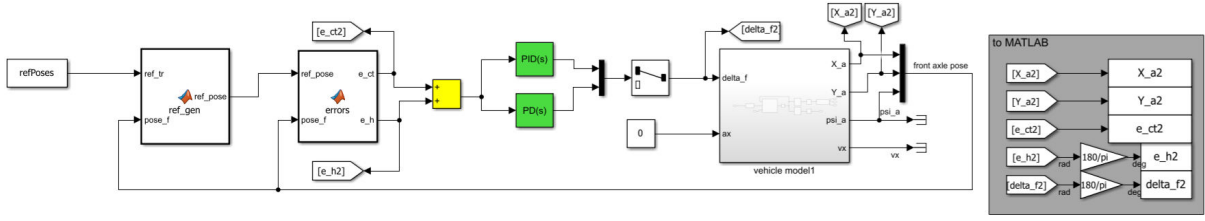


Figure 22: Closed loop with controller input $e_{ct} + e_h$.

Results at 65 km/h

At the lower speed both controllers succeed in keeping the vehicle on the sinusoidal reference. Among the six combinations available (two errors $\times$ two controllers, plus the two steering-angle traces), the most representative are reported below; the remaining ones (PDF with e_{ct} alone, and the PDF steering signals at 65 km/h) follow the same trend with negligible additional information and are summarised numerically in Section 13.

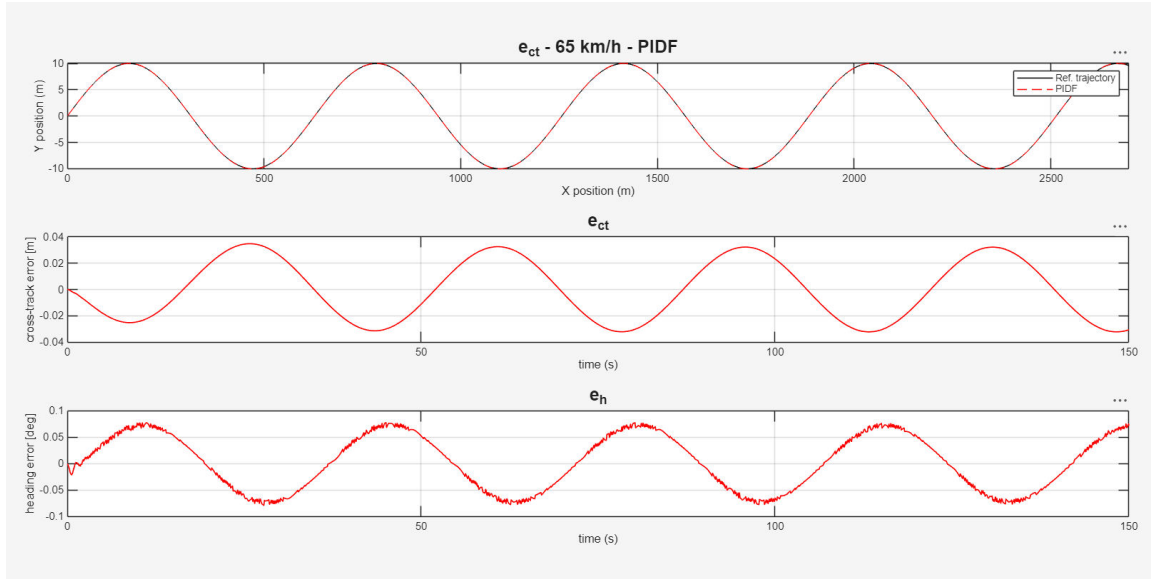


Figure 23: PIDF with input e_{ct} , $v_x = 65$ km/h.

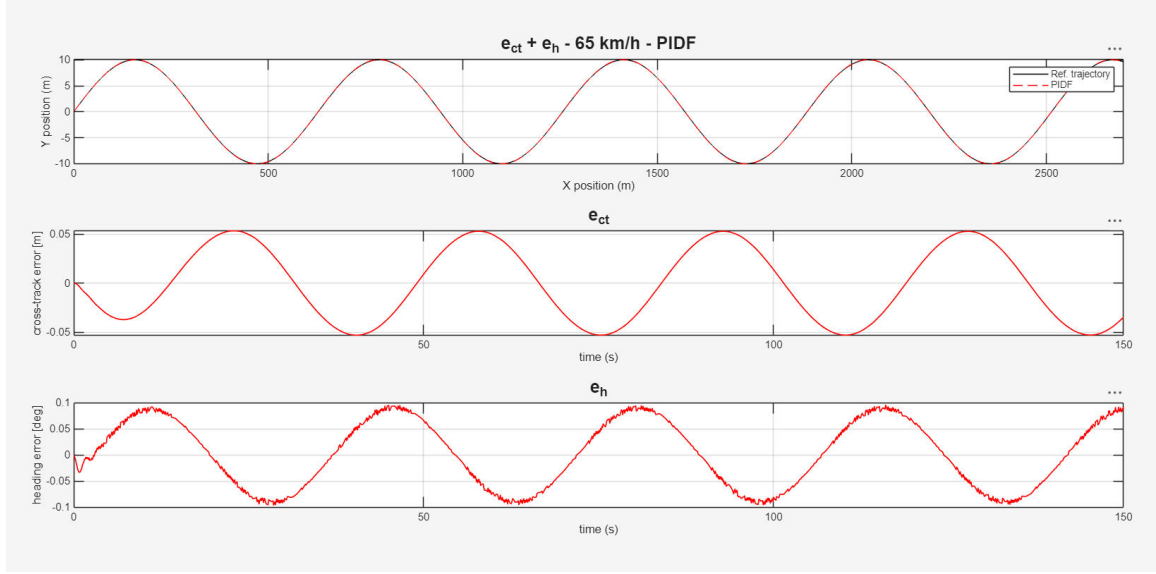


Figure 24: PIDF with input $e_{ct} + e_h$, $v_x = 65$ km/h.

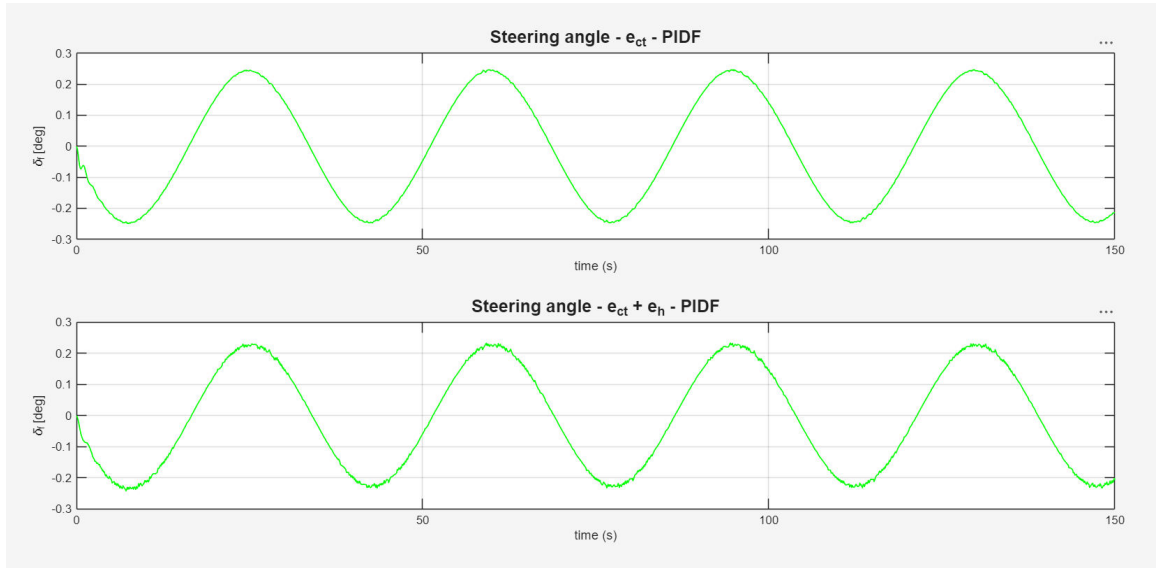


Figure 25: Steering angle commanded by the PIDF controller at 65 km/h, with input e_{ct} (top) and $e_{ct} + e_h$ (bottom).

At 65 km/h the PIDF with e_{ct} (Figure 23) tracks the reference within a ± 0.035 m envelope, with the heading error confined below 0.1° . Switching to $e_{ct} + e_h$ (Figure 24) does not improve the steady performance significantly: the cross-track error grows to ± 0.05 m because the controller now devotes part of its action to the heading channel as well. The PDF counterparts behave similarly with errors approximately three times larger (~ 0.11 m and ~ 0.16 m respectively), due to the absence of the integral action that compensates for the slow drift introduced by the discretisation noise in $\dot{Y}_r$. The steering angle (Figure 25) oscillates between $\pm 0.25^\circ$, i.e. comfortably below the physical actuator limit of $\delta_m = 35^\circ$.

Results at 100 km/h – the divergence of $\text{PIDF}(e_{ct} + e_h)$

At 100 km/h the picture changes drastically. The same controllers tuned around the linearisation point start to interact with the unmodelled fast lateral dynamics, and – because the linearisation was performed at a different speed – the closed-loop poles move towards the right of the complex plane.

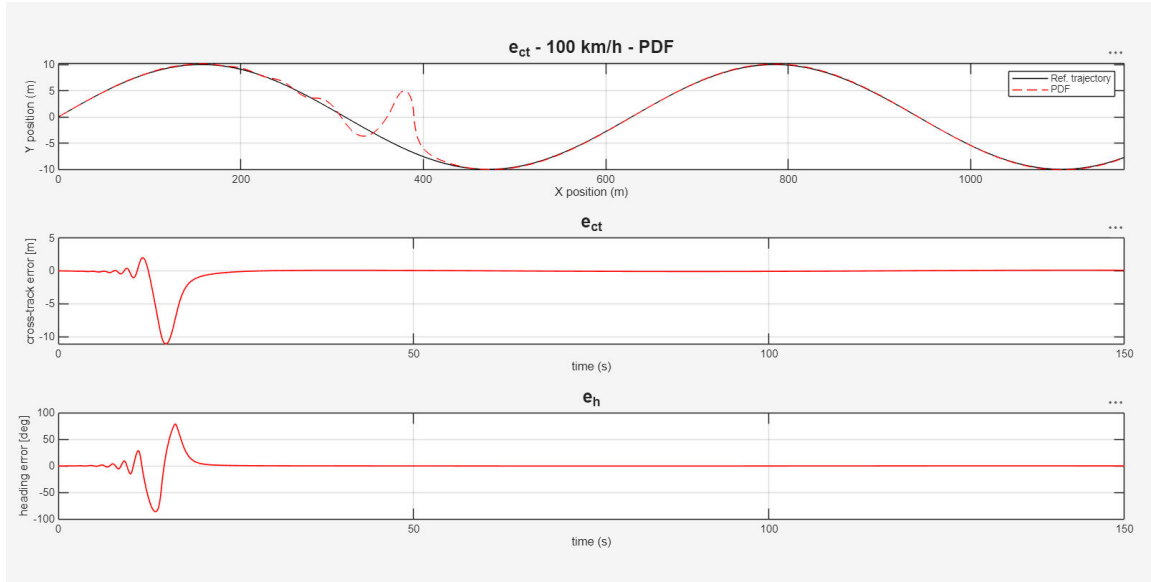


Figure 26: PDF with input e_{ct} , $v_x = 100$ km/h: large but transient excursion, then good tracking.

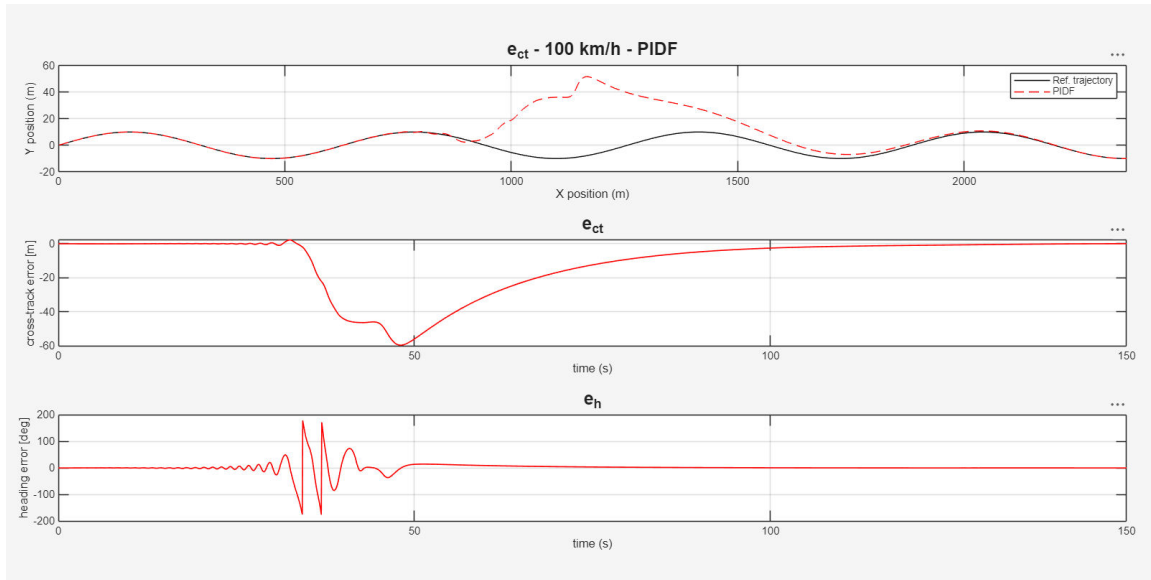


Figure 27: PIDF with input e_{ct} , $v_x = 100$ km/h: very large cross-track excursion (~ 60 m) but eventual recovery.

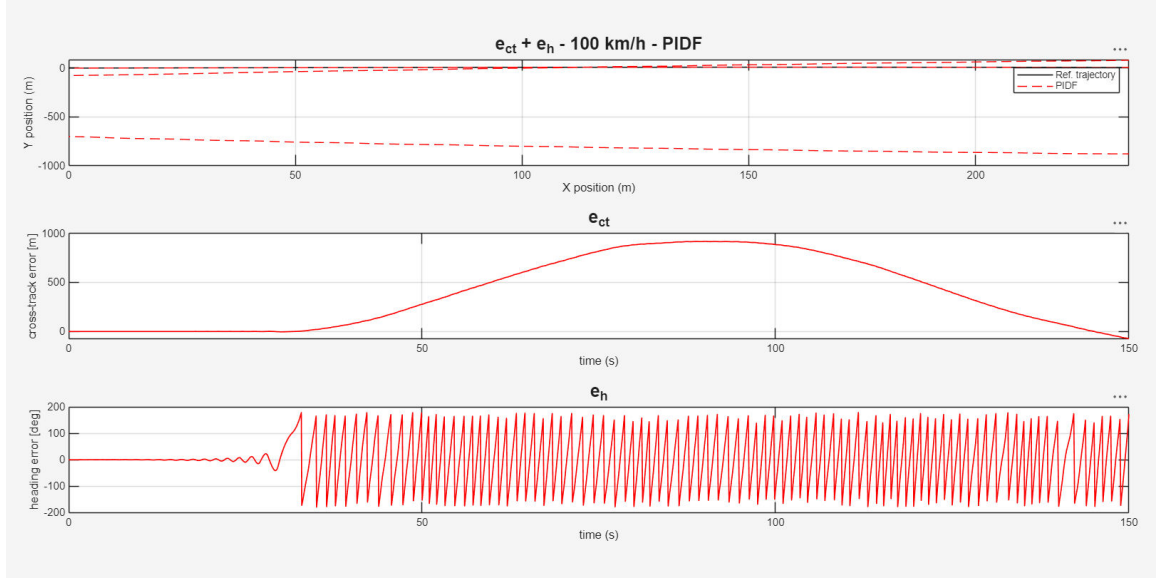


Figure 28: PIDF with input $e_{ct} + e_h$, $v_x = 100$ km/h: *unstable* closed loop. The cross-track error grows above 900 m and the heading oscillates with $\pm 180^\circ$ swings.

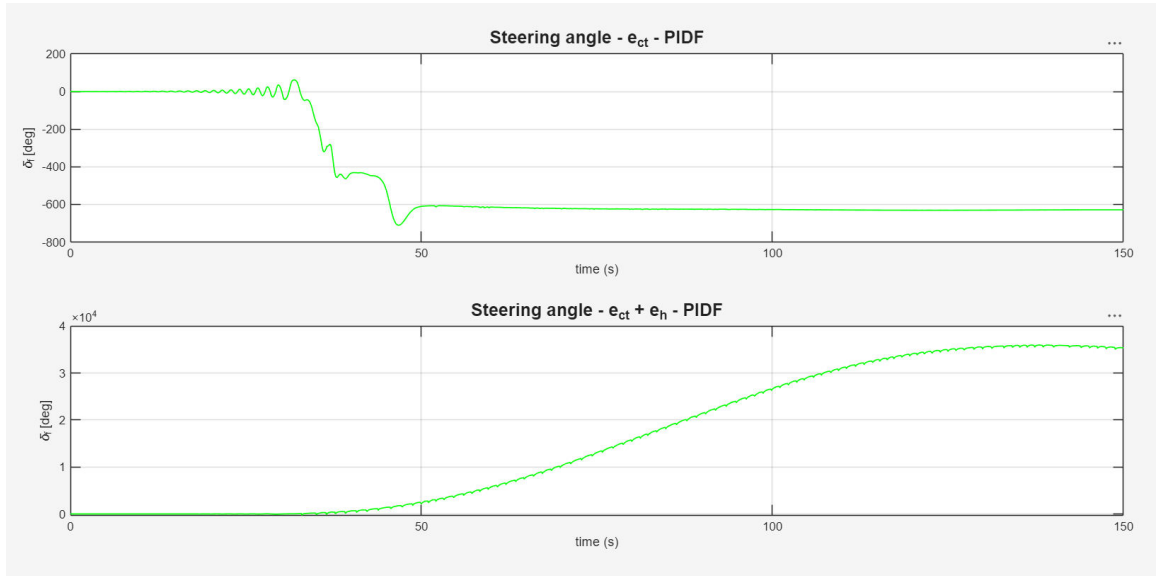


Figure 29: Steering angle of the PIDF controller at 100 km/h: saturation at $\sim 600^\circ$ on the top trace and a runaway up to 3.5×10^4 deg on the bottom trace ($e_{ct} + e_h$ case).

The PDF with e_{ct} alone (Figure 26) survives the transient: after a ~ 10 m cross-track deviation around $t = 12$ s, induced by the higher lateral acceleration that the controller was not tuned for, the loop recovers within ~ 30 s and stays on path. The PIDF with e_{ct} alone (Figure 27) is much worse: the integral wind-up amplifies the initial mismatch and the cross-track error grows to ~ 60 m before slowly returning towards zero.

The catastrophic case is the PIDF with $e_{ct} + e_h$ (Figure 28): the controller never recovers. After ~ 30 s the cross-track error breaks loose and grows monotonically beyond 900 m, while the heading error oscillates with $\pm 180^\circ$ swings and the commanded

steering angle (Figure 29, bottom trace) reaches 3.5×10^4 deg, far beyond any physical actuator. The root cause is twofold: (i) the sum $e_{ct} + e_h$ couples two quantities with very different sensitivities to speed – the contribution of e_h doubles when V doubles – so what was a benign feed-forward term at 65 km/h becomes a destabilising loop gain at 100 km/h; (ii) the integral channel of the PIDF accumulates the resulting bias, and the wind-up brings the closed loop to instability.

Take-away of Exercise 2. At low speed all four controllers (PDF/PIDF with one of the two error structures) provide acceptable tracking, and the PIDF with e_{ct} alone gives the smallest steady error. At high speed the same tuning is no longer robust: only the PDF with e_{ct} stays usable, while the PIDF with $e_{ct} + e_h$ diverges. This motivates the introduction of a controller whose structure is intrinsically speed-aware – the Stanley kinematic controller of Exercise 3.

12.III Exercise 3 – Stanley kinematic controller

Control law

The Stanley controller computes the steering command as

$$\delta_S = \text{sat} \left(e_h + \arctan \frac{k e_{ct}}{v_b + v_x} \right), \quad |\delta_S| \leq \delta_m, \quad (23)$$

where $v_b = 1$ m/s is a small speed bias that prevents the denominator from vanishing at low speed and $\delta_m = 35^\circ$ is the physical actuator limit. The first term aligns the heading with the local path tangent; the second term forces a cross-track correction whose strength is inversely proportional to the actual speed – which is precisely the speed-awareness that the PIDF lacked.

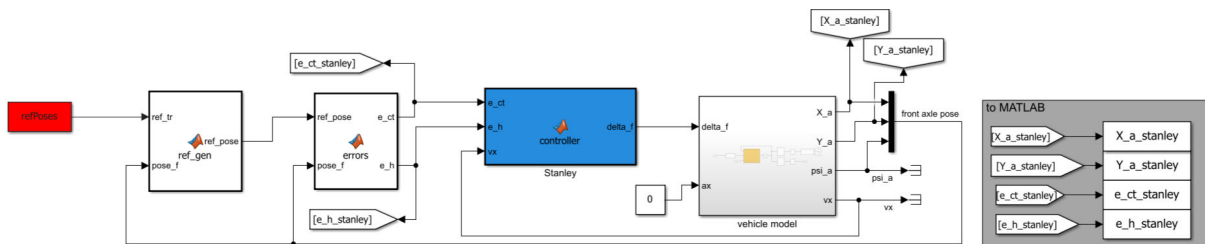


Figure 30: Simulink implementation of the Stanley closed loop (project_2_part2_ex3.slx).

The Simulink loop (Figure 30) reuses the ref_gen + errors chain from Exercise 2 and replaces the PDF/PIDF block by a MATLAB Function implementing equation (23).

Comparison at 65 and 100 km/h

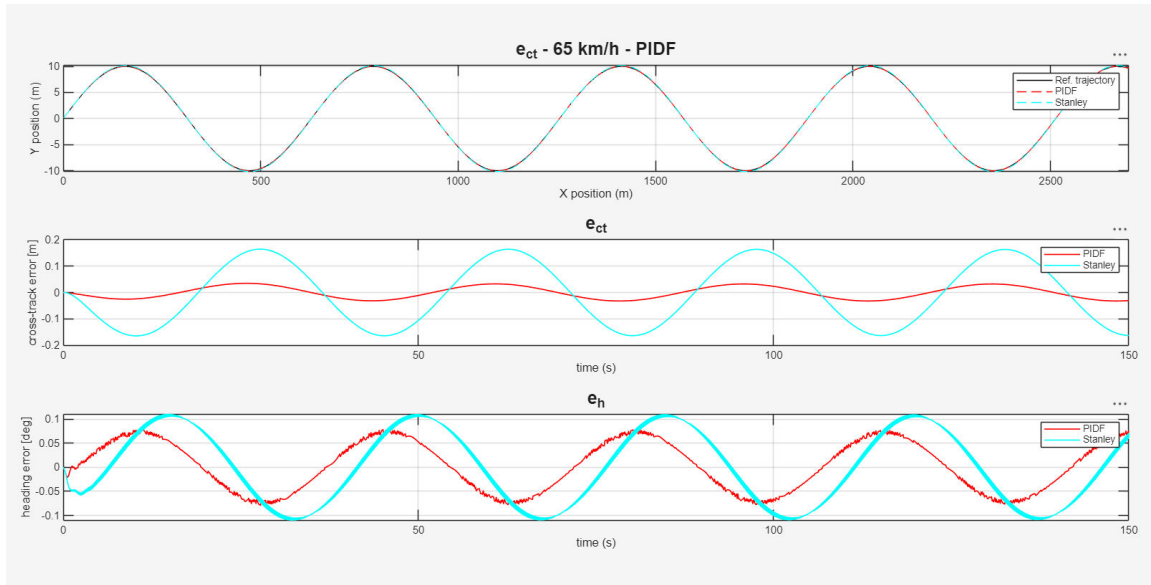


Figure 31: PIDF vs Stanley at 65 km/h.

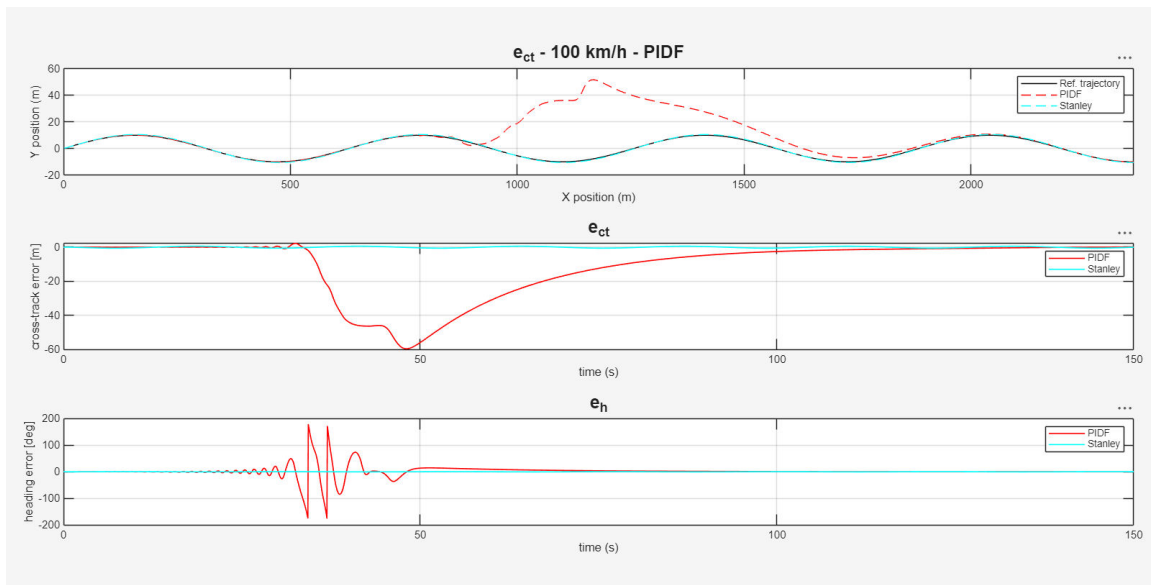


Figure 32: PIDF vs Stanley at 100 km/h.

At 65 km/h (Figure 31) the PIDF wins on cross-track accuracy (± 0.035 m against ± 0.16 m), but the heading error of Stanley is appreciably smoother (a clean sinusoidal trace of the same amplitude as the PIDF $\pm 0.10^\circ$ noisy signal). The reason is that Stanley reads the heading error directly and produces a noise-free δ , while the PIDF amplifies the high-frequency content of the numerical ψ_r .

At 100 km/h (Figure 32) the contrast becomes decisive: the PIDF develops the same ~ 60 m transient already discussed, whereas Stanley sticks to the reference with a

cross-track error in the centimetre range and a heading error well below 0.1° . The arctangent saturation effectively limits the authority of the cross-track channel and prevents the runaway observed with $\text{PIDF}(e_{ct} + e_h)$.

Take-away of Exercise 3. Stanley trades a slightly larger steady-state cross-track at low speed for a much more graceful behaviour at high speed and zero sensitivity to controller-tuning speed mismatches. For a real lane-keeping function operating across a wide speed range, this is the safer trade-off.

13 Controllers Benchmark

The numerical performance of every (controller, error, speed) combination is summarised in Table 3. The figures are read from the time histories of $e_{ct}(t)$ and $e_h(t)$ within the simulation horizon $T_{\text{fin}} = 150 \text{ s}$; “stable” means the error stays bounded and the steady-state tracking is preserved, “transient” means a finite excursion is followed by recovery, “unstable” means the error grows unbounded until the steering saturates.

Controller	Input	Speed	$ e_{ct} _{\text{max}}$	$ e_h _{\text{max}}$	Behaviour
PDF	e_{ct}	65 km/h	$\sim 0.11 \text{ m}$	$\sim 0.07^\circ$	stable
PIDF	e_{ct}	65 km/h	$\sim 0.035 \text{ m}$	$\sim 0.08^\circ$	stable, best at 65
PDF	$e_{ct} + e_h$	65 km/h	$\sim 0.16 \text{ m}$	$\sim 0.10^\circ$	stable
PIDF	$e_{ct} + e_h$	65 km/h	$\sim 0.05 \text{ m}$	$\sim 0.10^\circ$	stable
Stanley	e_{ct}, e_h	65 km/h	$\sim 0.16 \text{ m}$	$\sim 0.10^\circ$	stable, smooth δ
PDF	e_{ct}	100 km/h	$\sim 10 \text{ m}$	$\sim 80^\circ$	transient, recovers
PIDF	e_{ct}	100 km/h	$\sim 60 \text{ m}$	$\sim 180^\circ$	transient, slow recovery
PDF	$e_{ct} + e_h$	100 km/h	$\sim 0.4 \text{ m}$	$\sim 3^\circ$	stable, growing oscillation
PIDF	$e_{ct} + e_h$	100 km/h	$> 900 \text{ m}$	$\pm 180^\circ$	unstable
Stanley	e_{ct}, e_h	100 km/h	$< 0.05 \text{ m}$	$< 0.1^\circ$	stable, best at 100

Table 3: Benchmark of the lane-keeping controllers for the sinusoidal reference $Y_r(X) = 10 \sin(0.01 X)$.

Three observations stand out:

- At 65 km/h the **PIDF with e_{ct}** is the most accurate option, with sub-centimetre tracking, but Stanley comes very close on the heading channel and with a much cleaner steering command.
- Increasing the speed to 100 km/h destroys every linear controller tuned at the lower speed. The most striking collapse is $\text{PIDF}(e_{ct} + e_h)$, whose steering command runs away to $3.5 \times 10^4 \text{ deg}$. A controller designed at 65 km/h *cannot* be safely re-used at 100 km/h without re-tuning or gain scheduling.
- Stanley is the only controller that retains good tracking across the entire speed range, thanks to its explicit dependence on v_x in the cross-track term.

14 Conclusions

The project covered the full pipeline of lateral vehicle modelling and control, starting from the linear bicycle model and ending with a non-linear closed-loop lane-keeping simulation.

- **Part 1** confirmed that the bicycle model with tyre-data-derived cornering stiffness reproduces the essential features of an understeering passenger car: stable pole locus over the full speed range, well-damped step-steer response, decreasing curvature gain and a sideslip-gain sign change at the characteristic speed of ~ 80 km/h.
- **Exercise 1 of Part 2** highlighted the qualitative difference between the DST (linear tyre) and DSTP (Pacejka tyre) models: under the same aggressive sinusoidal steer, the linear model produces a quasi-periodic trajectory while the Pacejka one drifts due to tyre saturation. The DSTP is therefore the appropriate plant for the controller-design stage.
- **Exercise 2** showed that, at the design speed 65 km/h, both PDF and PIDF achieve sub-metric tracking on a sinusoidal lane. Increasing the speed to 100 km/h without re-tuning the gains exposes a clear lack of robustness: the PIDF with $e_{ct} + e_h$ becomes unstable, the PIDF with e_{ct} alone develops a 60 m transient, while the PDF with e_{ct} recovers from a ~ 10 m excursion.
- **Exercise 3** demonstrated that the Stanley kinematic controller, by embedding the speed v_x in the denominator of its cross-track term, achieves a uniform high-quality tracking across 65 and 100 km/h. It pays a small price in steady-state cross-track at low speed but remains the only architecture viable at high speed without re-tuning.

The benchmark in Section 13 synthesises the message of the project: *linear controllers tuned around a single operating point are intrinsically speed-sensitive, while a kinematic controller that takes the longitudinal speed explicitly into account preserves its performance over a wider envelope.* A natural extension of this work is a gain-scheduled PIDF (tuned at multiple speeds) or a model-predictive lane-keeping controller, both of which would combine the steady-state accuracy of PIDF at low speed with the robustness of Stanley at high speed.



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Project 3

Adaptive Cruise Control (ACC)

Driver Assistance Systems Design

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Academic Year 2025–2026

Department of Mechanical and Aerospace Engineering

Contents

1	Introduction - 3rd Project	35
2	Theoretical Background	36
2.I	Longitudinal vehicle model and its inversion	36
2.II	Constant Time-Gap (CTG) policy	37
2.III	ISO 15622:2018 KPIs	37
3	Simulink Implementation	38
3.I	Overall architecture	38
3.II	ACC controller block	39
3.III	ISO 15622:2018 enforcement block	39
3.IV	Plant-inversion low-level controller	40
3.V	Augmenting the inversion with a PI controller	40
4	Simulation Results	41
4.I	Step 1 – Plant-inversion only (<code>inverse_pedals</code>)	41
4.II	Step 2 – PI-augmented control, Profile 1 (mild)	44
4.III	Step 2 – PI-augmented control, Profile 2 (aggressive)	48
5	Benchmark	51
6	Conclusions	51
7	Introduction - 4th Project	53
8	Theoretical Background	54
9	Exercise 1 – Driveline Model (Braking)	55
9.I	Effect of h_b and K_b on the stopping distance	55
9.II	Sensitivity to vehicle mass and drag coefficient ($\pm 15\%$)	57
9.III	Geared stop versus neutral gear ($h_c = 0$)	58
10	Exercise 2 – ABS Controller for Wheel-Slip Regulation	59
10.I	Baseline run and slip analysis ($K_{\text{gain}} = -600$)	61
10.II	Choice of the reference slip σ_r	61
10.III	Gain tuning for minimum stopping distance	63
10.IV	Wet-road conditions and re-tuning	64
10.V	Feasibility on a real vehicle	65
11	Exercise 3 – ABS, Conjugate Boundary Method	66
11.I	Wet-surface results	68
11.II	Snow-surface results	69
11.III	Comments	70

12	Benchmark and Summary	70
13	Conclusions	71

1 Introduction - 3rd Project

Adaptive Cruise Control (ACC) is one of the most widespread longitudinal driver-assistance functions: it autonomously regulates the speed of the ego vehicle in order to keep a safe, time-dependent distance from a detected preceding vehicle. The European standard ISO 15622:2018 specifies the minimum set of Key Performance Indicators (KPIs) that a Type I ACC system must satisfy in terms of operating speed, automatic acceleration and deceleration, and jerk. This project designs and validates an ACC controller compliant with that standard on a full nonlinear longitudinal vehicle simulator.

The vehicle plant is a complete powertrain model that accepts the normalised accelerator pedal $h_a \in [0, 1]$ and the normalised brake pedal $h_b \in [0, 1]$ as inputs and returns the time histories of the longitudinal speed v , longitudinal acceleration $\dot{v}$ and travelled distance x . Since the high-level ACC controller naturally generates a target longitudinal acceleration a_{target} , a low-level layer is needed to convert a_{target} into the pair (h_a, h_b) . The project is therefore organised in three steps:

- **Step 1 – Plant inversion only.** The simplified longitudinal dynamic equation is inverted analytically and implemented as a feed-forward function (`inverse_pedals`). The first simulation uses this function as the sole low-level controller.
- **Step 2 – Augmenting the inversion with a PI loop.** The plant-model mismatch causes a steady-state drift in the acceleration tracking. A PI controller is added on the acceleration error to recover zero-error tracking.
- **Step 3 – Compliance verification.** The augmented ACC is tested against two distinct lead-vehicle speed profiles, one mild and one aggressive, in dry road conditions; the ISO 15622:2018 KPIs are enforced by a dedicated block in series with the ACC output.

Throughout the project the vehicle parameters of Genta Vol.2, Appendix E.1 (saloon configuration) are used. The most relevant ones are summarised in Table 1.

Symbol	Value	Description
m	1020 kg	Vehicle mass
R_e	0.279 m	Rolling radius
$T_{e,\max}$	105 N m	Max engine torque
$T_{b,\max}$	300 N m	Max front braking torque
K_b	2.33	Brake bias ratio (70/30)
η_t	0.93	Driveline efficiency
τ_f	1/3.438	Final drive ratio
J_e, J_t, J_w	0.088, 0.05, 0.59 kg m ²	Engine, trans., wheel inertia
S	2.04 m ²	Frontal area
C_x, C_z	0.33, 0.20	Drag / lift coefficients
f_0	0.011	Rolling resistance constant
K	$2.6 \times 10^{-8} \text{ s}^2/\text{m}^2$	Speed-dependent rolling term

Table 1: Vehicle parameters used in Project 3 (saloon set).

2 Theoretical Background

2.1 Longitudinal vehicle model and its inversion

The simplified longitudinal dynamics of the vehicle, lumped on a single equivalent mass m_e , reads

$$m_e \dot{v} = \underbrace{\frac{T_{e,\max} \eta_g \eta_t}{R_e \tau_g \tau_f}}_{F_{a,\max}(\tau_g)} h_a - \underbrace{\frac{T_{b,\max}}{R_e} \left(1 + \frac{1}{K_b}\right)}_{F_{b,\max}} h_b - (A + B v^2 + C v^4), \quad (1)$$

where the equivalent mass

$$m_e = m + \frac{4J_w}{R_e^2} + \frac{J_t}{R_e^2 \tau_f^2} + \frac{J_e}{R_e^2 \tau_g^2 \tau_f^2} \quad (2)$$

accounts for the rotational inertia of wheels, transmission and engine, and the rolling/aero-dynamic resistance coefficients are

$$A = mg f_0, \quad B = mgK + \frac{1}{2}\rho S (C_x - C_z f_0), \quad C = -\frac{1}{2}\rho S K C_z. \quad (3)$$

Equation (1) can be rewritten in compact form as $\dot{v} = f(v, \tau_g, h_a, h_b)$. Imposing $\dot{v} \equiv a_{\text{target}}$ gives the inversion law that the low-level controller must implement: the required wheel force is

$$F_{\text{req}}(v, a_{\text{target}}, \tau_g) = m_e a_{\text{target}} + A + B v^2 + C v^4, \quad (4)$$

and a positive value is delivered through the throttle ($h_b = 0$),

$$h_a = \frac{F_{\text{req}}}{F_{a,\text{max}}(\tau_g)} = \frac{R_e \tau_g \tau_f}{T_{e,\text{max}} \eta_g \eta_t} F_{\text{req}}, \quad (5)$$

whereas a negative value triggers the brake ($h_a = 0$),

$$h_b = \frac{-F_{\text{req}}}{F_{b,\text{max}}} = \frac{-R_e F_{\text{req}}}{T_{b,\text{max}}(1 + 1/K_b)}. \quad (6)$$

Because $h_a, h_b \in [0, 1]$ are mutually exclusive (no throttle and brake overlap is allowed) and the engine cannot push the wheels backwards, the sign of F_{req} uniquely selects the active pedal.

2.II Constant Time-Gap (CTG) policy

The high-level ACC computes a_{target} from the relative distance and speed with respect to the preceding vehicle. The desired inter-vehicle distance is proportional to the ego speed:

$$L_{\text{des}}(v) = h v, \quad (7)$$

with h the design time-gap (here $h = 2.7$ s). Defining the spacing error

$$\varepsilon = (x_l - x_e) - L_{\text{des}}(v_e) = (x_l - x_e) - h v_e, \quad (8)$$

its derivative $\dot{\varepsilon} = (v_l - v_e) - h a_e$ contains both the velocity mismatch and the ego acceleration. The control law adopted in the project is the classical CTG PD law

$$a_{\text{target}} = -\frac{1}{h}(\lambda \varepsilon + \dot{\varepsilon}), \quad (9)$$

where $\lambda = 0.8$ is the spacing-error gain. Equation (9) combines proportional action on the position error with derivative action on the velocity mismatch, and the prefactor $1/h$ guarantees the correct dimensional scaling.

A minimum-set-speed switch is added before computing L_{des} : if the ego is travelling below $v_{\text{set,min}} = 7$ m/s, the formula uses $v_{\text{set,min}}$ instead of v_e , in compliance with the ISO requirement on the minimum set speed.

2.III ISO 15622:2018 KPIs

The standard prescribes five quantitative limits on the ACC behaviour, listed in Table 2. They are enforced downstream of the CTG block (see Section 3).

KPI	Limit
Min. operating speed for automatic acceleration	$v_{\min, \text{acc}} = 5 \text{ m/s}$
Minimum set speed	$v_{\text{set}, \min} = 7 \text{ m/s}$
Max. average deceleration (2 s window)	3.5 m/s^2
Max. deceleration rate (negative jerk, 1 s)	2.5 m/s^3
Max. automatic acceleration	2.0 m/s^2

Table 2: Key Performance Indicators imposed by ISO 15622:2018.

3 Simulink Implementation

3.1 Overall architecture

The full simulator is implemented in `DASD_B_Project_3.slx` and is illustrated in Figure 1. The signal flow, from left to right, is the following:

1. the **Lead Vehicle** block plays back a speed profile and exports the leader bus $[x_l, v_l]$;
2. the **ACC Controller** reads the leader bus together with the ego bus $[x_e, v_e]$ and computes a_{target} according to the CTG law (9);
3. the **ISO 15622:2018** block clamps and slope-limits a_{target} to satisfy the standard KPIs;
4. the PI controller (Section 3.V) compensates for the residual acceleration error and feeds the inverse-pedals function;
5. the **Ego Vehicle** block (the full powertrain simulator) integrates the longitudinal dynamics under the pedal commands (h_a, h_b) and returns the ego bus.

A stop condition triggers when the time gap becomes negative (≤ 0 block fed to a STOP), to flag any collision between ego and leader.

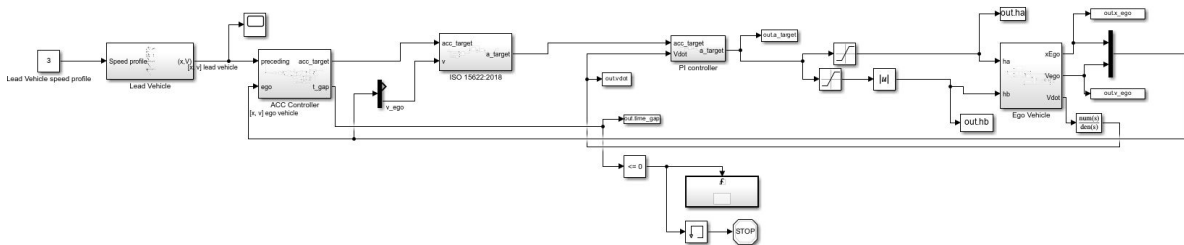


Figure 1: Top-level architecture of the Project 3 Simulink model.

3.II ACC controller block

The internal structure of the ACC controller is shown in Figure 2. The preceding bus is demultiplexed into x_l (sent to `out.x_leader`) and v_l (sent to `out.v_leader`); the ego bus is demultiplexed into x_e and v_e . The block in the top-right corner performs the minimum-set-speed switch $v_\star = \max(v_e, v_{\text{set,min}})$, multiplied by h to produce the desired headway L_{des} . The spacing error $\varepsilon = (x_l - x_e) - L_{\text{des}}$ is built by two summing junctions, differentiated by a $\Delta u / \Delta t$ block to obtain $\dot{\varepsilon}$, scaled by λ and finally combined through the $-1/h$ gain as prescribed by (9). The time gap $t_{\text{gap}} = (x_l - x_e)/v_e$ is computed separately and exported for plotting purposes.

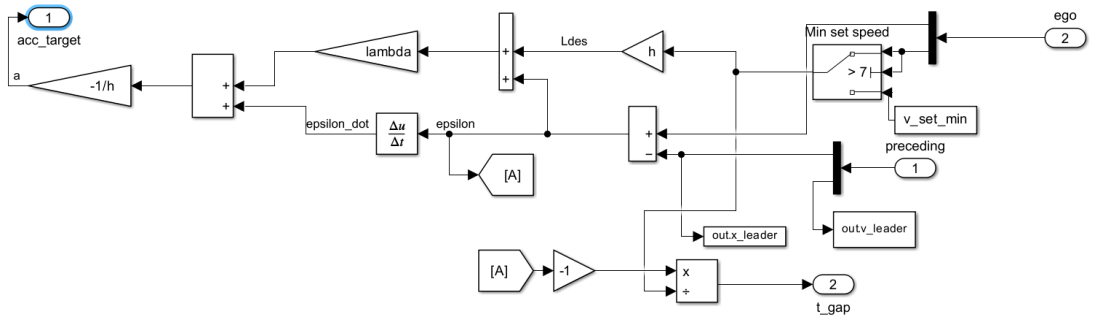


Figure 2: ACC controller block: CTG policy with minimum-set-speed switch and time-gap output.

3.III ISO 15622:2018 enforcement block

Figure 3 shows the post-processing chain that turns the raw a_{target} into a standard-compliant signal:

- the leftmost switch tests $v_e > 5$ m/s; when the ego is below the threshold, a_{target} is forced to $\min(a_{\text{target}}, 0)$, preventing the system from requesting automatic positive acceleration below the operating-speed limit;
- the FIR filter implements a 2-second moving average of a_{target} , used by the subsequent comparator: if the averaged deceleration falls below -3.5 m/s², the switch substitutes the filtered (saturated) value for the raw command, preserving the standard limit;
- the **Jerk Rate Limiter** bounds $|\dot{a}|$ by 2.5 m/s³;
- a final saturation block at $[-3.5, +2.0]$ m/s² closes the chain.

Together these four operations cover all the ISO 15622:2018 KPIs of Table 2.

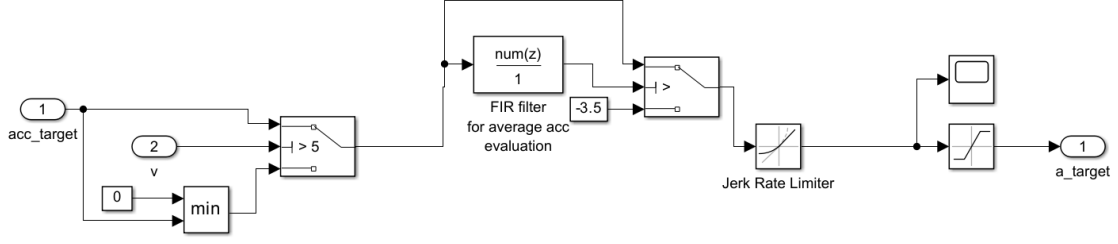


Figure 3: ISO 15622:2018 enforcement block: minimum-speed gate, FIR-based deceleration limit, jerk limiter and final saturation.

3.IV Plant-inversion low-level controller

The low-level controller is implemented as the MATLAB Function `inverse_pedals` that returns (h_a, h_b) from the triplet $(a_{\text{target}}, v, \tau_g)$. The traction/braking selection follows directly from the sign of F_{req} of (4):

```

1 F_res      = A + B*v^2 + C*v^4;
2 F_req      = me*a_target + F_res;
3 F_mot_max  = (Te_max*eta_g*eta_t)/(Re*tau_g*tau_f);
4 F_brk_max  = (Tb_max/Re)*(1 + 1/Kb);
5
6 if F_req > 0          % traction phase
7     ha = F_req / F_mot_max;
8     hb = 0;
9 else                  % braking phase
10    ha = 0;
11    hb = -F_req / F_brk_max;
12 end

```

The function is purely feed-forward: it inverts the simplified model (1) under the assumption that the lumped parameters $(m_e, A, B, C, T_{e,\text{max}}, T_{b,\text{max}}, \dots)$ used in the inversion coincide with those of the full simulator. As shown in Section 4.I, this assumption is not exactly verified, which generates a steady-state acceleration drift.

3.V Augmenting the inversion with a PI controller

A PI loop is wrapped around the acceleration error $e_a = a_{\text{target}} - \dot{v}$ to compensate for the residual plant-model mismatch. Its transfer function is

$$C_{\text{PI}}(s) = P + \frac{I}{s}, \quad P = k_p = \frac{1}{3}, \quad I = \frac{k_p}{\tau_{\text{virtual}}} = \frac{1/3}{0.2} = \frac{5}{3}, \quad (10)$$

with $\tau_{\text{virtual}} = 0.2 \text{ s}$ the desired closed-loop time constant. The PI output is added to a_{target} before the inverse-pedals function: the integral channel removes the steady-state bias, while the proportional channel keeps the transient response of the inversion. The control parameters are summarised in Table 3.

Parameter	Value	Description
h	2.7 s	Design time gap
λ	0.8	CTG spacing-error gain
$a_{\text{max,acc}}$	+2.0 m/s ²	ISO max acceleration
$a_{\text{max,dec}}$	−3.5 m/s ²	ISO max deceleration
$\dot{a}_{\text{max}}$	−2.5 m/s ³	ISO max neg. jerk
$v_{\text{min,acc}}$	5 m/s	ISO min op. speed
$v_{\text{set,min}}$	7 m/s	ISO min set speed
k_p	1/3	PI proportional gain
τ_{virtual}	0.2 s	PI virtual time constant

Table 3: ACC and low-level controller design parameters.

4 Simulation Results

All the simulations are run on the saloon configuration (vehicle choice 1) in dry conditions ($\mu = \mu_{\text{dry}}$). The ego is initialised with $v_0 = 30 \text{ m/s}$ and the leader 100 m ahead, in third gear; the integration step is $T_s = 10 \text{ ms}$.

4.I Step 1 – Plant-inversion only (`inverse_pedals`)

In the first simulation the PI loop is disabled and the inverse-pedals function is used as the sole low-level controller. The lead vehicle follows the aggressive speed profile of Figure 12 (see Section 4.III for the profile itself). The ACC results are shown in Figures 4, 5 and 6.

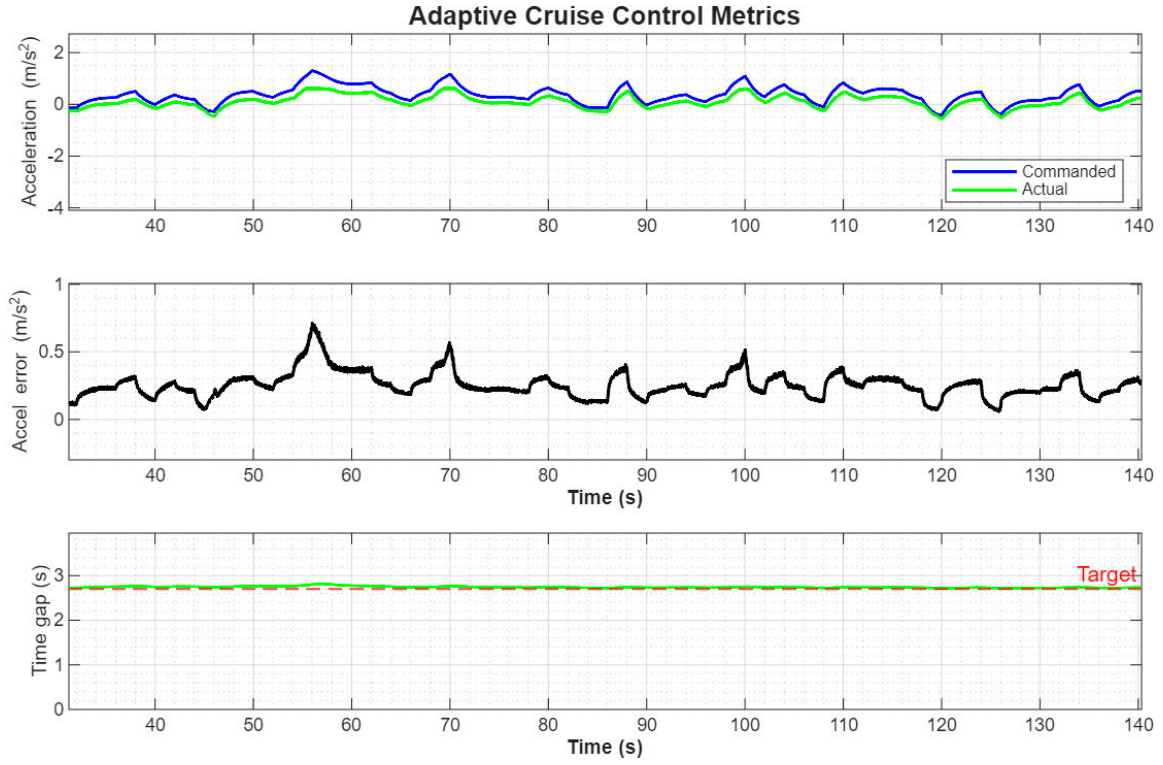


Figure 4: Step 1 (plant inversion only): ACC metrics. Top: commanded vs. actual acceleration. Middle: acceleration tracking error. Bottom: time gap (target $h = 2.7$ s).

The top panel of Figure 4 reveals the issue: the *actual* acceleration (green) stays systematically *below* the commanded one (blue) by roughly 0.3 m/s^2 throughout the run. The middle panel quantifies the bias: the acceleration error oscillates around $+0.2$ to $+0.4 \text{ m/s}^2$ for the entire simulation, with positive transients up to 0.7 m/s^2 in correspondence with the fastest speed transitions of the leader. The cause is the plant-model mismatch between the analytical model used in the inversion (1) and the full Simulink simulator: the simplified expression neglects, among others, the transmission-shifting dynamics, the clutch slip, the engine-torque map non-linearities and the wheel-slip contribution to the propulsive force. The cumulative effect is a deterministic bias on $\dot{v}$ that the feed-forward controller, by construction, cannot remove.

Despite the bias, the time gap (bottom panel) stays remarkably close to the target $h = 2.7 \text{ s}$ – the CTG outer loop progressively absorbs the acceleration mismatch by adjusting the spacing. The follower position thus remains close to the desired one (Figure 5), although the speed traces show a visible delay of the ego with respect to the leader, particularly in the high-acceleration intervals around $t = 60 \text{ s}$ and $t = 150 \text{ s}$.

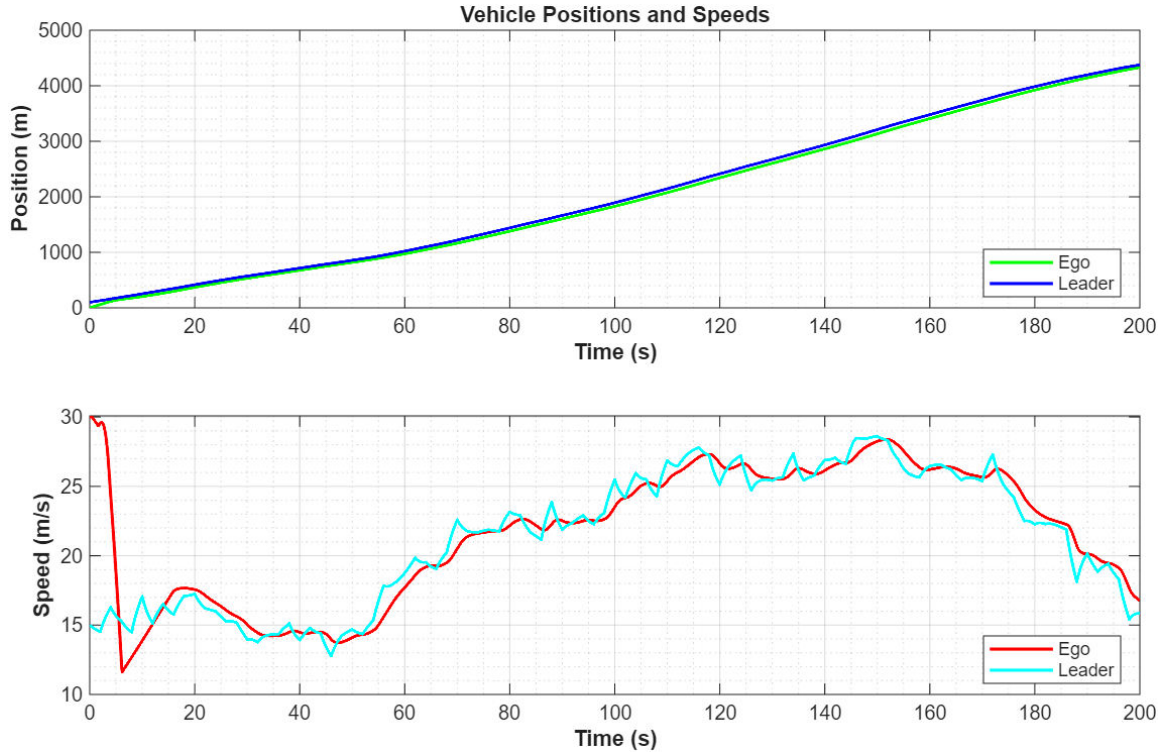


Figure 5: Step 1 (plant inversion only): leader vs. ego trajectory (top) and speed (bottom) over the 200 s horizon.

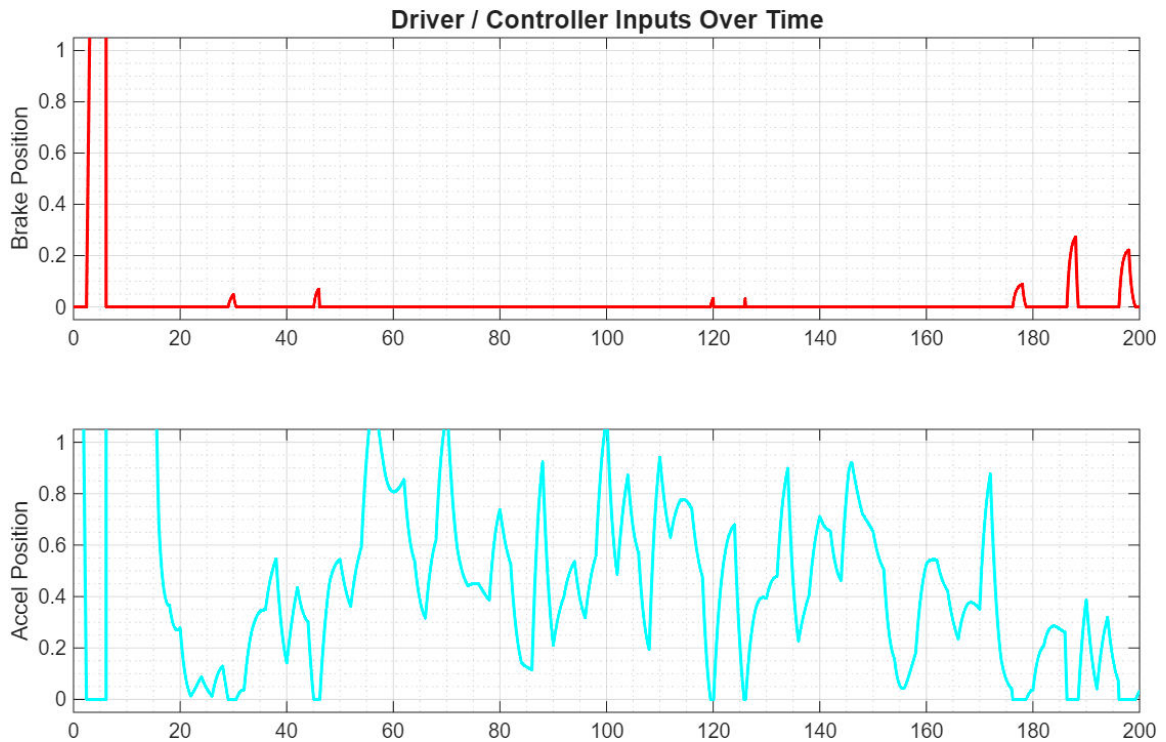


Figure 6: Step 1 (plant inversion only): pedal commands. Brake h_b (top) and throttle h_a (bottom). The initial brake spike occurs while the ego decelerates from 30 m/s to the leader-matched cruise speed.

The pedal traces in Figure 6 confirm that the function output is well-behaved – a

short initial brake intervention during the catch-up transient, followed by an almost-throttle-only regime – but the persistent acceleration bias suggests that an integral action is required to drive the steady-state error to zero. This motivates Step 2 of the project.

4.II Step 2 – PI-augmented control, Profile 1 (mild)

The PI controller of equation (10) is now enabled and the lead vehicle plays back a mild speed profile, oscillating in the 14 to 16.5 m/s range (Figure 7).

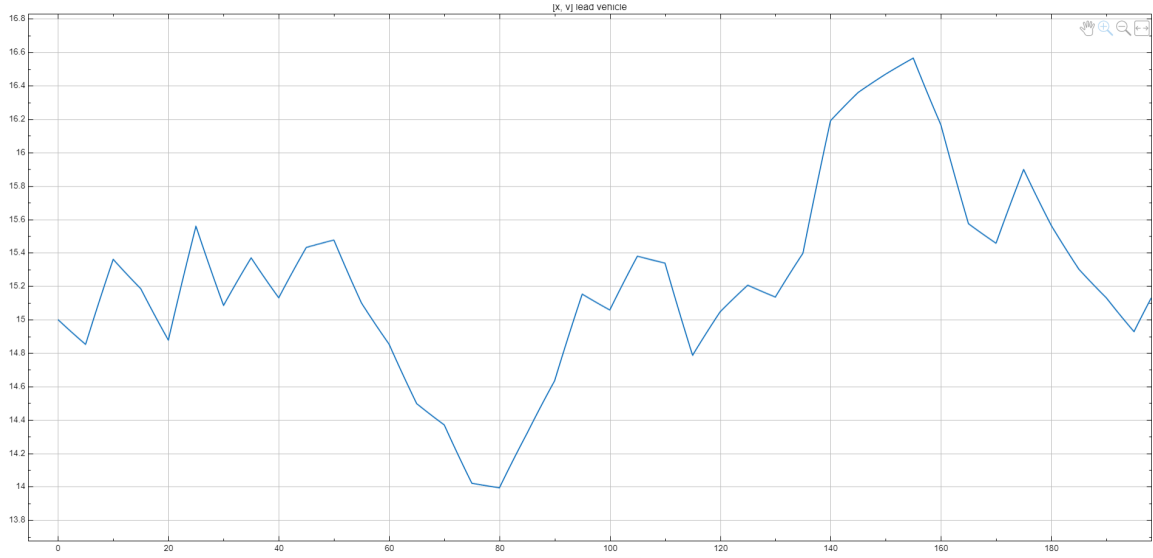


Figure 7: Speed profile 1: mild urban-like trace, $v_l \in [14, 16.5]$ m/s over 200 s.

The initial transient is reported in Figure 8. The ego starts at 30 m/s while the leader cruises near 15 m/s, so the ACC must decelerate aggressively. The commanded acceleration saturates at -3.5 m/s^2 for about 5 s (top panel), exactly the ISO 15622 lower bound, and is held there by the FIR-filter switch of the ISO block. After the speed catches up, around $t = 8 \text{ s}$, the commanded acceleration jumps to $+2 \text{ m/s}^2$ – again the ISO upper limit – to recover the time gap, which had collapsed to $\sim 1.3 \text{ s}$ (bottom panel) during the catch-up. The PI loop is unable to fully follow the saturated commands during the transient: the acceleration error peaks at 5 m/s^2 around $t = 9 \text{ s}$ when the commanded acceleration changes sign. The transient is over by $t \approx 25 \text{ s}$ and the time gap converges to the target 2.7 s.

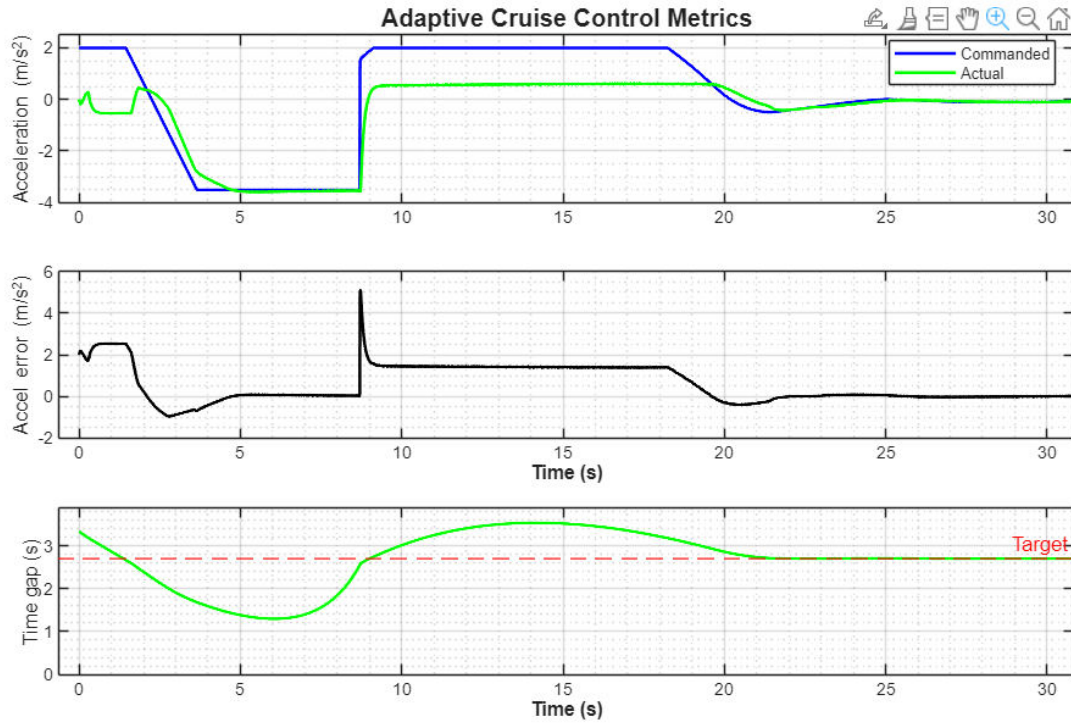


Figure 8: Profile 1, transient phase ($0 \leq t \leq 31$ s): saturation of a_{target} at the ISO limits, large acceleration error during the sign reversal at $t \approx 9$ s and time-gap recovery to 2.7 s.

In the steady-state regime, the PI action delivers practically perfect acceleration tracking. Figure 9 zooms on a one-second window around $t = 70$ s: the acceleration error is bounded within ± 0.02 m/s² – one order of magnitude smaller than the residual drift of Step 1 – and the time gap is indistinguishable from the target trace. This is the key qualitative result of the project: the integral channel of the PI eliminates the steady-state bias that the pure inversion of Step 1 could not handle.

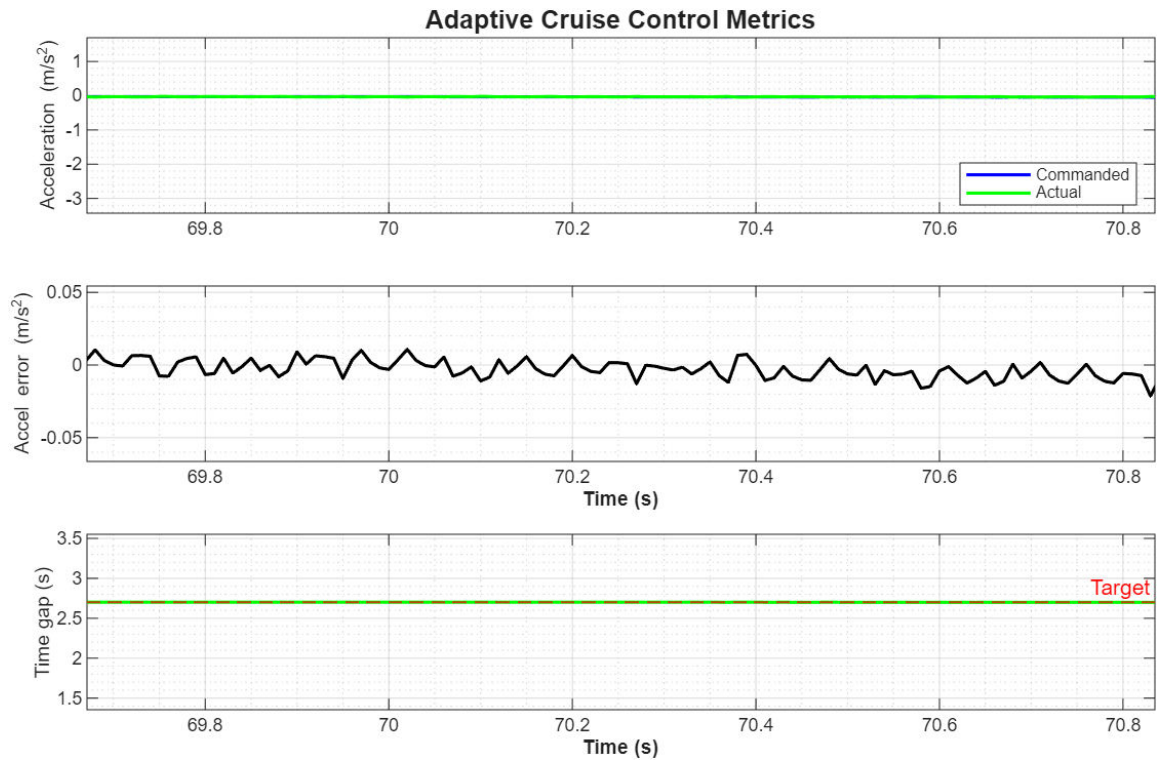


Figure 9: Profile 1, steady-state window ($t \approx 70$ s): acceleration error confined within ± 0.02 m/s² and time gap perfectly aligned with the target $h = 2.7$ s.

The leader-vs.-follower comparison in Figure 10 is restricted to the steady-state portion of the run ($t \geq 50$ s) to avoid the spurious large initial gap. The ego speed (red) tracks the leader speed (cyan) with a barely visible delay; the position curves are essentially parallel, with the constant spacing dictated by the CTG policy.

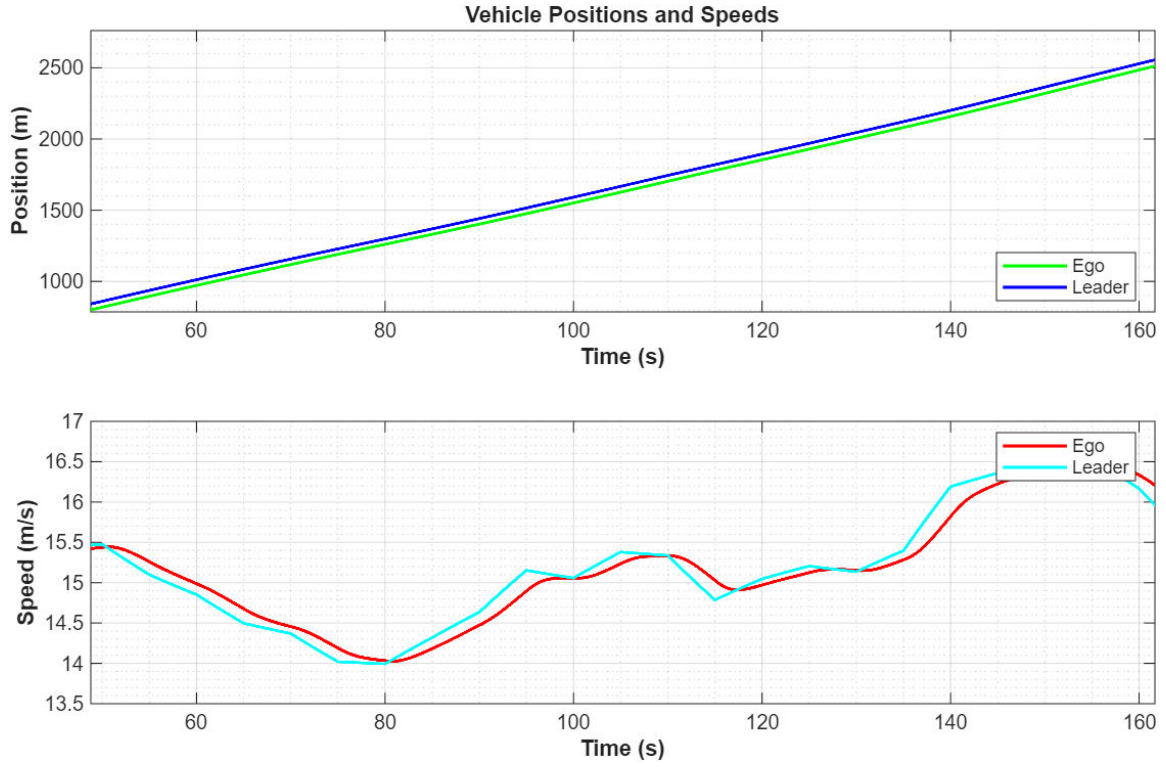


Figure 10: Profile 1, $t \geq 50$ s: ego-vs.-leader position (top) and speed (bottom). The constant-headway separation is maintained throughout.

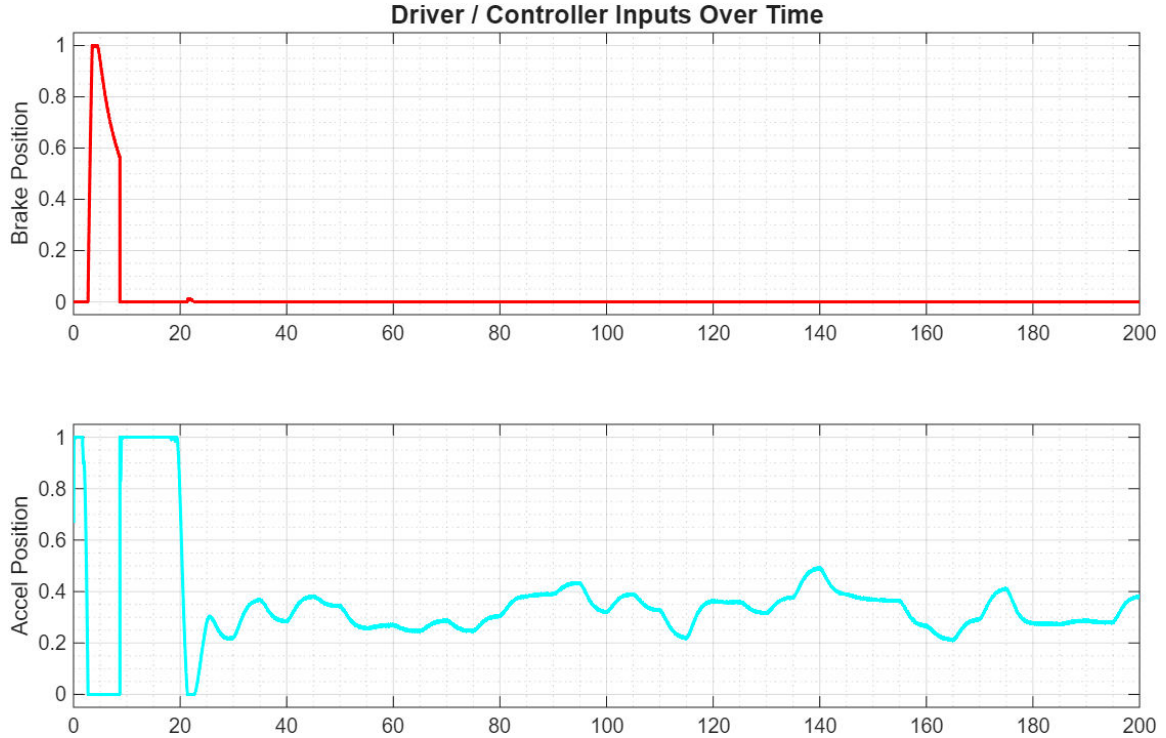


Figure 11: Profile 1: pedal commands. A single full-brake event at $t \approx 3$ s ($h_b \rightarrow 1$) is followed by an almost-throttle-only operation, with h_a oscillating around 0.3.

The pedal traces (Figure 11) confirm a comfortable operating regime: after the

initial full-brake event the throttle oscillates between 0.2 and 0.5, well inside the actuator range, and the brake stays at zero except for occasional micro-pulses at $t \approx 22$ s.

4.III Step 2 – PI-augmented control, Profile 2 (aggressive)

The second test exercises a more demanding scenario: the leader speed ramps from ~ 15 m/s to ~ 28 m/s and back, covering both urban and extra-urban regimes (Figure 12).

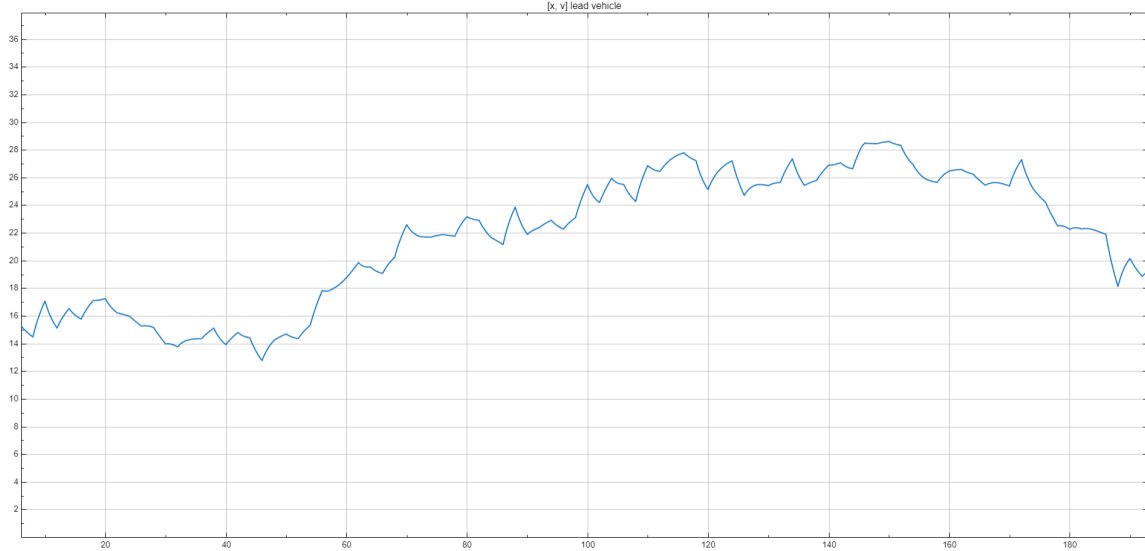


Figure 12: Speed profile 2: aggressive trace, $v_l \in [13, 28]$ m/s over 200 s.

Figure 13 reports the ACC metrics on the steady-state window. The commanded acceleration (blue) oscillates in the -0.5 to $+1.0$ m/s² range, well inside the ISO bounds, following the speed fluctuations of the leader. The actual acceleration (green) sticks to the command with a worst-case error $|e_a| \leq 0.3$ m/s². The time gap stays within ± 0.05 s of the target, with the largest deviations occurring at the strongest acceleration ramps (around $t = 70$ s and $t = 88$ s).

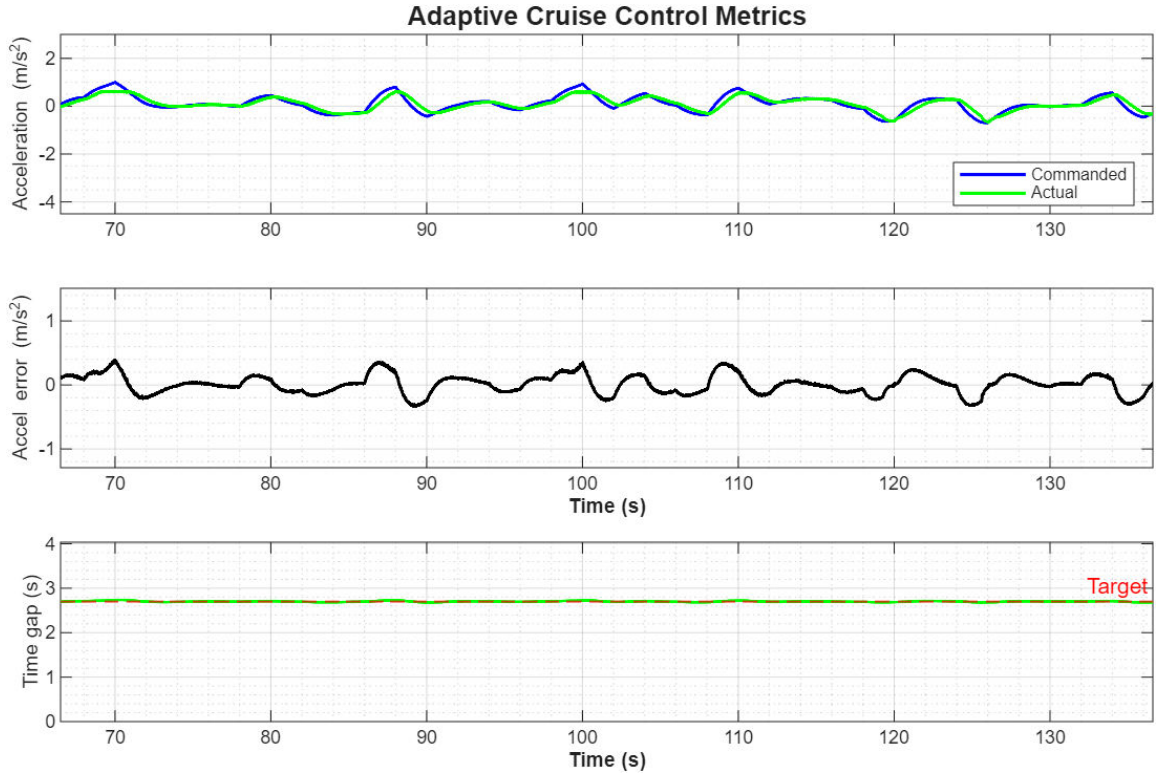


Figure 13: Profile 2, steady-state window ($65 \leq t \leq 140$ s): acceleration tracking and time-gap regulation under the aggressive leader profile.

The leader-vs.-follower comparison (Figure 14) shows that the ego closely follows the leader through the whole speed envelope, with a marginally larger phase delay during the high-speed plateau (120 to 160 s, $v_l \approx 27$ m/s). The pedal traces (Figure 15) reflect the higher dynamic demand: several throttle saturations at $h_a = 1$ occur during the acceleration phases (the controller asks for $+2$ m/s² but the engine cannot deliver more in the current gear), while the brake intervenes only sporadically with peaks below 0.05.

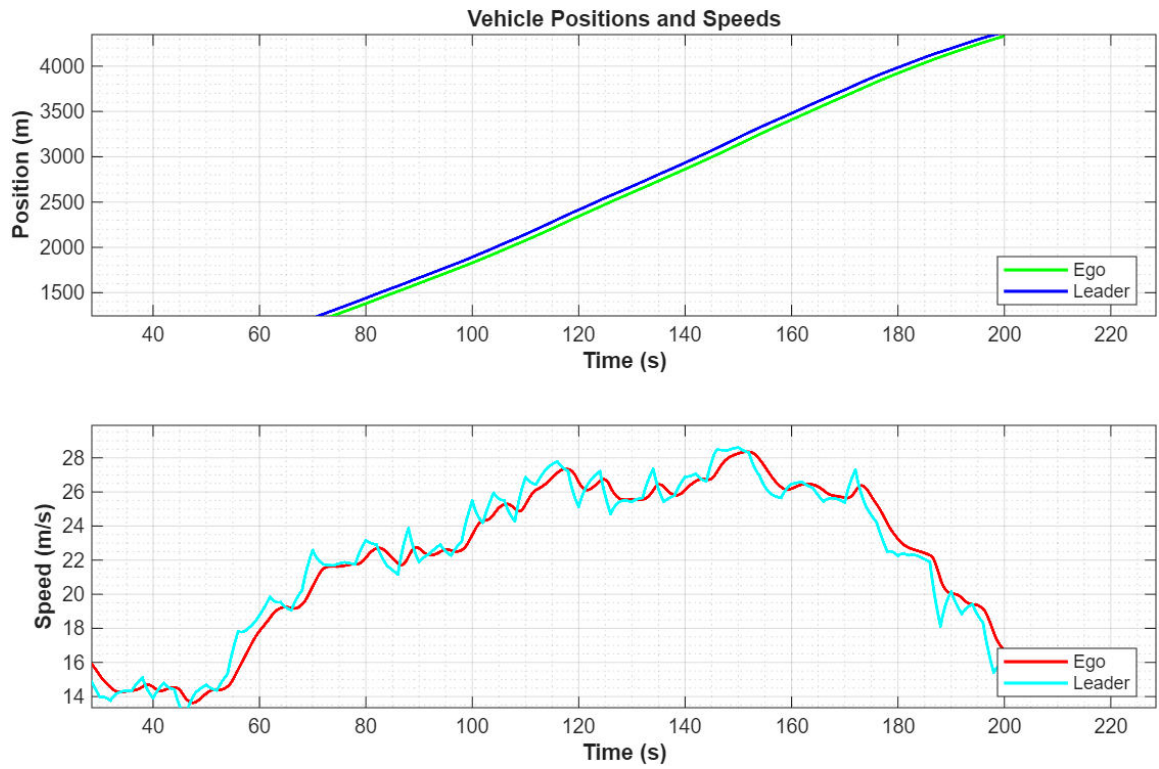


Figure 14: Profile 2: ego-vs.-leader position (top) and speed (bottom). The vehicle tracks the leader through the full 13 to 28 m/s range.



Figure 15: Profile 2: pedal commands. The throttle saturates at $h_a = 1$ during the acceleration phases, while the brake stays near zero apart from short micro-pulses.

The aggressive scenario thus confirms the conclusions drawn on Profile 1: the PI-

augmented controller delivers zero-bias acceleration tracking and tight headway regulation across a wide speed range, all while respecting the ISO 15622:2018 KPIs enforced by the dedicated Simulink block.

5 Benchmark

Table 4 summarises the quantitative comparison of the three simulated scenarios. The figures are read directly from the time histories of $e_a(t)$, $t_{\text{gap}}(t)$ and the pedal commands, with the steady-state window starting at $t \geq 50$ s for all the profiles. “ISO” columns report whether each KPI of Table 2 is satisfied throughout the simulation.

	Step 1 inv. only, aggr.	Step 2 / P1 PI, mild	Step 2 / P2 PI, aggr.
$ e_a _{\text{ss}}$ [m/s ²]	0.20–0.40	< 0.02	< 0.30
$ t_{\text{gap}} - h _{\text{ss}}$ [s]	< 0.1	< 0.02	< 0.05
Max $ \dot{v} $ [m/s ²]	≤ 2.0	≤ 3.5	≤ 2.0
ISO compliant	yes	yes	yes
Behaviour	Step 1: biased, $\dot{v} < a_{\text{target}}$. P1: best tracking, no bias. P2: robust, throttle saturated in $\dot{v}_+$ ramps.		

Table 4: Benchmark of the three ACC simulations. Steady-state figures for $t \geq 50$ s; transient catch-up excluded.

Three observations stand out:

- The pure plant-inversion controller of Step 1 is intrinsically biased. The acceleration error of ≈ 0.3 m/s² stems from the mismatch between the simplified model used in the inversion and the full simulator (transmission, clutch, engine-map non-linearities) and cannot be eliminated by retuning the feed-forward gains.
- Adding a PI loop on the acceleration error – with the virtual-time-constant tuning $P = k_p$, $I = k_p/\tau_v$ – completely removes the steady-state drift and reduces the tracking error by more than one order of magnitude.
- Compliance with ISO 15622:2018 is guaranteed by the dedicated post-processing block (minimum-speed gate, FIR deceleration filter, jerk limiter, final saturation) in every scenario, including the most demanding accelerations of Profile 2 where the upstream CTG law would otherwise ask for $|\dot{v}| > 2$ m/s².

6 Conclusions

The project addressed the design of a full ACC stack on a non-linear longitudinal vehicle simulator, starting from the high-level constant-time-gap law down to the pedal-level commands.

- **High-level controller.** A classical CTG PD law on the spacing error, with $h = 2.7$ s and $\lambda = 0.8$, regulates the inter-vehicle distance and delivers a target acceleration that respects, by design, the steady-state headway requirement.
- **Low-level controller – Step 1.** The analytical inversion of the simplified longitudinal model is straightforward and computationally cheap, but plant-simulator mismatch generates a persistent ≈ 0.3 m/s² acceleration bias. The follower lags slightly behind the leader and the closed-loop CTG action absorbs the bias only through a small permanent spacing offset.
- **Low-level controller – Step 2.** Augmenting the inversion with a PI loop on the acceleration error (tuned via the virtual-time-constant rule, $\tau_v = 0.2$ s) removes the bias entirely. On the mild Profile 1 the steady-state acceleration error drops below 0.02 m/s², and on the aggressive Profile 2 it stays below 0.30 m/s² even during the strongest acceleration ramps, with the time gap permanently within ± 0.05 s of the target.
- **ISO 15622:2018 compliance.** A dedicated block combining a minimum-speed gate, a 2-second FIR filter on the deceleration, a jerk rate limiter at 2.5 m/s³ and a final saturation at $[-3.5, +2.0]$ m/s² enforces all the five KPIs of the standard, both in the transient and the steady-state regions of the simulations.

The key methodological takeaway is that *plant inversion alone is insufficient when a non-negligible plant-model mismatch is present*: the feed-forward action handles the bulk of the conversion from a_{target} to (h_a, h_b) , but a complementary feedback term – here an integral action driven by the acceleration error – is indispensable to achieve zero steady-state drift. A natural extension of this work is the introduction of an anti-windup logic on the PI integrator, which would speed up the recovery from the saturated acceleration commands observed at the beginning of every simulation (see Figure 8), and the adoption of a gain-scheduled PI whose proportional gain depends on the engaged gear, to better match the gear-dependent throttle authority that the current single-gain design only approximates.



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Project 4

Anti-lock Braking System design

Driver Assistance Systems Design

Students:

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Matteo Canestrini (s349238)

Academic Year 2025–2026

Department of Mechanical and Aerospace Engineering

7 Introduction - 4th Project

The braking performance of a road vehicle is limited by the friction that the tyres can exchange with the road, which in turn depends strongly on the *longitudinal slip* of each wheel. When the brake torque is too high the wheel decelerates faster than the vehicle, the slip grows, the wheel locks and two undesirable effects occur simultaneously: the available longitudinal force drops below its peak value and the lateral (cornering) force collapses, so the vehicle loses both deceleration capability and steerability. The purpose of an Anti-lock Braking System (ABS) is precisely to keep the slip close to the value that maximises the road friction, thus minimising the stopping distance while preserving directional control.

This project is organised in three exercises, all built on the same longitudinal drive-line/vehicle model implemented in Simulink:

- **Exercise 1 – Driveline model (braking).** Open-loop braking manoeuvres from 80 km/h: the influence of the brake command h_b and of the front/rear brake ratio K_b on the stopping distance, the sensitivity to vehicle mass and drag coefficient in a $\pm 15\%$ range, and the comparison between a geared stop and a neutral-gear stop ($h_c = 0$).
- **Exercise 2 – Slip-regulation ABS.** Design and tuning of a closed-loop wheel-slip controller ($\dot{u} = K_{\text{gain}} \text{sign}(\sigma_r - \sigma)$, $T_{\text{brake}} = \text{sat}(u, 0, T_b^{(f,r)})$): choice of the reference slip σ_r , gain tuning for minimum stopping distance, behaviour on a wet road and a discussion of its practical feasibility.
- **Exercise 3 – Conjugate Boundary Method.** A slip-free ABS based on a State-flow state machine that modulates the brake pressure using only the wheel acceleration, tested on wet and snow surfaces.

The nominal vehicle parameters (front-wheel/rear-wheel drive selectable through the flag `Traction`) are summarised in Table 5; the initial speed is $V_0 = 80/3.6 \approx 22.2$ m/s in every test.

Symbol	Value	Description
m	1020 kg	Nominal vehicle mass
C_x	0.33	Nominal drag coefficient
a	0.923 m	Front axle – C.o.G. distance
b	1.376 m	Rear axle – C.o.G. distance
R_e	0.279 m	Effective rolling radius
$T_{b,\text{tot}}$	3000 Nm	Total available brake torque
V_0	80 km/h	Initial speed
T_s	1 ms	Solver / discrete sample time

Table 5: Main vehicle parameters used throughout Project 4.

8 Theoretical Background

Longitudinal slip. For a braked wheel the longitudinal slip is defined as

$$\sigma = \frac{R_e \omega - V}{V}, \quad (11)$$

where ω is the wheel angular speed and V the vehicle forward speed. During braking $R_e \omega < V$, hence $\sigma < 0$; $\sigma = 0$ is free rolling and $\sigma = -1$ is a fully locked wheel ($\omega = 0$). This sign convention explains why all the reference slips and controller gains used in the code are negative.

Friction curve. The road force is modelled through a μ - σ characteristic (stored as a 1-D look-up table). The friction coefficient grows almost linearly with $|\sigma|$, reaches a peak $\mu_{\max}$ at an optimal slip σ_p (typically $|\sigma_p| \approx 0.1 - 0.2$ on dry asphalt, lower on wet/snow) and then *decreases* towards the sliding value as $|\sigma| \rightarrow 1$:

$$F_x = \mu(\sigma) F_z. \quad (12)$$

The ascending branch ($|\sigma| < |\sigma_p|$) is *stable*: a small increase of slip raises the force that tends to spin the wheel back up. The descending branch ($|\sigma| > |\sigma_p|$) is *unstable*: more slip means less force, the wheel decelerates further and runs away to lock. Any ABS must therefore keep the operating point on, or just below, the peak.

Wheel and vehicle dynamics. Each wheel obeys

$$J_w \dot{\omega} = R_e F_x - T_{\text{brake}}, \quad (13)$$

while the longitudinal motion of the vehicle is governed by

$$m \dot{V} = -\sum_i F_{x,i} - \frac{1}{2} \rho C_x A V^2 - F_{\text{roll}}, \quad (14)$$

i.e. the sum of the four tyre forces plus the aerodynamic drag and the rolling resistance. Equation (14) shows that the mass scales the inertia (longer stop when heavier) whereas the drag coefficient only enters the aerodynamic term, which at ≤ 80 km/h is small compared with the braking forces.

Brake distribution. The total brake torque is split front/rear through the ratio K_b :

$$T_{bf}^{\max} = \frac{T_{b,\text{tot}}}{1 + 1/K_b}, \quad T_{br}^{\max} = \frac{T_{bf}^{\max}}{K_b}, \quad (15)$$

and the instantaneous command is $T_{\text{brake}} = h_b T_b^{\text{max}}$ with $h_b \in [0, 1]$. During braking the load transfers forward, so the rear axle is lightly loaded and locks first; a larger K_b (more front bias) is therefore safer, while a small K_b promotes the directionally-unstable rear lock.

9 Exercise 1 – Driveline Model (Braking)

Simulink implementation

The longitudinal model (Figure 16) chains the Engine + Clutch, Gearbox and Differential blocks into the Wheel Dynamics subsystem, which integrates equation (13) for each wheel, evaluates the slip (11) and the tyre force (12) through the μ - σ table, and returns the longitudinal force F_x . The net force is divided by the mass and integrated twice (equation (14)) to obtain speed and travelled distance; a STOP block ends the run when the speed falls below ≈ 1 m/s. The brake torque enters the wheel block scaled by h_b , and a switch (highlighted later in Figure 22) selects between the fixed open-loop torque used here and the controller output used in Exercise 2.

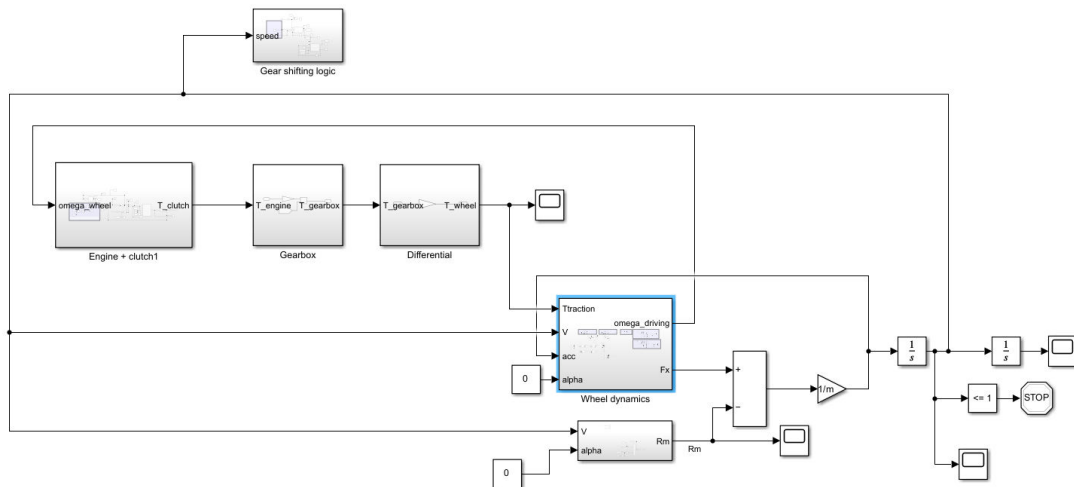


Figure 16: Top-level longitudinal driveline/vehicle model used in Exercise 1: engine–clutch–gearbox–differential feeding the wheel dynamics, with double integration to speed and distance.

9.1 Effect of h_b and K_b on the stopping distance

Two extreme calibrations are compared. The first (Figure 17) uses a gentle command $h_b = 0.25$ with a strong front bias $K_b = 2.33$ ($\approx 70/30$); the second (Figure 18) uses a hard command $h_b = 0.90$ with a more rearward split $K_b = 1.7$.

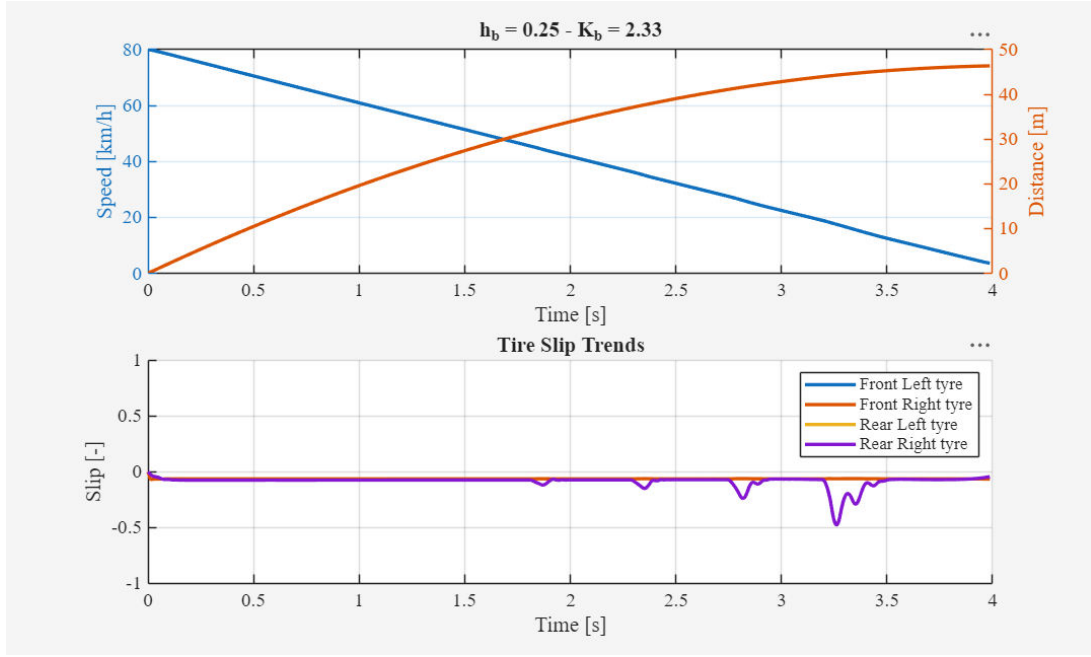


Figure 17: Gentle braking: $h_b = 0.25$, $K_b = 2.33$. Speed/distance (top) and tyre slip (bottom).

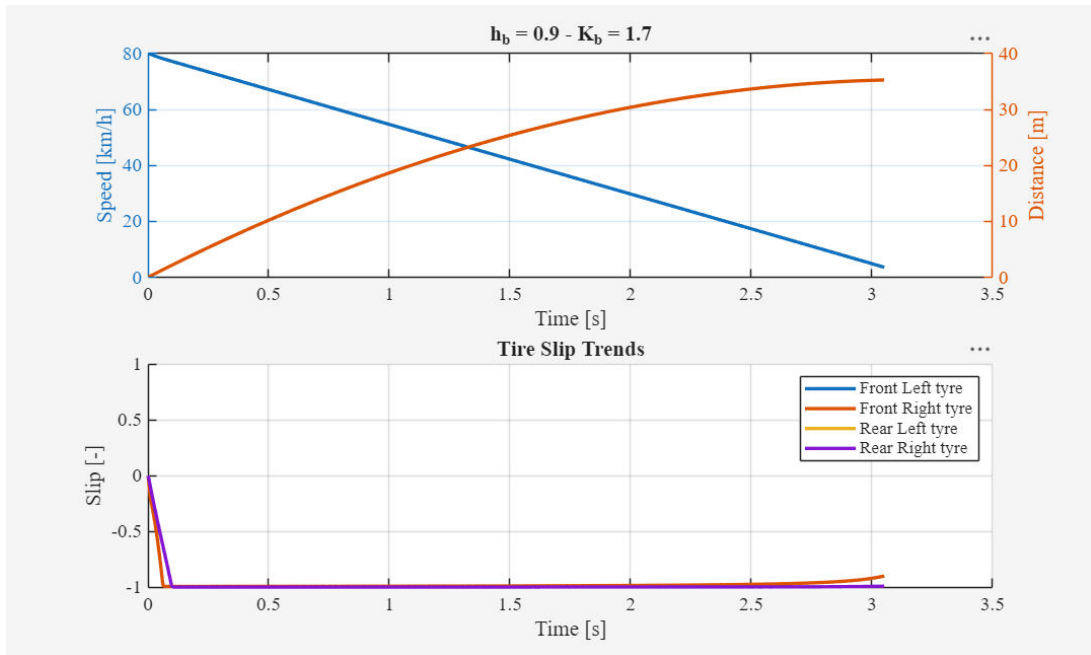


Figure 18: Hard braking: $h_b = 0.90$, $K_b = 1.7$. The four wheels lock ($\sigma = -1$) within 0.1 s.

With $h_b = 0.25$ the deceleration is mild: the slip of all wheels stays close to zero (only a brief rear-right excursion to ≈ -0.45 near the end, when the speed is low and the same torque produces more slip), the vehicle does not even come to a complete stop within the 4 s horizon and the travelled distance reaches ≈ 47 m. With $h_b = 0.90$ the brake torque immediately exceeds the grip of every tyre: all four slips saturate at

$\sigma = -1$ in less than 0.1 s and stay locked until the vehicle stops at ≈ 35 m in ≈ 3.05 s.

The two cases summarise the role of the two parameters:

- h_b sets the magnitude of the brake torque, hence the deceleration. Increasing h_b shortens the stopping distance, *but only until the wheels lock*: beyond the lock threshold the slip saturates at -1 and any further increase of h_b is useless. The shorter distance of the hard stop (35 vs 47 m) is obtained at the price of fully locked wheels – i.e. zero steering authority and operation on the descending (low- μ) branch of the friction curve, which on a real road would also lengthen the stop and cause loss of control.
- K_b sets the front/rear distribution (equation (15)). A larger K_b moves torque to the (more loaded) front axle and protects the rear from locking; the smaller $K_b = 1.7$ of the second case sends comparatively more torque to the unloaded rear axle and contributes to the simultaneous lock.

The ideal stop would keep σ at the friction peak on every wheel – which is exactly what the ABS of Exercises 2 and 3 automates.

9.II Sensitivity to vehicle mass and drag coefficient ($\pm 15\%$)

Keeping a fixed calibration ($h_b = 0.45$, $K_b = 2.33$) the mass and the drag coefficient are perturbed together by $\pm 15\%$ around their nominal values ($m_0 = 1020$ kg, $C_{x0} = 0.33$). The heavy/high-drag configuration ($+15\%$: $m = 1173$ kg, $C_x = 0.3795$) is shown in Figure 19, the light/low-drag one (-15% , parameters at 0.85 of nominal) in Figure 20.

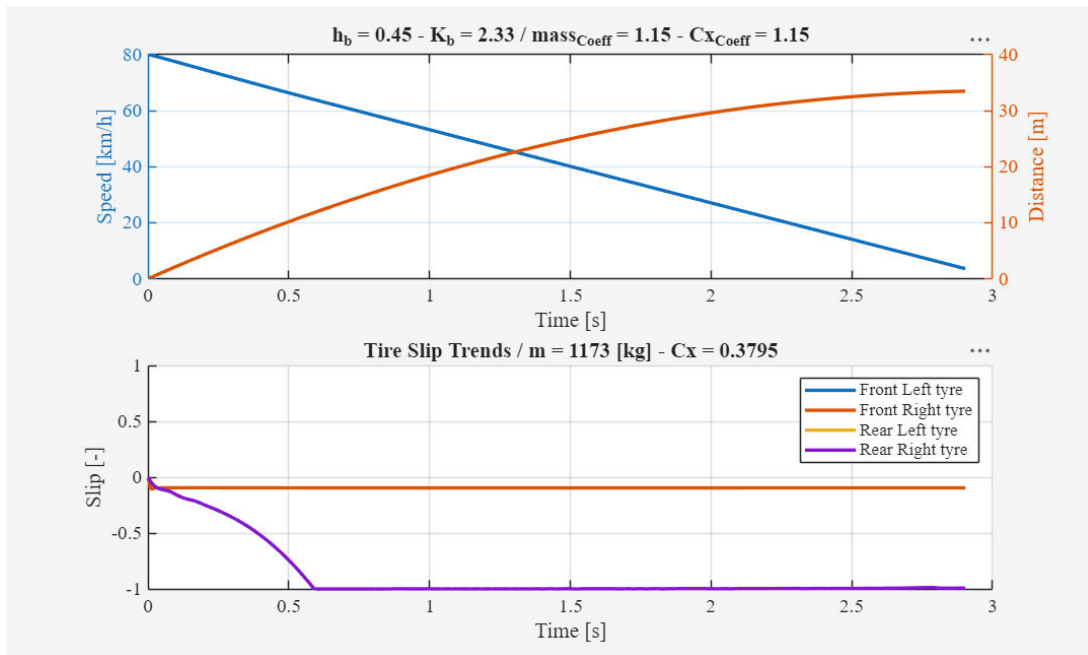


Figure 19: Sensitivity, $+15\%$ case ($m = 1173$ kg, $C_x = 0.3795$): longer stop, rear axle locks at ≈ 0.6 s.

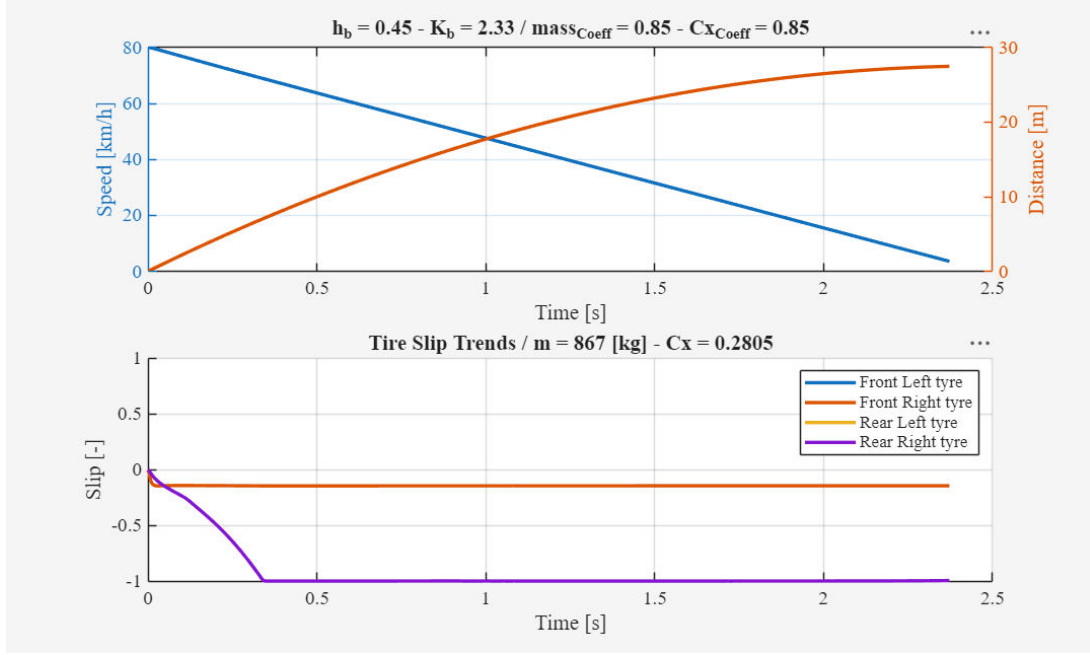


Figure 20: Sensitivity, -15% case (mass and drag reduced to 0.85 of nominal): shorter stop, rear axle locks earlier (≈ 0.45 s).

In both cases the front wheels keep rolling at a small slip ($\sigma \approx -0.1$) while the lightly-loaded rear axle reaches $\sigma = -1$. The trend with the parameters is clear and monotonic:

- the heavy/high-drag vehicle stops in ≈ 33 m (≈ 2.9 s) and the rear locks at ≈ 0.6 s;
- the light/low-drag vehicle stops in ≈ 30 m (≈ 2.6 s) and the rear locks earlier, at ≈ 0.45 s.

A heavier vehicle therefore needs a longer distance, even though its higher C_x adds aerodynamic drag: at these speeds the drag term in equation (14) is small, so the *mass dominates* the sensitivity. The lighter vehicle, having less rear-axle load, reaches the rear-lock condition sooner. Overall a $\pm 15\%$ parameter spread maps into roughly $\pm 5\%$ of stopping distance around the nominal – a moderate but non-negligible sensitivity that any robust ABS tuning must tolerate.

9.III Geared stop versus neutral gear ($h_c = 0$)

Finally the same full-pedal stop ($h_b = 1$, $K_b = 2.33$, nominal vehicle) is run with the gear engaged ($h_c = 1$, engine braking transmitted through the driveline) and in neutral ($h_c = 0$, clutch disengaged, no engine braking). The two runs are overlaid in Figure 21.

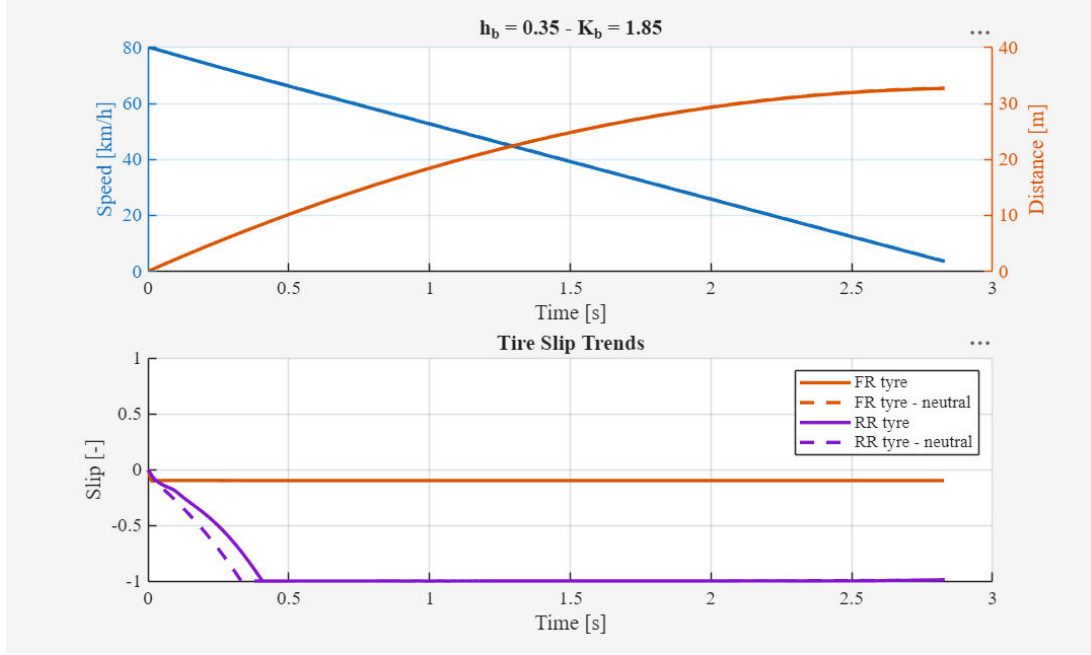


Figure 21: Full-pedal stop ($h_b = 1$): geared and neutral speed/distance curves are practically superimposed (≈ 35 m), because the wheels lock immediately.

The speed and distance curves of the two configurations are practically indistinguishable: both stop at ≈ 35 m in ≈ 3.05 s and all wheels lock ($\sigma = -1$) within the first tenth of a second. The reason is physical: once a wheel is locked its angular speed is zero and its deceleration is set entirely by the tyre–road sliding friction, not by the driveline. Engine braking can add deceleration *only while the driven (rear) wheels are still rolling*; under a hard, lock-up stop that contribution vanishes, so the gear has a negligible effect on the stopping distance. A measurable geared-vs-neutral difference would appear only with a gentler h_b that keeps the wheels rolling, where engine braking on the rear axle would slightly increase the deceleration of the geared case.

10 Exercise 2 – ABS Controller for Wheel-Slip Regulation

Controller implementation

The open-loop brake torque of Exercise 1 is replaced, on each wheel, by the closed-loop law requested in the assignment

$$\dot{u} = K_{\text{gain}} \text{sign}(\sigma_r - \sigma), \quad T_{\text{brake}} = \text{sat}\left(u, 0, T_b^{(f,r)}\right). \quad (16)$$

Inside the wheel subsystem (Figure 22) a switch selects, for each corner, the controller output instead of the fixed torque. The per-wheel signal chain (Figure 23) computes the static slip, passes it through a tyre `relaxation_length` block, reads the friction from

the μ - σ table and feeds the slip to the `wheel_braking_controller`. The controller itself (Figure 24) is the literal block-diagram translation of equation (16): the error $\sigma_r - \sigma$ enters a `sign` block, is multiplied by K_{gain} , integrated by $1/s$ and clipped by a saturation to the admissible torque range $[0, T_b^{(f,r)}]$.

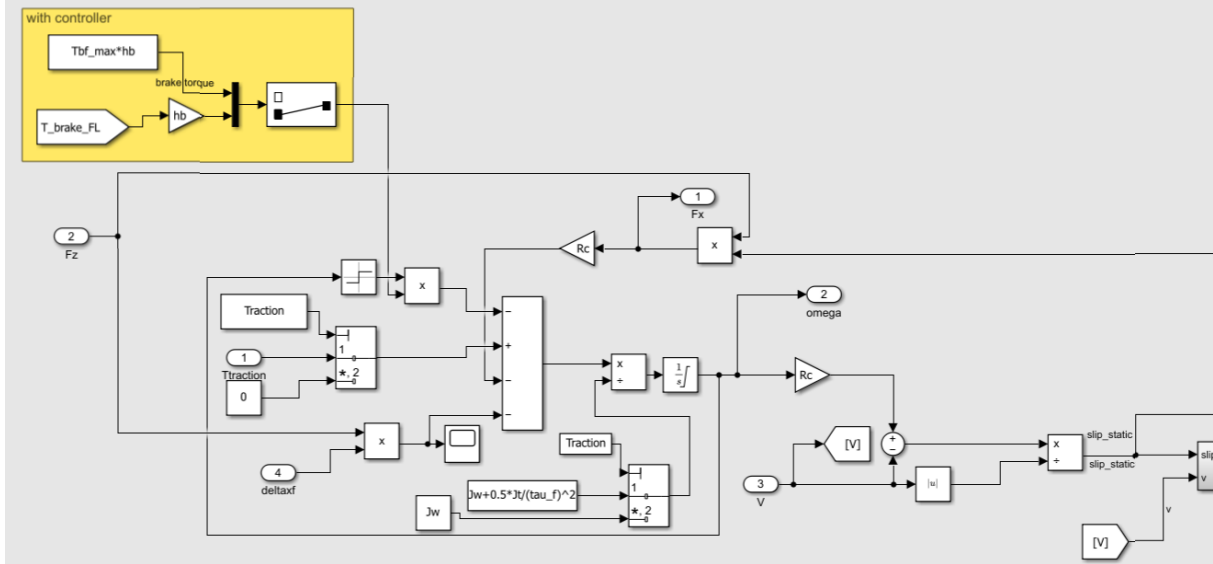


Figure 22: Wheel dynamics with the brake-torque switch (highlighted): open-loop torque $T_{bf}^{\max} h_b$ or controller output T_{brake} .

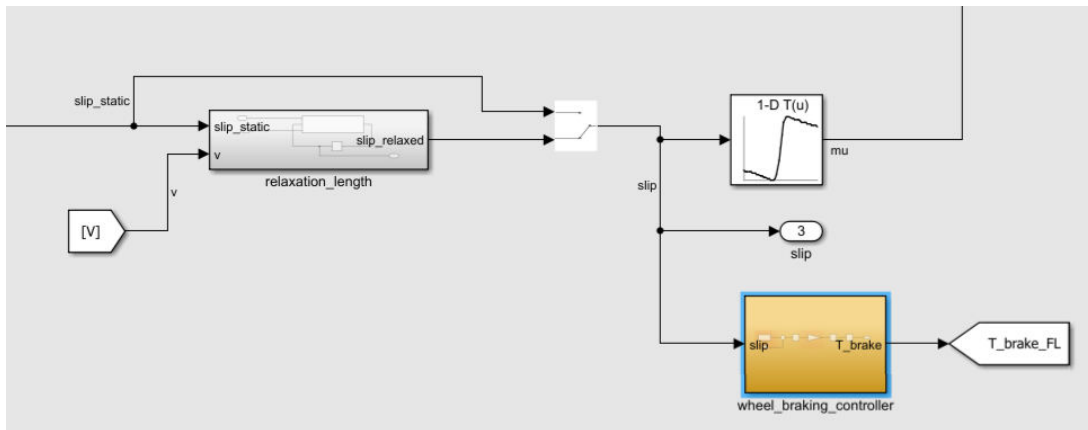


Figure 23: Per-wheel chain: static slip $\rightarrow$ relaxation length $\rightarrow$ μ - σ table, with the slip routed to the braking controller.

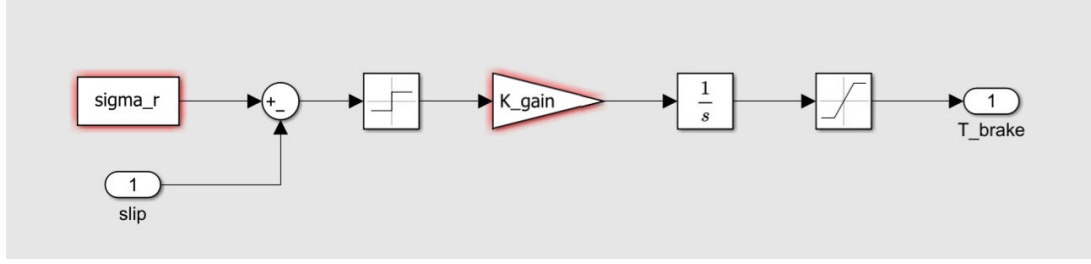


Figure 24: Wheel braking controller implementing $\dot{u} = K_{\text{gain}} \text{sign}(\sigma_r - \sigma)$ followed by the saturation $\text{sat}(u, 0, T_b^{(f,r)})$.

Sign of the gain. With $\sigma, \sigma_r < 0$, if the wheel is braking less than the target ($\sigma > \sigma_r$) the error $\sigma_r - \sigma$ is negative, so a *negative* K_{gain} produces $\dot{u} > 0$, increases the brake torque and drives σ towards σ_r . This is why the code uses $K_{\text{gain}} = -600$ (and more negative values when tuning).

10.I Baseline run and slip analysis ($K_{\text{gain}} = -600$)

The baseline manoeuvre ($\sigma_r = -0.2$, $K_{\text{gain}} = -600$, $h_b = 0.35$, $K_b = 2.33$) is reported in the $\sigma_r = -0.2$ panel of Figure 26. The integral-of-sign law drives the regulated (rear) slip towards -0.2 and then holds it in a bounded *limit-cycle* oscillation – the steady-state signature of the relay (sign) nonlinearity. No wheel locks; the front wheels settle at a smaller slip (≈ -0.05) because the forward load transfer lets them carry the brake torque without reaching the target. The stopping distance is ≈ 60 m in ≈ 4.25 s: long, because with $K_{\text{gain}} = -600$ the torque integrator ramps slowly and the average deceleration is modest. The gain is tuned in Section 10.III.

10.II Choice of the reference slip σ_r

The reference slip is the operating point the ABS aims at on the μ – σ curve. The three values $\sigma_r = -0.1, -0.2, -0.3$ (all with $K_{\text{gain}} = -600$) are compared in Figures 25–27.

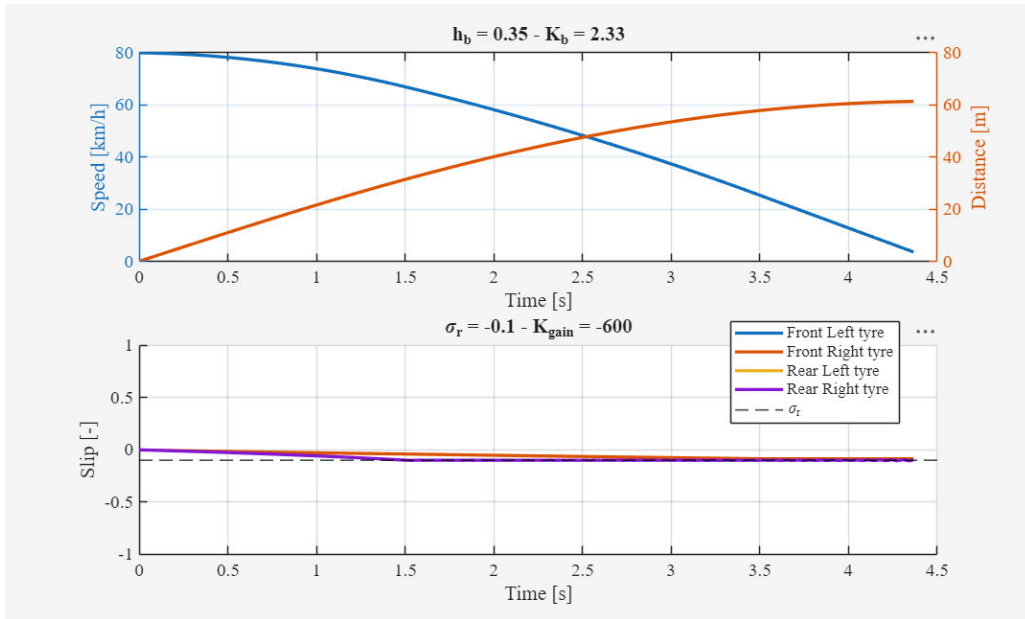


Figure 25: $\sigma_r = -0.1$: slip stays shallow and smooth, but the grip is under-used (operating point below the friction peak).

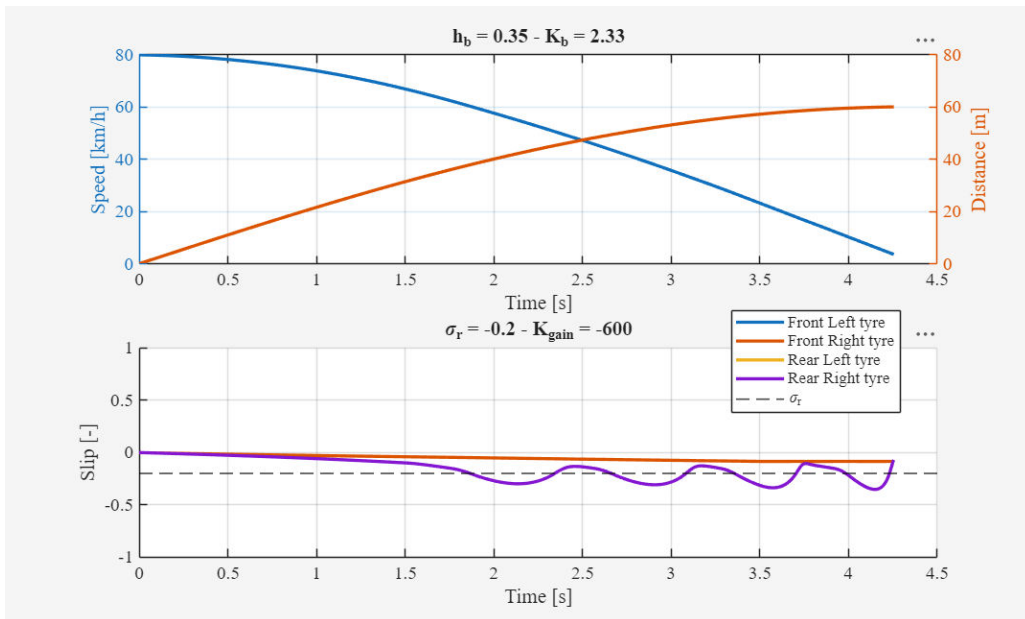


Figure 26: $\sigma_r = -0.2$: the slip oscillates in a bounded limit cycle around the target, close to the friction peak – best compromise.

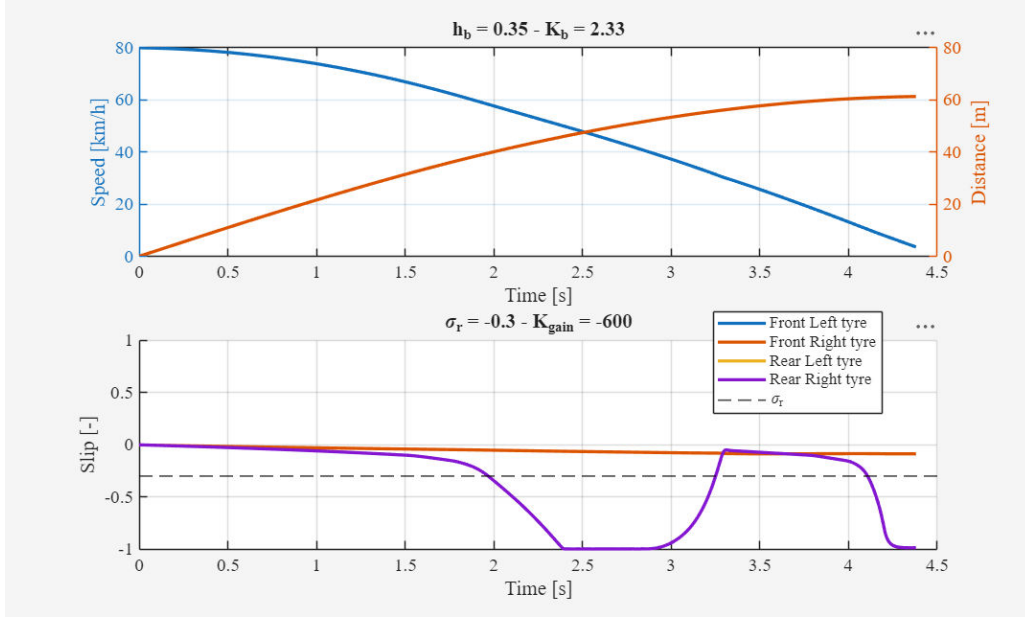


Figure 27: $\sigma_r = -0.3$: the target lies on the unstable branch; the rear slip repeatedly dives to -1 (intermittent lock) and recovers.

- $\sigma_r = -0.1$ – the target sits on the steep, *stable* part of the curve, well below the peak. The slip is smooth and free of oscillation, but the friction coefficient is not fully exploited, so the deceleration is lower and the stop is the longest (≈ 61 m). It is the safest but least efficient choice.
- $\sigma_r = -0.2$ – the target is at/near the peak of the friction curve. The slip oscillates in a small bounded limit cycle around -0.2 , the friction coefficient is close to $\mu_{\max}$ and the stop is slightly shorter (≈ 60 m) at the same gain. This is the best trade-off between braking force and stability, which is why -0.2 is selected.
- $\sigma_r = -0.3$ – the target is *beyond* the peak, on the unstable descending branch (Section 8). As soon as the slip exceeds the peak the friction drops, the wheel decelerates further and runs away: the rear slip dives to -1 for sustained intervals (around $t \approx 2.4 - 3$ s and again near $t \approx 4.2$ s) before the controller manages to recover it. This intermittent locking gives no distance benefit (≈ 61 m) while losing braking force and steerability.

The slip trends confirm the rule: the reference must be placed at the friction peak; a smaller magnitude wastes grip, a larger one destabilises into locking.

10.III Gain tuning for minimum stopping distance

Keeping $\sigma_r = -0.2$, the gain magnitude is increased so that the torque integrator reaches the regulating level sooner. Figure 28 shows the result for $K_{\text{gain}} = -1450$.

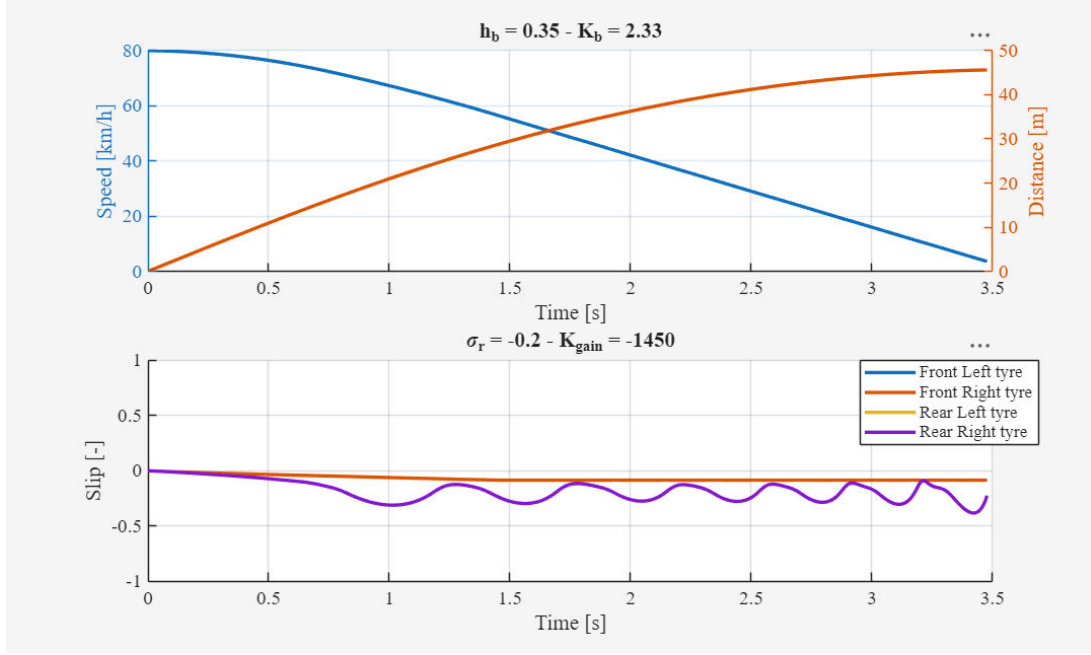


Figure 28: Tuned controller, $\sigma_r = -0.2$, $K_{\text{gain}} = -1450$: the stopping distance drops to ≈ 46 m while the slip still limit-cycles around the target without locking.

Raising $|K_{\text{gain}}|$ from 600 to 1450 shortens the stop from ≈ 60 m (4.25 s) to ≈ 46 m (≈ 3.5 s): the faster integrator builds the brake torque more quickly, so the wheels reach the high-friction slip region earlier. The slip keeps oscillating in a bounded cycle around $-0.2/-0.3$ but never locks. Increasing the gain further would enlarge the limit-cycle amplitude and risk overshooting past the friction peak into locking, so $K_{\text{gain}} = -1450$ is a good practical minimum-distance setting on dry road.

10.IV Wet-road conditions and re-tuning

The $\mu-\sigma$ block is switched to the wet characteristic and the gain is re-tuned. Figure 29 compares the dry calibration ($K_{\text{gain}} = -1450$) with the wet one ($K_{\text{gain}} = -700$), at $h_b = 0.5$.

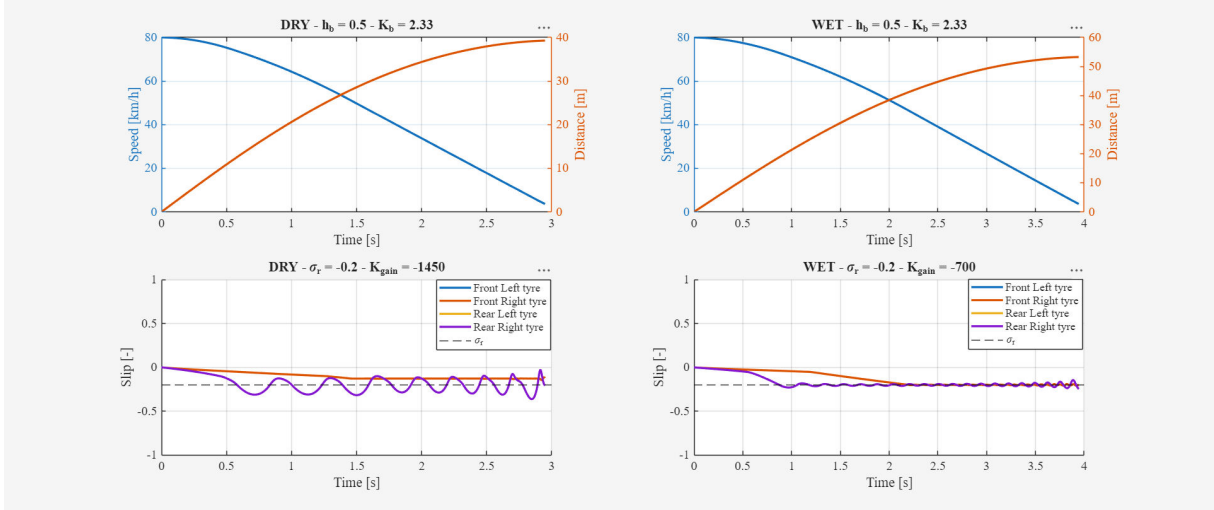


Figure 29: Dry (left, $K_{\text{gain}} = -1450$) versus wet (right, $K_{\text{gain}} = -700$) braking with $\sigma_r = -0.2$: speed/distance (top) and slip (bottom).

Three observations follow:

- **Distance.** The lower wet friction lengthens the stop from ≈ 39 m (≈ 3.0 s) on dry to ≈ 53 m (≈ 3.9 s) on wet – the unavoidable consequence of the reduced μ_{max} .
- **Re-tuning is necessary.** With the aggressive dry gain the dry slip already shows a *growing* limit cycle (bottom-left); on the lower-grip wet surface the same gain would overshoot the (lower, sharper) friction peak even more easily and drive the wheel towards locking. Reducing the gain to -700 yields a well-damped slip that settles cleanly at $\sigma_r = -0.2$ (bottom-right).
- **Surface dependence.** The optimal gain is therefore surface-dependent: a single fixed gain tuned on dry is too aggressive on wet. This lack of robustness of the fixed-gain slip controller motivates the threshold-based logic of Exercise 3.

10.V Feasibility on a real vehicle

The controller of equation (16) regulates the slip $\sigma = (R_e \omega - V)/V$, which requires the true vehicle ground speed V . While the wheel speed ω is readily measured by wheel-speed sensors, V is *not* directly measurable precisely when it matters: during hard braking the wheels slip, so the wheel speed no longer equals the vehicle speed. Obtaining V would need additional sensors (optical ground-speed sensor, GPS, inertial integration) or a dedicated state estimator, so the slip must be *estimated* rather than measured. In addition, the `sign`-based law produces chattering – continuous high-frequency brake modulation – which stresses the hydraulic valves and degrades comfort.

For these reasons this exact strategy cannot be installed as-is. Its practical relevance is nonetheless high: it defines the ideal control objective (hold the slip at the

friction peak), it is a valuable reference/benchmark for simulation and Hardware-in-the-Loop testing, and it makes explicit why production ABS infers the wheel state from the wheel *acceleration* instead of the slip – which is exactly the approach of Exercise 3.

11 Exercise 3 – ABS, Conjugate Boundary Method

Control logic and implementation

The conjugate-boundary ABS avoids any slip measurement and acts only on the wheel acceleration $\dot{\omega}$. It is implemented as the Stateflow machine of Figure 30, embedded in the wiring of Figure 31: the requested torque $T_{\text{req}} = T_{bf}^{\text{max}} h_b$, the wheel acceleration $\dot{\omega}$, the previous output (through a unit delay $1/z$, giving T_{in}) and a `wheel_lock` flag (= 1 when $|\dot{\omega}| < 0.1$) feed the controller, whose output T_{out} drives the brake.

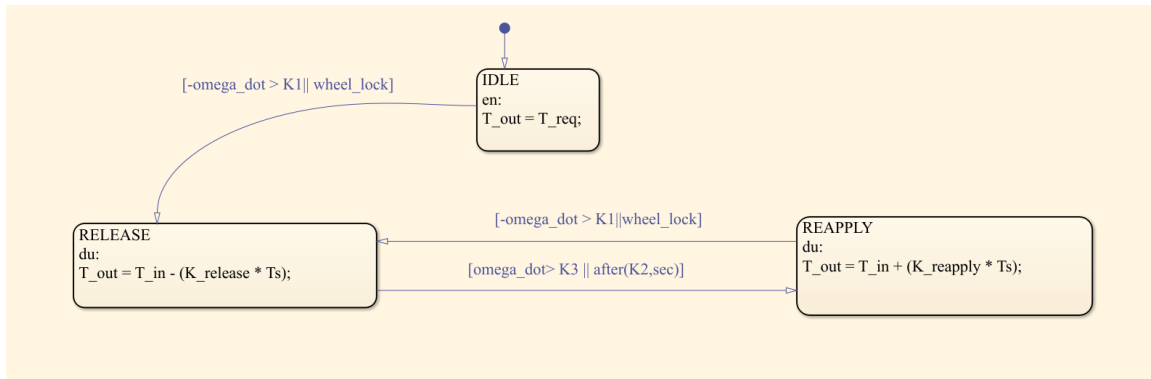


Figure 30: Stateflow machine: IDLE, RELEASE and REAPPLY with the corresponding output-torque updates and transition guards.

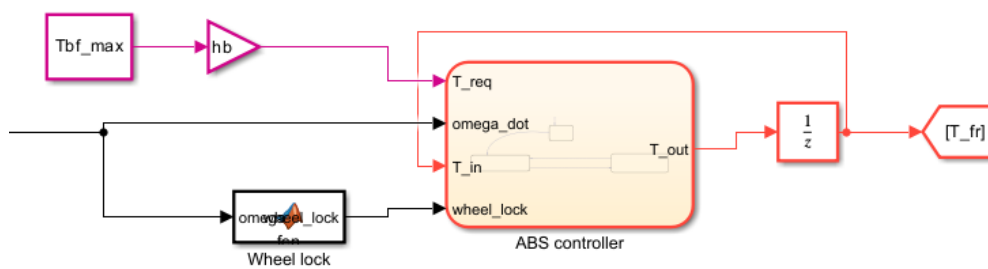


Figure 31: ABS-controller wiring: requested torque, wheel acceleration $\dot{\omega}$, delayed torque T_{in} and the wheel-lock flag drive the state machine.

The three states implement

$$\text{IDLE: } T_{\text{out}} = T_{\text{req}}, \quad (17)$$

$$\text{RELEASE: } T_{\text{out}}(k) = T_{\text{out}}(k-1) - K_{\text{release}} T_s, \quad (18)$$

$$\text{REAPPLY: } T_{\text{out}}(k) = T_{\text{out}}(k-1) + K_{\text{reapply}} T_s, \quad (19)$$

with the guards

- $\rightarrow$ **RELEASE** when $-\dot{\omega} > K_1$ (the wheel decelerates too fast) *or* wheel_lock;
- **RELEASE** $\rightarrow$ **REAPPLY** when $\dot{\omega} > K_3$ (the wheel spins back up) *or* after(K_2 , sec) (time-out).

The system thus dumps brake pressure as soon as a skid is detected and rebuilds it – more slowly – once the wheel recovers, producing the characteristic saw-tooth pressure modulation that keeps the slip near the friction peak *without ever computing it*. Following the assignment hints, the triggers are started from $K_1 \approx K_3$, the charge/discharge rates are kept constant within each scenario, and $K_{\text{reapply}} < K_{\text{release}}$ to avoid chattering. The two surfaces use the settings of Table 6.

Parameter	Wet	Snow	Unit
K_1 (decel. trigger)	40	40	m/s ²
K_3 (accel. trigger)	35	35	m/s ²
K_2 (max time RELEASE)	0.06	0.06	s
K_{release}	4500	5500	Nm/s
K_{reapply}	500	200	Nm/s
h_b / K_b	0.6/1.8	0.5/1.8	–

Table 6: Conjugate-boundary ABS settings for the two surfaces.

11.1 Wet-surface results

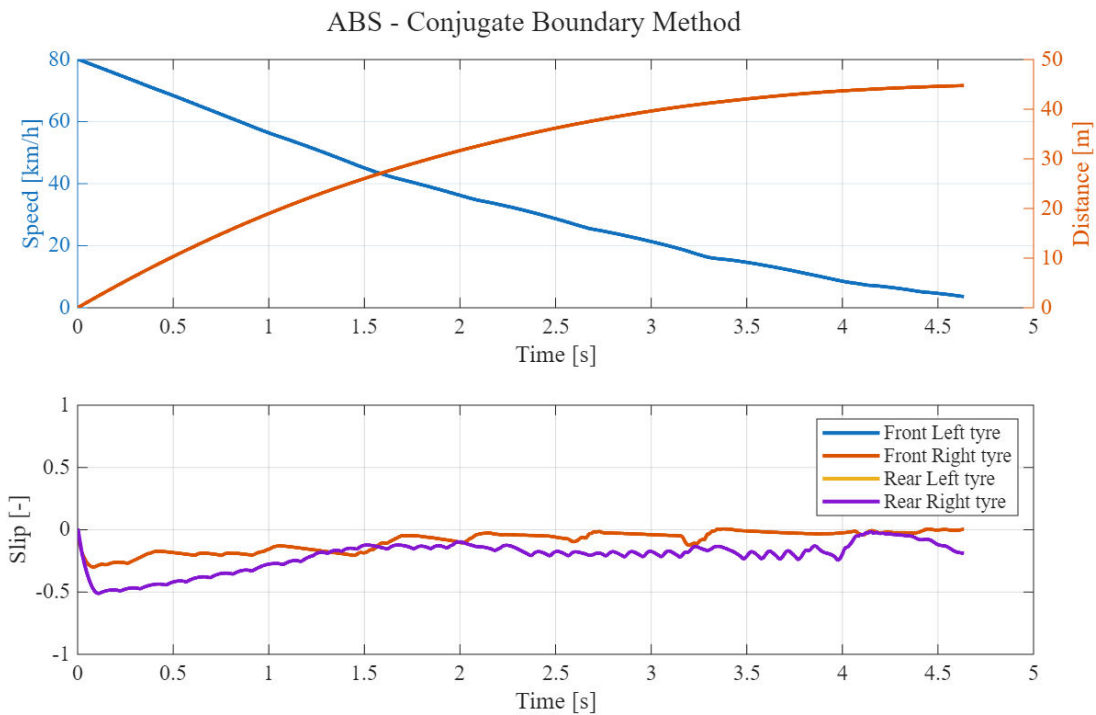


Figure 32: Wet surface: speed/distance (top) and tyre slip (bottom). The slip is held in the useful band after a brief initial spike.

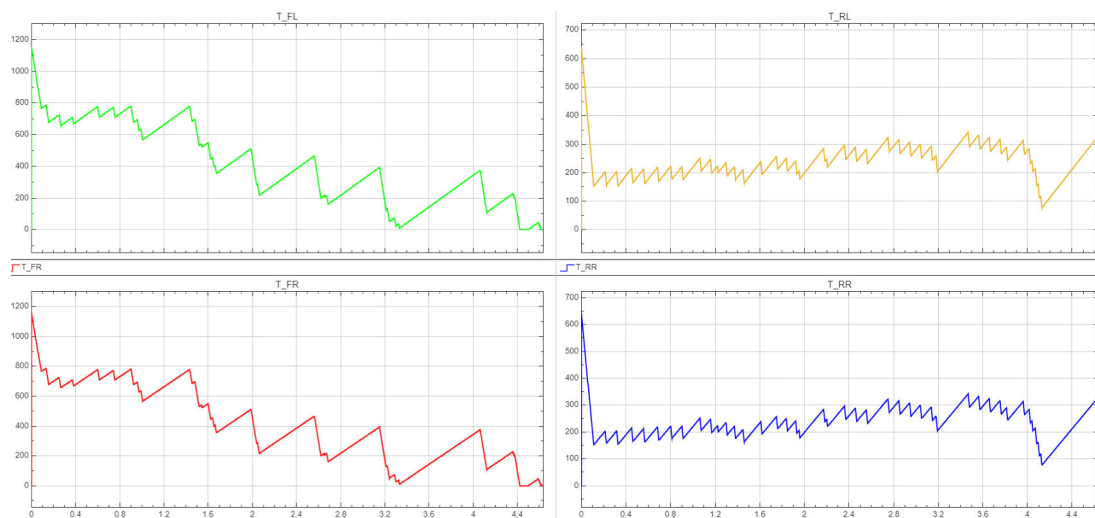


Figure 33: Wet surface: brake-torque saw-tooth on the four wheels (front $\sim$ up to 1100 Nm, rear $\sim$ up to 600 Nm).

On the wet road the vehicle stops in ≈ 4.6 s over ≈ 45 m. After a short initial dip to $\sigma \approx -0.5$ – the transient before the state machine catches the first skid – the slip of every wheel is held in the useful band (≈ -0.1 to -0.3), oscillating around the friction peak instead of locking. The brake torque (Figure 33) shows the expected saw-tooth: a fast drop in RELEASE followed by a slower ramp in REAPPLY, repeated as a limit

cycle; the more loaded front wheels carry the higher torque (~ 1100 Nm peak) than the rear (~ 600 Nm).

11.II Snow-surface results

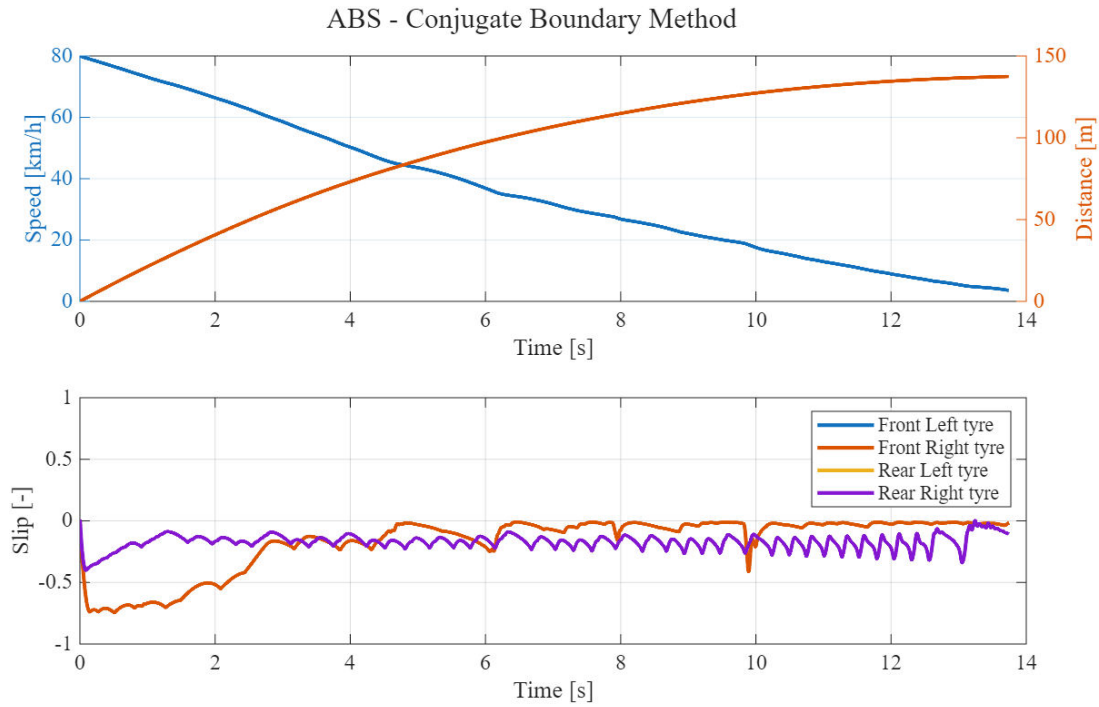


Figure 34: Snow surface: speed/distance (top) and tyre slip (bottom). Very long stop, slip still kept bounded by the modulation.

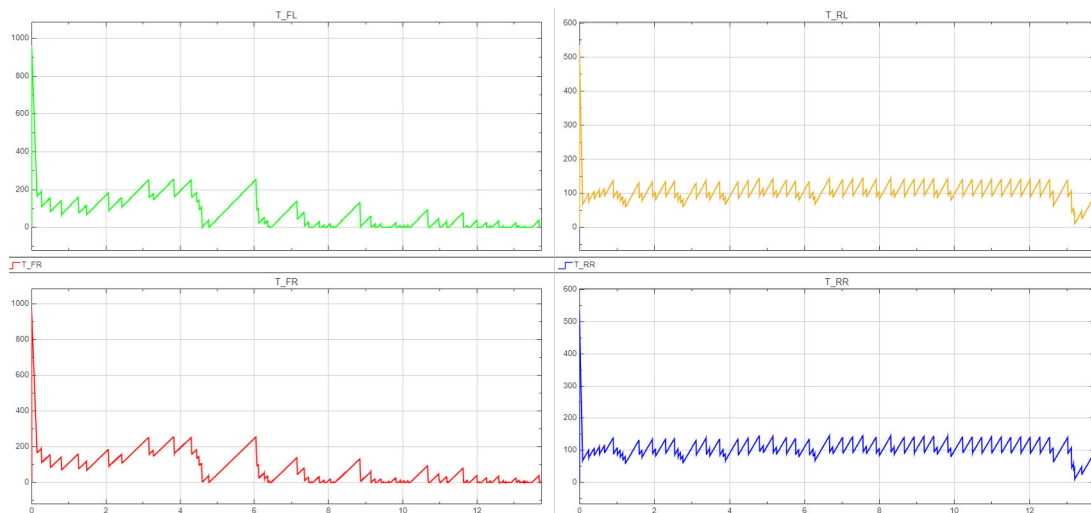


Figure 35: Snow surface: finer and more frequent torque modulation at lower levels (front $\sim$ up to 950 Nm, rear ~ 100 –150 Nm).

On snow the very low friction stretches the stop to ≈ 13.7 s and ≈ 140 m. Despite the extreme conditions the controller still prevents sustained lock: after an initial front

excursion to $\sigma \approx -0.75$ the slip is brought back and modulated around -0.1 to -0.4 for the rest of the manoeuvre. The torque (Figure 35) cycles more frequently and at much lower levels than on wet; the faster discharge ($K_{\text{release}} = 5500$) and the gentler recharge ($K_{\text{reapply}} = 200$) are precisely what keeps the slippery wheel from re-locking and avoids chattering.

11.III Comments

Using only the wheel acceleration and a lock flag, the conjugate-boundary ABS keeps the slip in the high-friction region on both low- μ surfaces – no slip estimation is required, which is the whole point of the method and its key practical advantage over the controller of Exercise 2. The stopping distance scales directly with the surface friction (snow $\gg$ wet), and the saw-tooth brake-torque trace reproduces the pressure modulation of real production ABS. The price to pay is a coarser, oscillatory regulation around the peak rather than the precise slip tracking of the (non-deployable) Exercise-2 controller.

12 Benchmark and Summary

Table 7 collects the stopping distances of all the manoeuvres. The figures are read from the speed/distance histories; distances of different exercises are not directly comparable because the brake command h_b differs, but within each block the trends are consistent.

Exercise / case	Settings	Dist. [m]	Time [s]	Wheel behaviour
Ex1 – gentle	$h_b=0.25$, $K_b=2.33$	≈ 47	> 4	no lock (rolls)
Ex1 – hard	$h_b=0.90$, $K_b=1.7$	≈ 35	3.05	all wheels locked
Ex1 – +15%	$m=1173$, $C_x=0.38$	≈ 33	2.9	rear lock @ 0.6 s
Ex1 – -15%	$0.85 \times$ nominal	≈ 30	2.6	rear lock @ 0.45 s
Ex1 – geared/neutral	$h_b=1$	≈ 35	3.05	identical, all locked
Ex2 – $\sigma_r=-0.1$	$K_{\text{gain}}=-600$	≈ 61	4.35	shallow, stable
Ex2 – $\sigma_r=-0.2$	$K_{\text{gain}}=-600$	≈ 60	4.25	limit cycle @ peak
Ex2 – $\sigma_r=-0.3$	$K_{\text{gain}}=-600$	≈ 61	4.35	intermittent lock
Ex2 – tuned	$\sigma_r=-0.2$, $K_{\text{gain}}=-1450$, $h_b=0.35$	≈ 46	3.5	best dry slip ABS
Ex2 – dry	$K_{\text{gain}}=-1450$, $h_b=0.5$	≈ 39	3.0	limit cycle (growing)
Ex2 – wet	$K_{\text{gain}}=-700$, $h_b=0.5$	≈ 53	3.9	well-damped @ -0.2
Ex3 – wet	conjugate boundary	≈ 45	4.6	bounded, no lock
Ex3 – snow	conjugate boundary	≈ 140	13.7	bounded, no lock

Table 7: Stopping-distance benchmark across the three exercises.

13 Conclusions

The project followed the braking problem from the open-loop driveline model to two complementary ABS designs.

- **Exercise 1** quantified the open-loop behaviour: increasing h_b shortens the stop only until the wheels lock, a larger K_b (front bias) protects the rear axle, the stopping distance is dominated by the vehicle mass (with a $\pm 15\%$ swing giving $\approx \pm 5\%$ distance) and weakly by the drag, and – under a hard lock-up stop – the engine braking of the engaged gear makes no practical difference with respect to neutral.
- **Exercise 2** showed that a slip-regulating controller recovers the lost braking by holding σ at the friction peak. The reference $\sigma_r = -0.2$ is the right target (below it the grip is under-used, beyond it the wheel locks), increasing $|K_{\text{gain}}|$ minimises the distance up to the onset of locking, and the optimal gain is surface-dependent (it had to be reduced from -1450 on dry to -700 on wet). The strategy is, however, not directly deployable because the slip is not measurable on a real vehicle.
- **Exercise 3** delivered a deployable ABS that uses only the wheel acceleration. The IDLE/RELEASE/REAPPLY state machine modulates the brake pressure in a saw-tooth limit cycle and keeps the slip in the useful band on both wet (≈ 45 m) and snow (≈ 140 m) surfaces, with no slip estimation and no sustained lock.

The overall message is consistent with real ABS practice: locking the wheels is never optimal – the shortest *and* controllable stop is obtained by keeping the longitudinal slip near the friction peak. A slip controller defines that ideal target but needs an unmeasurable quantity, whereas the acceleration-based conjugate-boundary method approximates the same behaviour with quantities that are actually available on board, which is why it is the approach adopted in practice.